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Schupp et al.

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(54) **METHODS AND ASSAYS FOR DETECTION AND SUBTYPING OF MICROBIAL PATHOGENS**

(71) Applicants: **The Translational Genomics Research Institute**, Phoenix, AZ (US); **Arizona Board of Regents on Behalf of Northern Arizona University**, Flagstaff, AZ (US)

(72) Inventors: **James Schupp**, Flagstaff, AZ (US); **Jason Sahl**, Phoenix, AZ (US); **Paul Keim**, Flagstaff, AZ (US); **David Engelthaler**, Flagstaff, AZ (US); **David Wagner**, Phoenix, AZ (US); **Viacheslav Fofanov**, Phoenix, AZ (US)

(73) Assignees: **The Translational Genomics Research Institute**, Phoenix, AZ (US); **Arizona Board of Regents on Behalf of Northern Arizona University**, Flagstaff, AZ (US)

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(51) **Int. Cl.**

C12Q 1/689 (2018.01)

C12Q 1/04 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **C12Q 1/689** (2013.01); **C12Q 1/04**

(2013.01); **C12Q 1/6844** (2013.01); **C12Q**

1/686 (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC **C12Q 1/689**

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2004/0259226 A1* 12/2004 Robey **C12Q 1/689**
435/252.3

2006/0003352 A1* 1/2006 Lipkin **C12Q 1/686**
435/6.14

(Continued)

FOREIGN PATENT DOCUMENTS

WO 2015/070187 A2 5/2015

OTHER PUBLICATIONS

Genbank CP019726.1—*Bacillus anthracis* strain Sterne 34F2 genome—
NCBI (Year: 2017).*

(Continued)

Primary Examiner — Aaron A Priest

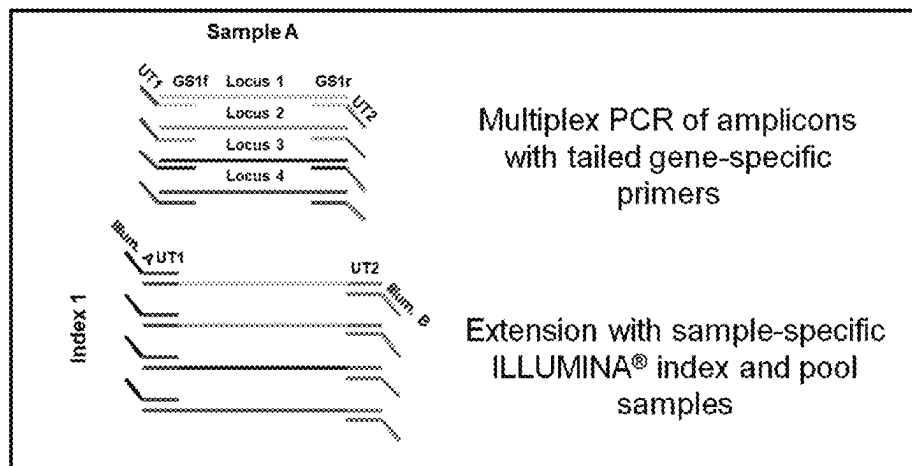
Assistant Examiner — Tian Yu

(74) *Attorney, Agent, or Firm* — FULLER IP LAW LLC;
Rodney J. Fuller

(57) **ABSTRACT**

The present invention provides methods of detecting a biothreat agent in a sample, comprising detecting at least one biothreat-specific amplicon in the sample. The methods also

(Continued)



encompass confirming the absence of the biothreat agent by detecting Near Neighbor specific amplicons to avoid false positive results.

20 Claims, 27 Drawing Sheets

Specification includes a Sequence Listing.

(51) Int. Cl.

C12Q 1/6844 (2018.01)

C12Q 1/686 (2018.01)

C12R 1/07 (2006.01)

(52) U.S. Cl.

CPC ... *C12Q 2600/156* (2013.01); *C12Q 2600/16* (2013.01); *C12R 2001/07* (2021.05)

(56) References Cited

U.S. PATENT DOCUMENTS

2006/0057603	A1 *	3/2006	Radnedge	<i>C12Q 1/689</i> 435/6.12
2013/0149708	A1	6/2013	Riojas et al.	
2016/0326572	A1	11/2016	Schupp et al.	
2018/0173843	A1	6/2018	Lemmer et al.	

OTHER PUBLICATIONS

Afshinnakoo, E., et al. Geospatial Resolution of Human and Bacterial Diversity with City-Scale Metagenomics. *Cell Systems* 2015; 1(1):72-87.

Agaisse, H., et al. PlcR is a pleiotropic regulator of extracellular virulence factor gene expression in *Bacillus thuringiensis*. *Microbiology* 1999; 32(5):1043-1053.

Slamti, L., et al. Distinct mutations in PlcR explain why some strains of *Bacillus cereus* group are nonhemolytic. *Journal of Bacteriology* 2004; 186(11):3531-3538.

Kolsto, A., et al. What Sets *Bacillus anthracis* Apart from Other *Bacillus* Species? *Annual Review of Microbiology* 2009; 63(1):451-476.

Lau, S. Y., et al. Cell Division Defects in *Escherichia coli* Deficient in the Multidrug Efflux Transporter AcrEF-TolC. *J Bacteriol* 2005; 187(22):7815-7825.

Nagakubo, S., et al. The putative response regulator BaeR stimulates multidrug resistance of *Escherichia coli* via a novel multidrug exporter system, MdtABC. *J Bacteriol* 2002; 184(15):4161-4167.

Tanabe, H., et al. Growth phase-dependent transcription of *emrKY*, a homolog of multidrug efflux *emrAB* genes of *Escherichia coli*, is induced by tetracycline. *J Gen Appl Microbiol* 1997; 43(5):257-263.

Sulavik, M. C., et al. Antibiotic susceptibility profiles of *Escherichia coli* strains lacking multidrug efflux pump genes. *Antimicrob Agents Chemother* 2001; 45(4):1126-1136.

Bharat, A., et al. Effect of Variants of Penicillin-Binding Protein 2 on Cephalosporin and Carbapenem Susceptibilities in *Neisseria gonorrhoeae*. *Antimicrob Agents Chemother* 2015; 59(8):5003-5006.

Sun, S., et al. Resistance to β -Lactam Antibiotics Conferred by Point Mutations in Penicillin-Binding Proteins PBP3, PBP4 and PBP6 in *Salmonella enterica*. *PLoS One* 2014; 9:e97202.

Scholz, P., et al. Complete nucleotide sequence and gene organization of the broad-host-range plasmid RSF1010. *Gene* 1989; 75(2):271-288.

Agerso, Y., et al. Identification of Tet 39, a novel class of tetracycline resistance determinant in *Acinetobacter* spp. of environmental and clinical origin. *J Antimicrob Chemother* 2005; 55(4):566-569.

Liu, B., et al. ARDB—Antibiotic Resistance Genes Database. *Nucl Acids Res* 2009; 37 (suppl 1): D443-D447.

McArthur, A. G., et al. The comprehensive antibiotic resistance database. *Antimicrob Agents Chemother* 2013; 57(7):3348-3357.

Goodwin, S., et al. Coming of age: ten years of next-generation sequencing technologies. *Nat Rev Genet* 2016; 17(6):333-351.

GenBank submission EF410354, Feb. 12, 2007 (online); Retrieved May 27, 2020 from the internet <URL: <https://www.ncbi.nlm.nih.gov/nucleotide/EF410354>>nt 1344-1363.

GenBank submission CP003187, Jan. 30, 2014 (online); Retrieved May 27, 2020 from the internet <URL: <https://www.ncbi.nlm.nih.gov/nucleotide/CP003187>>nt 4396310-4396329, nt 4396521-4396542.

* cited by examiner

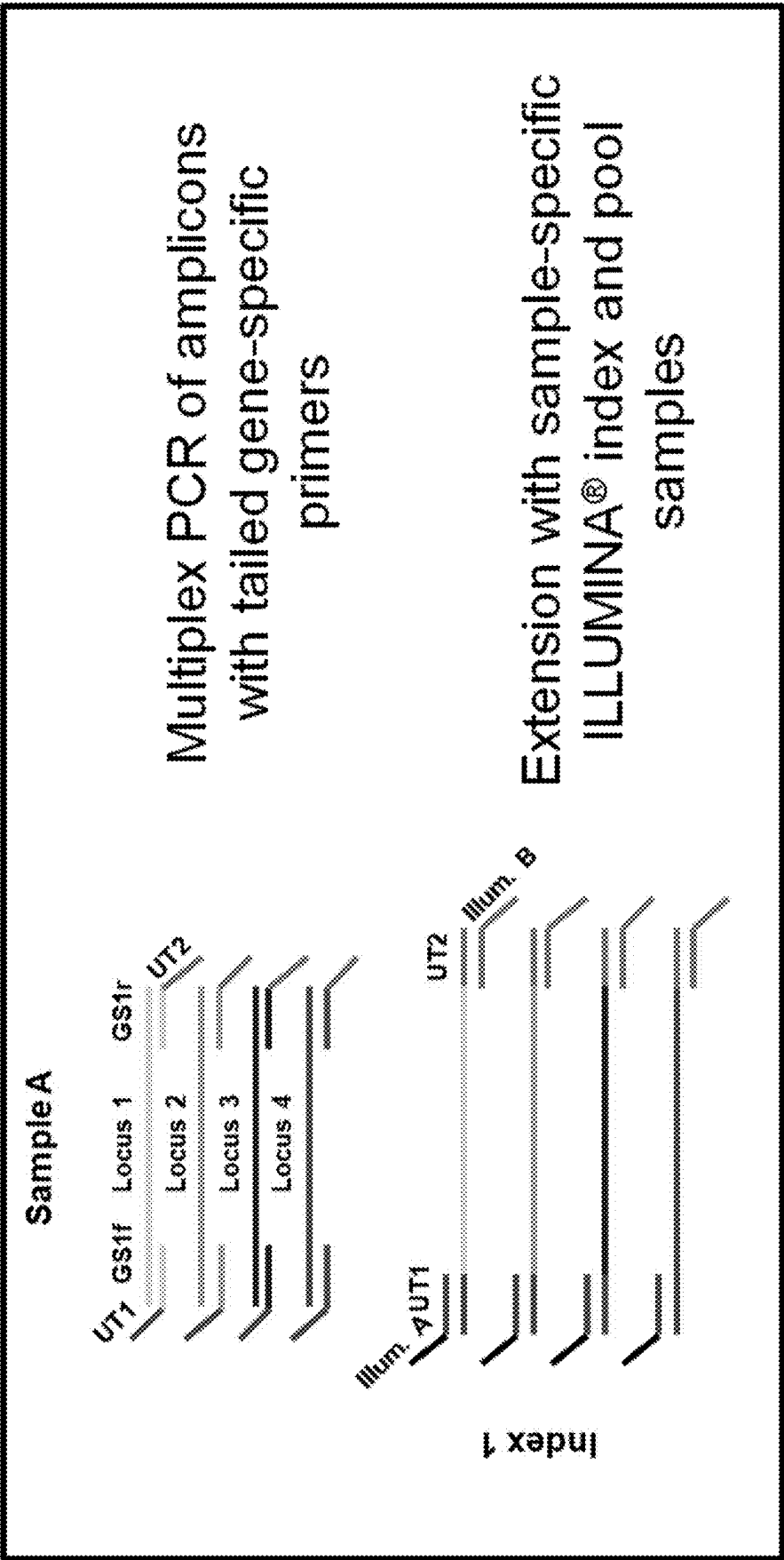


FIG. 1A

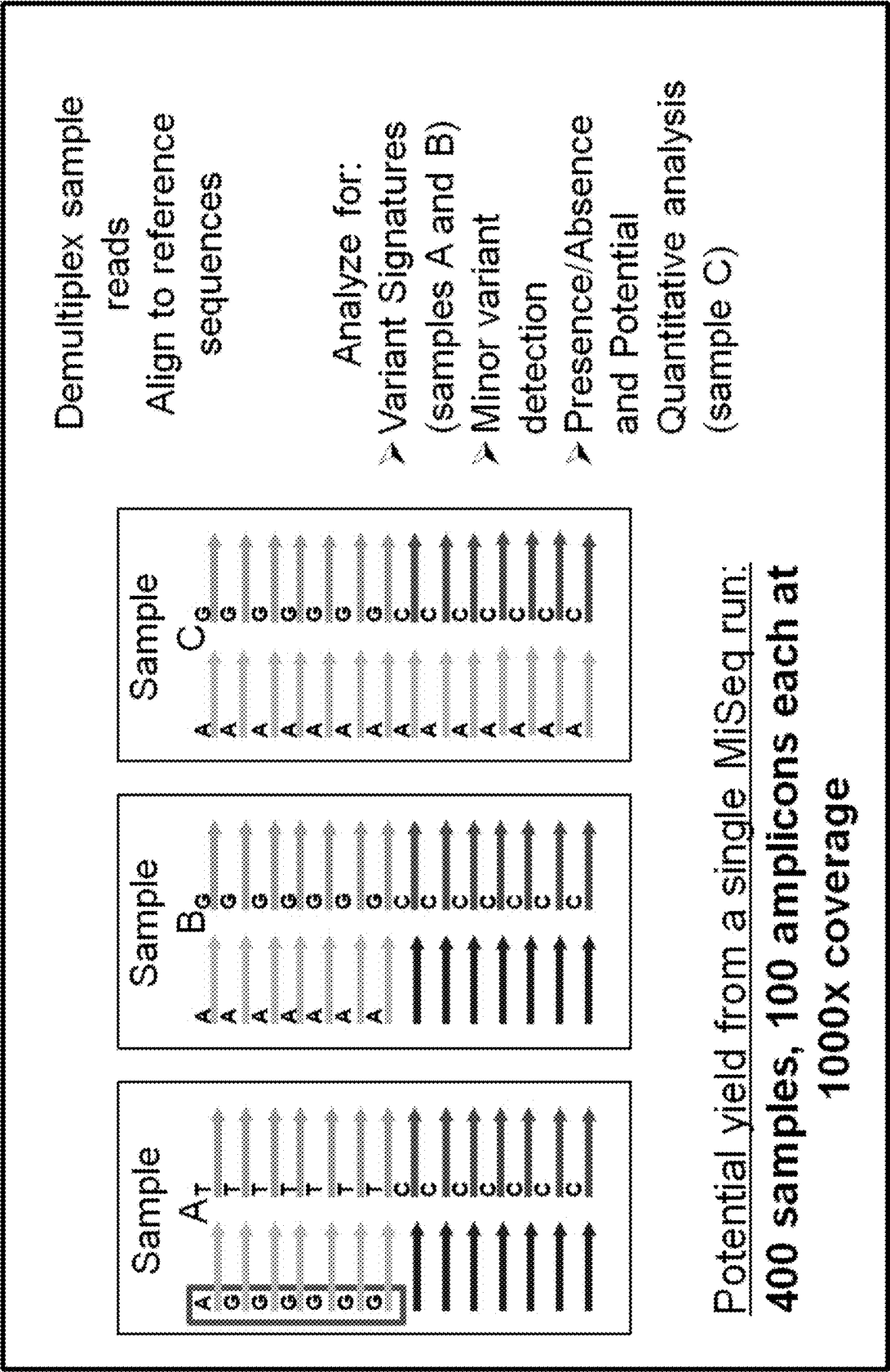


FIG. 1B

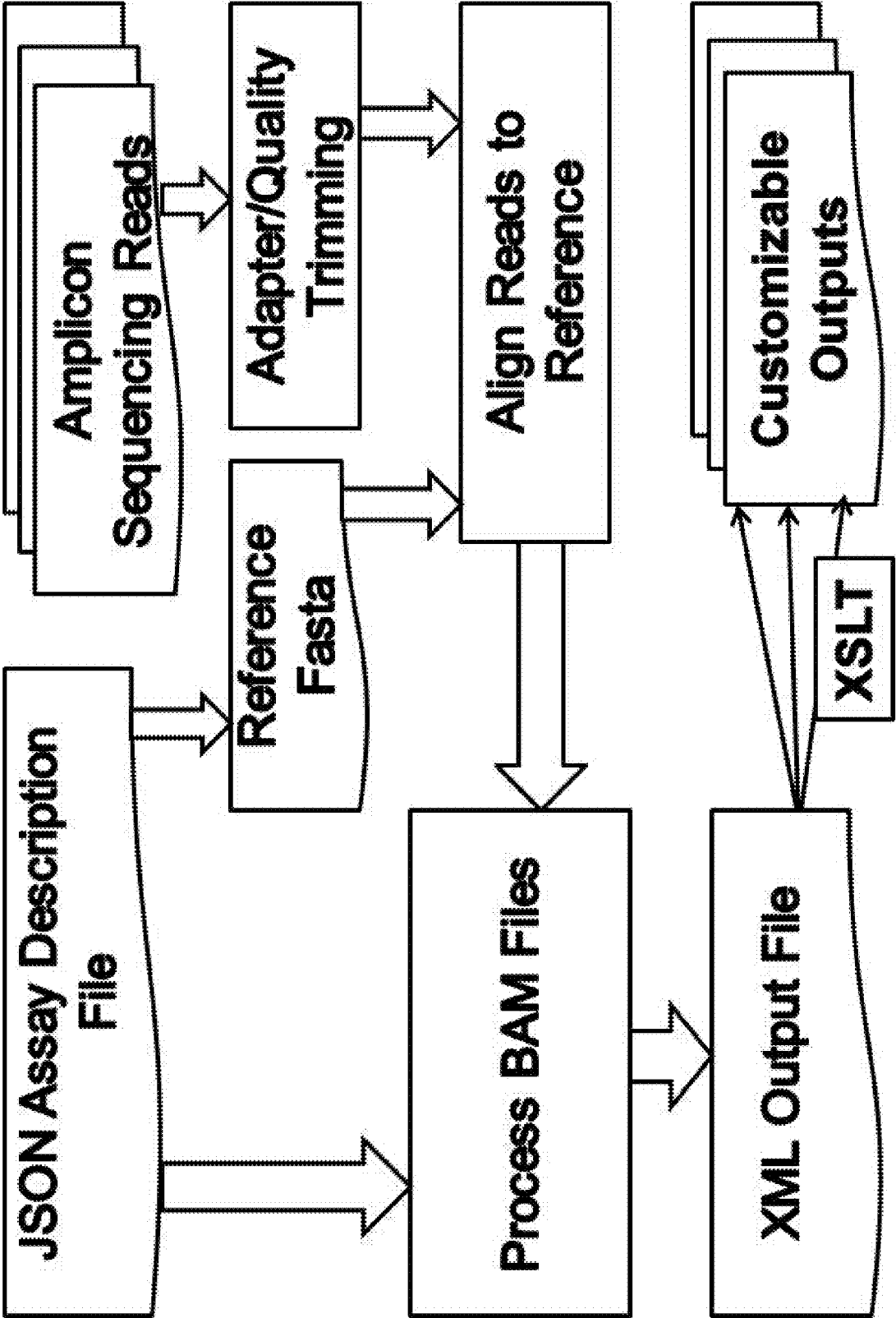


FIG. 2

[illegible]

FIG. 3C

FIG. 4D

[illegible]

FIG. 5A

[illegible]

FIG. 5C

FIG. 5D

Species	TTS1	pseudomallet				pseudomallet mallet				mallet		pseudomallet complex SNP	repacta complex SNPs	Bpe MSS		AMR		
pseudomallet	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
pseudomallet	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
pseudomallet	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
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pseudomallet	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
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mallet	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
humphreensis	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
humphreensis	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
humphreensis	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
humphreensis	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
MS148254-IIIa	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
MS148264-IIIa	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+

FIG. 6A

[illegible]

FIG. 6B

[illegible]

FIG. 6C

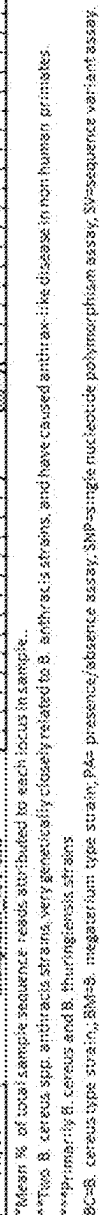


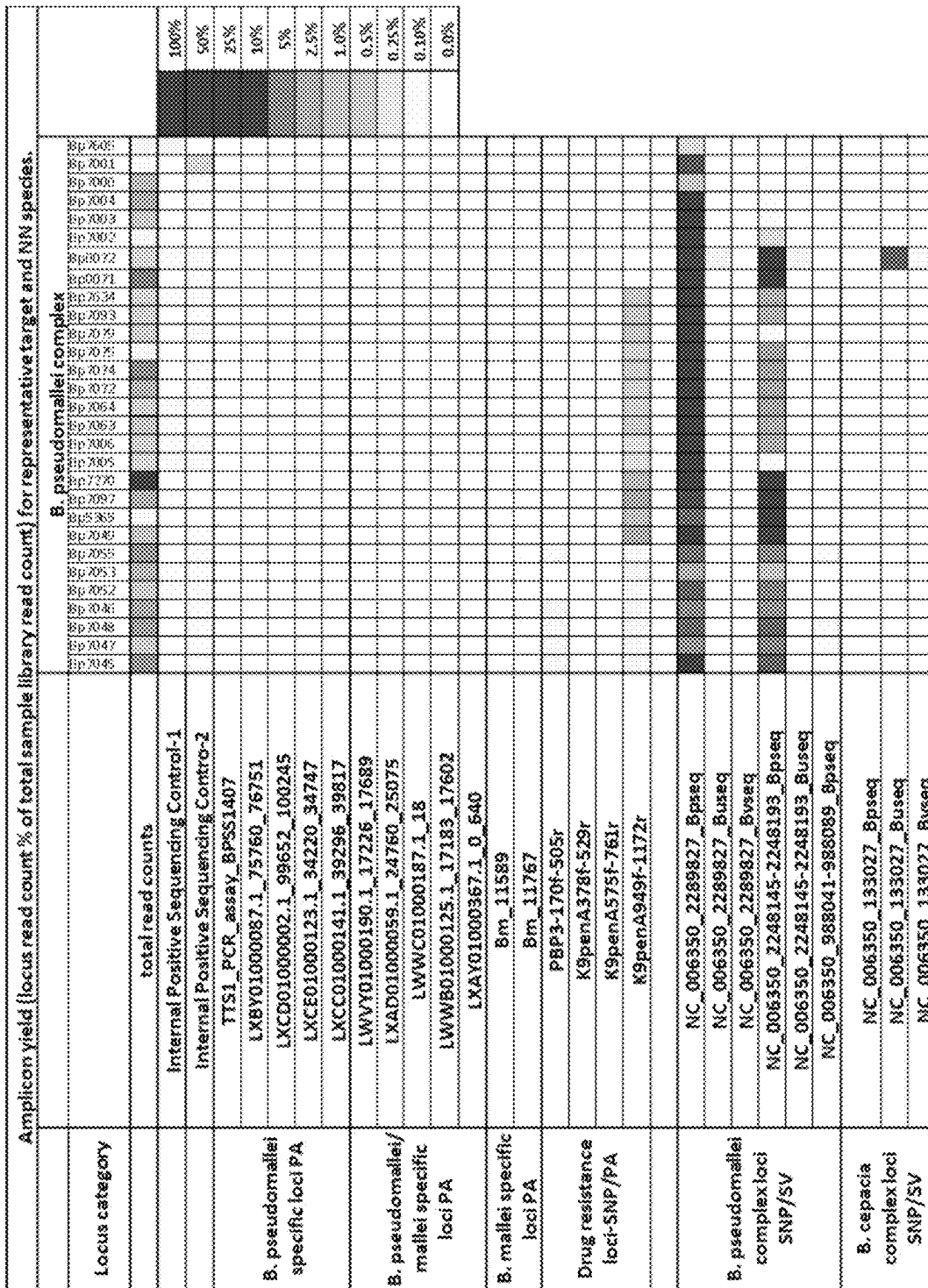
FIG. 7

Amplicon yield (focus read count % of total sample library read count) for representative target and NN species.																									
Locus category	total read counts	B. pseudomallei / mallei																							Mean %*
		Bp00021	Bp00113	Bp00112	Bp73821	Bp6903	Bp6339	Bp6093	Bp5710	Bp4118	Bp4058	Bp4042	Bp4003	Bp1002	Bp3839	Bp1972	Bp1837	Bp0938	Bp0704	Bp0668	Bp0371	Bp0347	Bp0245		
	Internal Positive Sequencing Control-1																								100%
	Internal Positive Sequencing Control-2																								50%
	TTS1_PCR_assay_BPSS1407																								25%
B. pseudomallei specific loci PA	LXBY01000087.1_75760_76751																								10%
	LXCD01000002.1_99652_100245																								5%
	LXCE01000123.1_34220_34747																								2.5%
	LXCC01000141.1_39296_39817																								1.0%
B. pseudomallei/ mallei specific loci PA	LWVY01000190.1_17226_17689																								0.5%
	LXAD01000059.1_24760_25075																								0.25%
	LWAWC01000187.1_18																								0.10%
	LWWB01000125.1_17183_17602																								0.0%
B. mallei specific loci PA	LXAY01000367.1_0_640																								
	Bm_11589																								
	Bm_11767																								
	PBP3-170f-505r																								
Drug resistance loci-SNP/PA	K9penA378f-529r																								
	K9penA575f-761r																								
	K9penA949f-1172r																								
B. pseudomallei complex loci SNP/SV	NC_006350_2289827_Bpseq																								
	NC_006350_2289827_Buseq																								
	NC_006350_2289827_Bvseq																								
	NC_006350_2248145-2248193_Bpseq																								
B. cepacia complex loci SNP/SV	NC_006350_2248145-2248193_Buseq																								
	NC_006350_988041-988089_Bpseq																								
	NC_006350_133027_Bpseq																								
	NC_006350_133027_Buseq																								
	NC_006350_133027_Bvseq																								

*Mean % of total sample sequence reads attributed to each locus in sample.

5pA=presence/absence assay; 5NP=single nucleotide polymorphism assay; 5V=sequence variant assay.

FIG. 8A



**Mean % of total sample sequence reads attributed to each locus in sample.

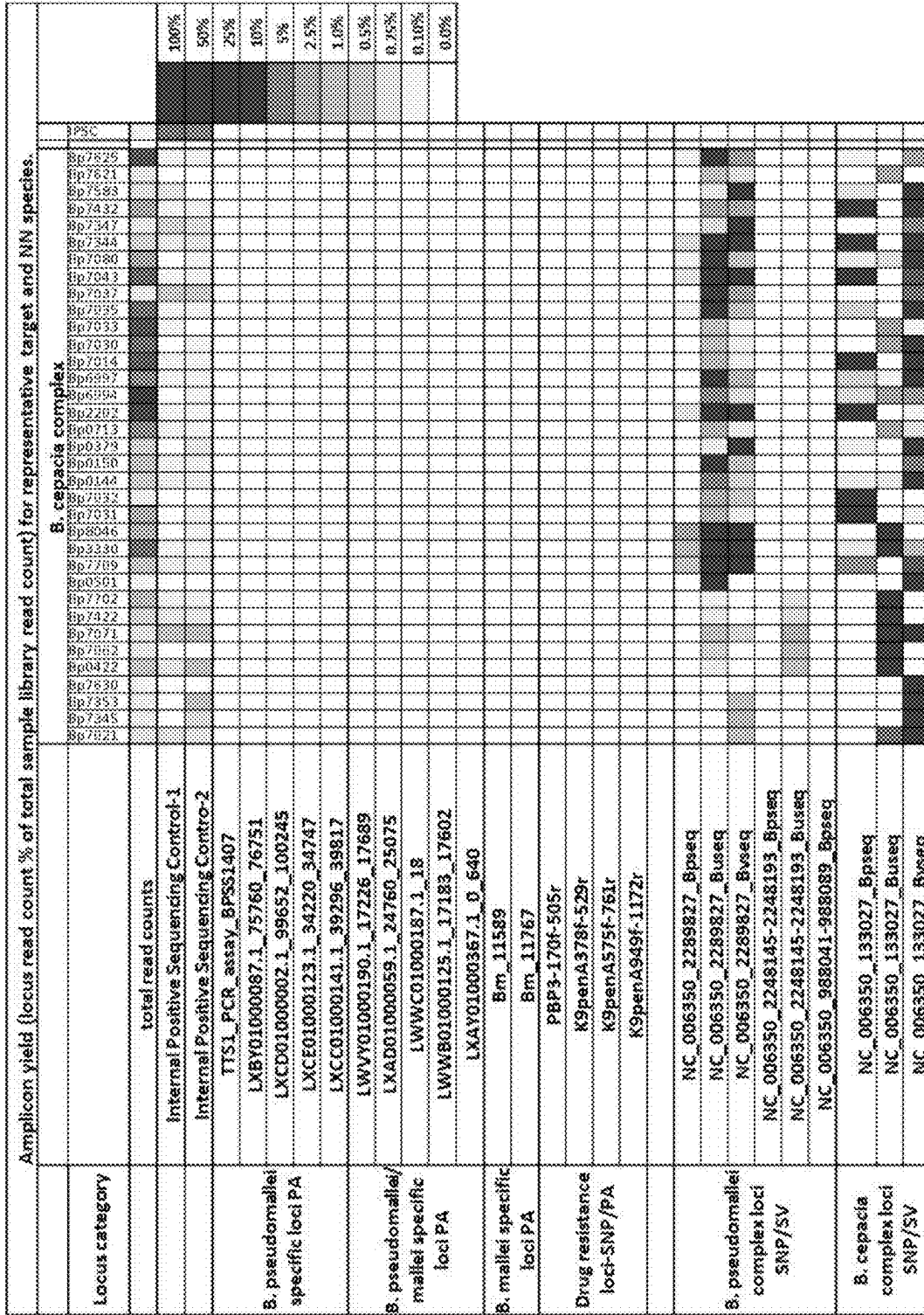
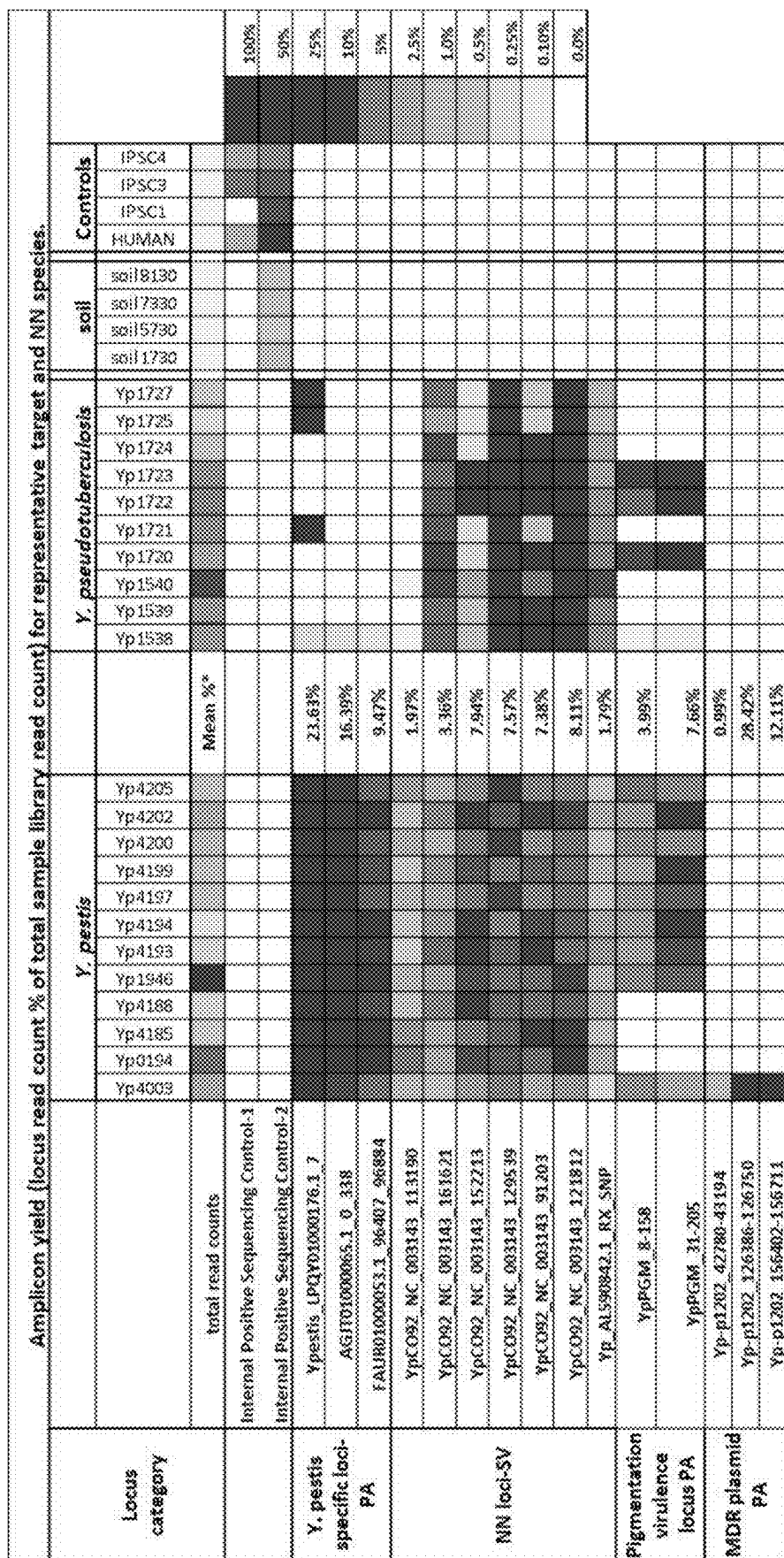


FIG. 8C



*Mean % of total sample sequence reads attributed to each locus in sample.

PA= presence/absence assay; SNP=single nucleotide polymorphism assay; SV=sequence variant assay.

Locus category	Amplicon yield (locus read count % of total sample library read count) for representative target and NN species.																				
	F. tularensis										F. tularensis spp. novicida/novicida-like					F. philomiragia					
	F0003	F0005	F0007	F0016-h	F0017-h	F0080	F0309	F0311	F0319	F0325	F1074	F1106	F1107	F1108	F1110	F0364	F0528	F0051	F0052	F0771	
F. tularensis specific loci PA	Total read counts										Mean %*										
	IPSC 2																				
	Ft_dup_CP000915.1_197										6.52%										
	Ftularensis_CP000915.1_1762										11.72%										
	Ftularensis_CP000915.1_731										0.02%										
NN loci SNP/SV	FLVS_AM233362_1562618										19.24%										
	FLVS_AM233362_1643765										13.72%										
	FLVS_AM233362_1648546										0.88%										
	FtA										14.18%										
	FtA1										16.13%										
NN species loci PA	FtA2										3.81%										
	FtB										13.21%										
	Ftnovicida_CP009607.1																				
	Fphilum_CP009444.1_285																				
	Fphilum_CP009444.1_569																				

*Mean % of total sample sequence reads attributed to each locus in sample.

PA= presence/absence assay; SNP=single nucleotide polymorphism assay; SV=sequence variant assay.

FIG. 10

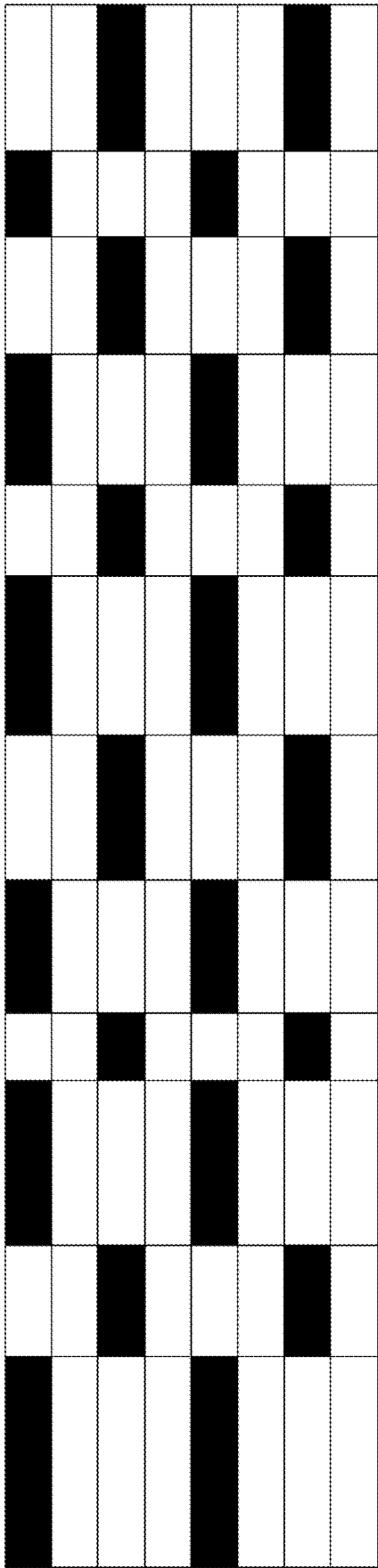


FIG. 11

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METHODS AND ASSAYS FOR DETECTION AND SUBTYPING OF MICROBIAL PATHOGENS

CROSS REFERENCE TO RELATED APPLICATIONS

This application is the U.S. National Stage of International Application No. PCT/US2020/015395, filed on Jan. 28, 2020, which claims priority to U.S. Provisional Patent Application No. 62/798,463, filed on Jan. 29, 2019, the contents of each of which are incorporated herein by reference in their entireties.

REFERENCE TO SEQUENCE LISTING SUBMITTED ELECTRONICALLY

The official copy of the sequence listing is submitted electronically via EFS-Web as an ASCII-formatted sequence listing with a file named “91482_239PCT_SeqList_ST25.txt” created on Jan. 27, 2020 and having a size of 83.2 kilobyte, and is filed concurrently with the specification. The sequence listing contained in this ASCII-formatted document is part of the specification and is herein incorporated by reference in its entirety.

TECHNICAL FIELD

This invention relates to methods, primers, assays, and kits for detecting the presence of microbial pathogens in a sample.

BACKGROUND

Throughout recent history, various aggressor nation states and terrorist groups have shown the willingness and/or capability to develop and use biological weapons against war fighters and civilian populations. The ability to detect the agents being developed as well as their virulence and antibiotic resistance profiles, in environmental and clinical materials, would further our capability to detect the development of these agents and their use.

The goal of several federal biosurveillance projects has been the early detection of biothreat agents to prevent or curtail mass civilian or military casualties. These systems have relied upon real-time-PCR to give a binary answer of presence or absence of the target. One challenge has been the complexity of the environmental samples, where tens of thousands of microorganisms exists, many of which are highly similar to the target pathogens. BioWatch is an example where numerous false positive results have been generated due to poorly known near-neighbor species confusing individual assays. While our knowledge of near-neighbors and of the target Biothreat agents is rapidly increasing, it is unrealistic to ever expect complete knowledge of either. DNA sequencing offers great potential, and there is a need for primers, methods, assays, and kits with greater ability to discriminate microbial pathogens in complex environmental and clinical sample matrices.

SUMMARY

Timely and accurate detection and characterization of bacterial biothreat agents is vital for our nation's safety. Current systems for early detection of these agents rely upon single locus Polymerase Chain Reaction (PCR) methods,

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giving only presence/absence results. This methodology can and has led to false positives due to limited signature validation. The Inventors have developed a multi-agent multi-locus amplicon sequencing protocol encompassing 79 targets aimed at detecting the presence or absence of 5 biothreat agents, as well as the presence and sequence of plasmids, virulence factors, antimicrobial resistance factors, and sequence variant loci for Near Neighbor species differentiation. The agents targeted are *Burkholderia pseudomallei*, *Burkholderia mallei*, *Bacillus anthracis*, *Yersinia pestis*, and *Francisella tularensis*.

The multi-agent assay, consisting of two multiplex amplification reactions, was validated against a diverse subset of target agent and near neighbor panels that were previously used to validate assays targeting individual agents. These panels consisted of 10-14 target agent strains and 11-48 NN strains. Sensitivity was 100% for all target agents, specificity was 91-100%. Targeted amplicon sequencing utilizing a universal amplicon indexing scheme provides a superior alternative to the current single locus PCR systems and enables the detection of multiple biothreat agents across multiple samples with a single sequencing run.

In certain aspects, the present invention provides A method of detecting *Bacillus anthracis* in a sample, comprising detecting at least one *B. anthracis*-specific amplicon in the sample using at least one primer pair selected from the group consisting of: SEQ ID NO: 1 and SEQ ID NO: 2; SEQ ID NO: 3 and SEQ ID NO: 4; SEQ ID NO: 5 and SEQ ID NO: 6; SEQ ID NO: 7 and SEQ ID NO: 8; SEQ ID NO: 9 and SEQ ID NO: 10; SEQ ID NO: 11 and SEQ ID NO: 12; SEQ ID NO: 13 and SEQ ID NO: 14; SEQ ID NO: 15 and SEQ ID NO: 16; SEQ ID NO: 17 and SEQ ID NO: 18; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein the presence of the *B. anthracis*-specific amplicon indicates the presence of *B. anthracis* in the sample, and the absence of the *B. anthracis*-specific amplicon indicates the absence of *B. anthracis* from the sample.

In other aspects, the method further comprises confirming the absence of *B. anthracis* by detecting at least one *B. anthracis* Near Neighbor-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 19 and SEQ ID NO: 20; SEQ ID NO: 21 and SEQ ID NO: 22; SEQ ID NO: 23 and SEQ ID NO: 24; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *B. anthracis* Near Neighbor-specific amplicon in the sample confirms the absence of *B. anthracis*.

In yet other aspects, the method further comprises confirming the absence of *B. anthracis* by detecting at least one *B. anthracis* Near Neighbor-specific sequence variant (SV) or single nucleotide polymorphism (SNP) using at least one primer pair selected from the group consisting of: SEQ ID NO: 25 and SEQ ID NO: 26; SEQ ID NO: 27 and SEQ ID NO: 28; SEQ ID NO: 29 and SEQ ID NO: 30; SEQ ID NO: 31 and SEQ ID NO: 32; SEQ ID NO: 33 and SEQ ID NO: 34; SEQ ID NO: 35 and SEQ ID NO: 36; SEQ ID NO: 37 and SEQ ID NO: 38; SEQ ID NO: 39 and SEQ ID NO: 40; SEQ ID NO: 41 and SEQ ID NO: 42; SEQ ID NO: 43 and SEQ ID NO: 44; SEQ ID NO: 45 and SEQ ID NO: 46; SEQ ID NO: 47 and SEQ ID NO: 48; SEQ ID NO: 49 and SEQ ID NO: 50; SEQ ID NO: 51 and SEQ ID NO: 52; SEQ ID NO: 53 and SEQ ID NO: 54; SEQ ID NO: 55 and SEQ ID NO: 56; SEQ ID NO: 57 and SEQ ID NO: 58; SEQ ID NO: 59 and SEQ ID NO: 60; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *B. anthracis* Near Neighbor-specific SV in the sample confirms the absence of *B. anthracis*.

In some aspects, the method further comprises detecting a virulence locus or virulence plasmid in the sample by detecting a virulence-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 29 and SEQ ID NO: 30; SEQ ID NO: 31 and SEQ ID

NO: 32; SEQ ID NO: 33 and SEQ ID NO: 34; SEQ ID NO: 35 and SEQ ID NO: 36; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein the presence of the virulence-specific amplicon indicates the presence of the virulence locus or virulence plasmid in the sample.

In other aspects, the method further comprises detecting at least one drug resistance single nucleotide polymorphism (SNP) from *B. anthracis* in the sample using at least one primer pair selected from the group consisting of: SEQ ID NO: 37 and SEQ ID NO: 38; SEQ ID NO: 39 and SEQ ID NO: 40; SEQ ID NO: 41 and SEQ ID NO: 42; SEQ ID NO: 43 and SEQ ID NO: 44; SEQ ID NO: 45 and SEQ ID NO: 46; SEQ ID NO: 47 and SEQ ID NO: 48; a pair of sequences which are at least 85% identical thereto; and RNA equivalents. In other aspects, the method further comprises detecting *Burkholderia pseudomallei* and/or *Burkholderia mallei* in the sample by detecting at least one *B. pseudomallei* or *B. mallei*-specific amplicon uses at least one primer pair selected from the group consisting of: SEQ ID NO: 61 and SEQ ID NO: 62; SEQ ID NO: 63 and SEQ ID NO: 64; SEQ ID NO: 65 and SEQ ID NO: 66; SEQ ID NO: 67 and SEQ ID NO: 68; SEQ ID NO: 69 and SEQ ID NO: 70; SEQ ID NO: 71 and SEQ ID NO: 72; SEQ ID NO: 73 and SEQ ID NO: 74; SEQ ID NO: 75 and SEQ ID NO: 76; SEQ ID NO: 77 and SEQ ID NO: 78; SEQ ID NO: 79 and SEQ ID NO: 80; SEQ ID NO: 81 and SEQ ID NO: 82; SEQ ID NO: 83 and SEQ ID NO: 84; SEQ ID NO: 85 and SEQ ID NO: 86; SEQ ID NO: 87 and SEQ ID NO: 88; SEQ ID NO: 89 and SEQ ID NO: 90; SEQ ID NO: 91 and SEQ ID NO: 92; SEQ ID NO: 93 and SEQ ID NO: 94; SEQ ID NO: 95 and SEQ ID NO: 96; SEQ ID NO: 97 and SEQ ID NO: 98; SEQ ID NO: 99 and SEQ ID NO: 100; SEQ ID NO: 101 and SEQ ID NO: 102; SEQ ID NO: 103 and SEQ ID NO: 104; SEQ ID NO: 103 and SEQ ID NO: 104; SEQ ID NO: 105 and SEQ ID NO: 106; SEQ ID NO: 107 and SEQ ID NO: 108; SEQ ID NO: 117 and SEQ ID NO: 118; SEQ ID NO: 119 and SEQ ID NO: 120; SEQ ID NO: 121 and SEQ ID NO: 122; SEQ ID NO: 123 and SEQ ID NO: 124; SEQ ID NO: 125 and SEQ ID NO: 126; a pair of sequences which are at least 85% identical thereto; and RNA equivalents wherein the presence of the *B. pseudomallei* or *B. mallei*-specific amplicon indicates the presence of *B. pseudomallei* and/or *B. mallei* in the sample, and an absence of the *B. pseudomallei* or *B. mallei*-specific amplicon indicates an absence of *B. pseudomallei* and *B. mallei* in the sample.

In certain aspects, the method further comprises confirming the absence of *B. pseudomallei* and *B. mallei* by detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 177 and SEQ ID NO: 178; SEQ ID NO: 179 and SEQ ID NO: 180; SEQ ID NO: 181 and SEQ ID NO: 182; SEQ ID NO: 183 and SEQ ID NO: 184; SEQ ID NO: 185 and SEQ ID NO: 186; SEQ ID NO: 187 and SEQ ID NO: 188; SEQ ID NO: 189 and SEQ ID NO: 190; SEQ ID NO: 191 and SEQ ID NO: 192; SEQ ID NO: 193 and SEQ ID NO: 194; SEQ ID NO: 195 and SEQ ID NO: 196; SEQ ID NO: 197 and SEQ ID NO: 198; SEQ ID NO: 199 and SEQ ID NO: 200; SEQ ID NO: 201 and SEQ ID NO: 202; SEQ ID NO: 203 and SEQ ID NO: 204; SEQ ID NO: 205 and SEQ ID NO: 206; SEQ ID NO: 207 and SEQ ID NO: 208; SEQ ID NO: 207 and SEQ ID NO: 208; SEQ ID NO: 209 and SEQ ID NO: 210; SEQ ID NO: 211 and SEQ ID NO: 212; SEQ ID NO: 213 and SEQ ID NO: 214; SEQ ID NO: 215 and SEQ ID NO: 216; SEQ ID NO: 217 and SEQ ID NO: 218; SEQ ID NO: 219 and SEQ ID NO: 220; SEQ ID NO: 221 and SEQ ID NO:

222; SEQ ID NO: 223 and SEQ ID NO: 224; SEQ ID NO: 225 and SEQ ID NO: 226; SEQ ID NO: 227 and SEQ ID NO: 228; SEQ ID NO: 229 and SEQ ID NO: 230; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *B. pseudomallei* or *B. mallei* Near Neighbor-specific amplicon in the sample confirms the absence of *B. pseudomallei* and *B. mallei*.

In yet other aspects, the method further comprises confirming the absence of *B. pseudomallei* and *B. mallei* by detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific SNP or SV using at least one primer pair selected from the group consisting of: SEQ ID NO: 109 and SEQ ID NO: 110; SEQ ID NO: 111 and SEQ ID NO: 112; SEQ ID NO: 113 and SEQ ID NO: 114; SEQ ID NO: 115 and SEQ ID NO: 116; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *B. pseudomallei* or *B. mallei* Near Neighbor-specific SNP or SV in the sample confirms the absence of *B. pseudomallei* and *B. mallei*.

In some aspects, the method further comprises detecting at least one drug resistance SNP or SV from *Burkholderia* spp. in the sample using at least one primer pair selected from the group consisting of: SEQ ID NO: 127 and SEQ ID NO: 128; SEQ ID NO: 129 and SEQ ID NO: 130; SEQ ID NO: 131 and SEQ ID NO: 132; SEQ ID NO: 133 and SEQ ID NO: 134; SEQ ID NO: 135 and SEQ ID NO: 136; SEQ ID NO: 137 and SEQ ID NO: 138; SEQ ID NO: 145 and SEQ ID NO: 146; SEQ ID NO: 147 and SEQ ID NO: 148; SEQ ID NO: 149 and SEQ ID NO: 150; SEQ ID NO: 151 and SEQ ID NO: 152; SEQ ID NO: 153 and SEQ ID NO: 154; a pair of sequences which are at least 85% identical thereto; and RNA equivalents.

In other aspects, the method further comprises detecting *Francisella tularensis* in the sample by detecting at least one *F. tularensis*-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 265 and SEQ ID NO: 266; SEQ ID NO: 267 and SEQ ID NO: 268; SEQ ID NO: 269 and SEQ ID NO: 270; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein the presence of the *F. tularensis*-specific amplicon indicates that *F. tularensis* is present in the sample, and an absence of the *F. tularensis*-specific amplicon indicates that *F. tularensis* is absent in the sample.

In yet other aspects, the method further comprises confirming the absence of *F. tularensis* by detecting at least one *F. tularensis* Near Neighbor-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 285 and SEQ ID NO: 286; SEQ ID NO: 287 and SEQ ID NO: 288; SEQ ID NO: 289 and SEQ ID NO: 290; SEQ ID NO: 291 and SEQ ID NO: 292; SEQ ID NO: 293 and SEQ ID NO: 294; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *F. tularensis* Near Neighbor-specific amplicon in the sample confirms the absence of *F. tularensis*.

In one aspect, the method further comprises confirming the absence of *F. tularensis* by detecting at least one *F. tularensis* Near Neighbor-specific SNP or SV using at least one primer pair selected from the group consisting of: SEQ ID NO: 271 and SEQ ID NO: 272; SEQ ID NO: 273 and SEQ ID NO: 274; SEQ ID NO: 275 and SEQ ID NO: 276; SEQ ID NO: 277 and SEQ ID NO: 278; SEQ ID NO: 279 and SEQ ID NO: 280; SEQ ID NO: 281 and SEQ ID NO: 282; SEQ ID NO: 283 and SEQ ID NO: 284; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *F. tularensis* Near Neighbor-specific SNP or SV in the sample confirms the absence of *F. tularensis*.

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In another aspect, the method further comprises detecting *Yersinia pestis* in the sample by detecting at least one *Y. pestis*-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 231 and SEQ ID NO: 232; SEQ ID NO: 233 and SEQ ID NO: 234; SEQ ID NO: 235 and SEQ ID NO: 236; SEQ ID NO: 237 and SEQ ID NO: 238; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein the presence of the *Y. pestis*-specific amplicon indicates the presence of *Y. pestis* in the sample, and an absence of the *Y. pestis*-specific amplicon indicates an absence of *Y. pestis* in the sample.

In still another aspect, the method further comprises confirming the absence of *Y. pestis* by detecting at least one *Y. pestis* Near Neighbor-specific SNP or SV using at least one primer pair selected from the group consisting of: SEQ ID NO: 249 and SEQ ID NO: 250; SEQ ID NO: 251 and SEQ ID NO: 252; SEQ ID NO: 253 and SEQ ID NO: 254; SEQ ID NO: 255 and SEQ ID NO: 256; SEQ ID NO: 257 and SEQ ID NO: 258; SEQ ID NO: 259 and SEQ ID NO: 260; SEQ ID NO: 261 and SEQ ID NO: 262; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *Y. pestis* Near Neighbor-specific SNP or SV confirms the absence of *Y. pestis*.

In certain aspects, the method further comprises confirming the absence of *Y. pestis* by detecting at least one *Y. pestis* Near Neighbor-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 263 and SEQ ID NO: 264; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *Y. pestis* Near Neighbor-specific amplicon confirms the absence of *Y. pestis*.

In other aspects, the method further comprises characterizing and/or subtyping *Y. pestis* in the sample by detecting at least one amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 239 and SEQ ID NO: 240; SEQ ID NO: 241 and SEQ ID NO: 242; SEQ ID NO: 243 and SEQ ID NO: 244; SEQ ID NO: 245 and SEQ ID NO: 246; SEQ ID NO: 247 and SEQ ID NO: 248; a pair of sequences which are at least 85% identical thereto; and RNA equivalents.

In some aspects, the amplicons are generated with at least one multiplex amplification reaction. In other aspects, the amplicons are generated with at least two, at least three, at least four, or at least five multiplex amplification reactions.

In other aspects, the amplicon, SNP or SV is determined using next-generation sequencing. In one aspect, each primer in the at least one primer pair comprises a universal tail sequence. In some aspects, the universal tail sequence comprises SEQ ID NO: 301 or SEQ ID NO: 303.

In certain aspects, the amplicon is present when a locus read count of the amplicon is at least 10 sequence reads covering at least 75% of a corresponding amplicon reference sequence.

In other aspects, sequence analysis of sequence read alignments is performed to determine whether a target species, Near Neighbor species, virulence or antibiotic resistance allele is present in the sample, wherein the target species is *Bacillus anthracis*, *Burkholderia pseudomallei*, *Burkholderia mallei*, *Francisella tularensis*, or *Yersinia pestis*.

In one embodiment, the sample is an environmental sample. In another embodiment, the sample is a biological sample obtained from a subject.

In certain embodiments, the method further comprises administering an effective amount of at least one antibiotic to the subject, wherein the at least one antibiotic is selected

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from the group consisting of a fluoroquinolone, an aminoglycoside, a glycopeptide, a lincosamide, a macrolide/ketolide, a cephalosporin, a monobactam, a nitroimidazole, a penicillin, a streptogramin, a tetracycline, and a physiologically acceptable salt, prodrug, or combination thereof.

In another embodiment, the at least one antibiotic is not a fluoroquinolone if a *gyrA* drug resistance SNP is detected; and/or the at least one antibiotic is not a fluoroquinolone if a *parC* drug resistance SNP is detected; and/or the at least one antibiotic is not a fluoroquinolone or an aminocoumarin if a *gyrB* drug resistance SNP is detected; and/or the at least one antibiotic is not a rifamycin if a *rpoB* drug resistance SNP is detected; and/or the at least one antibiotic is not a β -lactam if a *penA* drug resistance SNP is detected; and/or the at least one antibiotic is not a trimethoprim and sulfamethoxazole combination, co-trimoxazole, if a *folM* drug resistance SV is detected; and/or the at least one antibiotic is not a trimethoprim and sulfamethoxazole combination, co-trimoxazole, if a *bpeT* drug resistance SV is detected; and/or the at least one antibiotic is not a trimethoprim and sulfamethoxazole combination, co-trimoxazole, if a *bpeS* drug resistance SV is detected.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A and 1B depict a universal index multiplex amplicon sequencing assay (UI-AmpSeq). Gene-specific primers containing universal tails (UT) amplify gene-specific targets. An ILLUMINA® extension PCR is run on these tailed PCR amplicons using ILLUMINA® indices containing complementary sequences to the UT, which allow for amplicon sequencing on an ILLUMINA® platform. FIG. 1A depicts sequence ready multi-locus amplification. The universal indexing strategy comprising universal tails is described in U.S. Publication No. 2016/0326572, the contents of which are incorporated herein by reference. FIG. 1B depicts sequencing and bioinformatic analysis.

FIG. 2 depicts an ASAP analysis. Enhanced Amplicon Sequencing Analysis pipeline (ASAP) allows the user to quickly analyze read data for specific targets and provides a detailed report output file. The ASAP bioinformatics method is described in U.S. Publication No. 2018/0173843, the contents of which are incorporated herein by reference.

FIGS. 3A-3E. An ASAP output result table, showing a subset of results from a large diverse sample set (693 isolates). The ASAP output demonstrates a general presence/absence for each select agent isolate or near neighbor isolate tested using UI-AmpSeq assays specific for *B. anthracis*, known virulence determinants, near neighbor (NN) species, and antimicrobial resistance (AMR) targets.

FIGS. 4A-4D An ASAP output result table, showing a subset of results from a large diverse sample set (693 isolates). The ASAP output demonstrates a general presence/absence for each select agent isolate or near neighbor isolate tested using UI-AmpSeq assays specific for *Y. pestis*, known virulence determinants, near neighbor (NN) species, and antimicrobial resistance (AMR) targets.

FIGS. 5A-5E An ASAP output result table, showing a subset of results from a large diverse sample set (693 isolates). The ASAP output demonstrates a general presence/absence for each select agent isolate or near neighbor isolate tested using UI-AmpSeq assays specific for *F. tularensis*, known virulence determinants, near neighbor (NN) species, and antimicrobial resistance (AMR) targets.

FIGS. 6A-6C An ASAP output result table, showing a subset of results from a large diverse sample set (693 isolates). The ASAP output demonstrates a general presence/

absence for each select agent isolate or near neighbor isolate tested using UI-AmpSeq assays specific *B. pseudomallei*, *B. mallei*, known virulence determinants, near neighbor (NN) species, and antimicrobial resistance (AMR) targets.

FIG. 7. Sensitivity and specificity results for *B. anthracis* assay, including amplicon yield (locus read count % of total sample library read count) for representative target and NN species.

FIGS. 8A-8C. Sensitivity and specificity results for *B. pseudomallei/mallei* assay, including amplicon yield (locus read count % of total sample library read count) for representative target and NN species.

FIG. 9. Sensitivity and specificity results for *Y. pestis* assay, including amplicon yield (locus read count % of total sample library read count) for representative target and NN species.

FIG. 10. Sensitivity and specificity results for *F. tularensis* assay, including amplicon yield (locus read count % of total sample library read count) for representative target and NN species.

FIG. 11 An Example Clinical Plate layout. The black wells are samples.

DETAILED DESCRIPTION

Definitions

As used herein, the verb “comprise” as is used in this description and in the claims and its conjugations are used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded.

As used herein, reference to an element by the indefinite article “a” or “an” does not exclude the possibility that more than one of the elements are present, unless the context clearly requires that there is one and only one of the elements. The indefinite article “a” or “an” thus usually means “at least one.”

The sample in this method is preferably a biological sample from a subject. The term “sample” or “biological sample” or “environmental sample” is used in its broadest sense. Depending upon the embodiment of the invention, for example, a sample may comprise a bodily fluid including whole blood, serum, plasma, urine, saliva, cerebral spinal fluid, semen, vaginal fluid, pulmonary fluid, tears, perspiration, mucus and the like; an extract from a cell, chromosome, organelle, or membrane isolated from a cell; a cell; genomic DNA, RNA, or cDNA, in solution or bound to a substrate; a tissue; a tissue print, or any other material isolated in whole or in part from a living subject or organism. Biological samples may also include sections of tissues such as biopsy and autopsy samples, and frozen sections taken for histologic purposes such as blood, plasma, serum, sputum, stool, tears, mucus, hair, skin, and the like. Biological samples also include explants and primary and/or transformed cell cultures derived from patient tissues. In some embodiments, sample may comprise a portion of a non-animal organism, such as a plant (e.g., castor beans or derivatives thereof). In other embodiments, the sample comprises soil, water, air or other environmental material.

In some embodiments, sample or biological sample may include a bodily tissue, fluid, or any other specimen that may be obtained from a living organism that may comprise additional living organisms. By way of example only, in some embodiments, sample or biological sample may include a specimen from a first organism (e.g., a human) that may further comprise an additional organism (e.g., bacteria,

including pathogenic or non-pathogenic/commensal bacteria, viruses, parasites, fungi, including pathogenic or non-pathogenic fungi, etc.). In some embodiments, the additional organism may be separately cultured after isolation of the sample to provide additional starting materials for downstream analyses, in some embodiments, the sample or biological sample may comprise a direct portion of the additional, non-human organism and the host organism (e.g., a biopsy or sputum sample that contains human cells and bacteria).

Some embodiments of the invention may comprise a multiplex assay. As used herein, the term “multiplex” refers to the production of more than one amplicon, PCR product, PCR fragment, amplification product, etc. in a single reaction vessel. In other words, multiplex is to be construed as the amplification of more than one target-specific sequences within a PCR reaction or assay within the same PCR assay mixture (e.g., more than one amplicon is produced within a single vessel that contains all of the reagents necessary to perform a PCR reaction). In some embodiments, a step prior to performing the PCR (or RT-PCR, quantitative RT-PCR, etc.) reaction can occur such that sets of primers and/or primers and probes are designed, produced, and optimized within a given set of reaction conditions to ensure proper amplicon production during the performance of the PCR. BioThreatSeq

Currently, there are two approaches being used to identifying specimens in the environment. The first approach, metagenomics, tries to sequence “all” of the DNA in a sample and then to unravel its content computationally. Sequencing “all” of the DNA is difficult, slow, and very expensive. The computational approaches are improving but still contain many flaws that lead to false conclusions. A recent sensationalized report of DNA from anthrax and plague bacteria in the NYC subways illustrates the pitfalls of such endeavors (Mason et al., Cell Systems). Significant amounts of data are produced with metagenomics but frequently not enough for informative signatures to differentiate pathogens from near neighbors. Deep sequencing of single specimens may cost nearly \$1,000 and take several weeks to generate. It is clearly not ready at this time for implementation for biosurveillance.

The second approach, amplicon sequencing, involves the deep (>5,000×) sequencing of the 16S gene PCR amplicon to identify individual components of mixed bacterial communities. While the PCR primers are not specific, the intervening sequences can be highly informative and can be used to discriminate among bacterial taxa. Unfortunately for biothreat detection, the 16S gene has insufficient discrimination power to differentiate biothreat pathogens from their near neighbors. Discrimination power is a function of gene diversity and the 16S has low or no diversity among closely related bacteria. It cannot effectively identify a biothreat agent or distinguish from near-neighbor species.

Increasing the amplicon discrimination power can be accomplished through comparative genomic analysis to identify diverse genomics regions. This is most effective when large genome databases are available and can be highly predictive of success once implemented. In clinical diagnostics, this approach is being used to identify multiple pathogens and to predict their virulence and resistance to antibiotics. PCR primers specific to a pathogen genera or species is sufficient, if there is additional DNA sequence information that can be leveraged for precise agent identification. Multiplex systems of several hundred amplicons are becoming common and provide coverage for dozens of pathogens. Sample preparation that works for real-time-PCR

also works for this technology, so currently sampling schemes would adapt well to this type of assay. A multiple amplicon sequencing system would be easily adapted to changing targets with addition of new amplicons. Because of the multiplex nature of the assay, redundant amplicons can easily be included to verify the identification of a biothreat agent and even provide a differential identification of a near-neighbor species. Variation within the amplicons can be analyzed to identify drug resistance, virulence factors and subtype to the strain level.

The ideal multiple amplicon sequencing system for identifying major biothreat agents should distinguish between the biothreat agent and its near-neighbor species using both amplification positive/negative criteria and qualitative analysis of sequence within the amplicons. This latter analysis provides strain identification and drug susceptibility identification. The analytical system should be supported by an automated interpretive software that generates actionable reports. Such a system includes quality assurance data to identify sample and/or process issues rapidly, to limit the effect of QC issues on final results.

“BioThreatSeq” detects a presence, an absence, and/or a clinically important characteristic of nucleic acids from one or more microbial pathogens. BioThreatSeq is based upon very discriminating genetic regions bioinformatically identified using public and private genome sequences from microbial pathogens including but not limited to *Bacillus anthracis*, *Burkholderia pseudomallei*, *Burkholderia mallei*, *Francisella tularensis*, *Yersinia pestis* and Near Neighbor (NN) species. BioThreatSeq can be used to screen environmental samples for presence of target agent DNA, as well as war fighter and civilian patients for target agent carriage in the event of suspected exposure.

In an embodiment, BioThreatSeq comprises a highly multiplexed amplicon sequencing assay. The assay is a highly informative screening tool capable of simultaneously detecting a presence of a microbial pathogen and a clinically important characteristic of the microbial pathogen without a live culturing step. Non-limiting examples of the clinically important characteristics include: virulence, or antibiotic resistance genetic signatures, etc.

The utility of this assay has been demonstrated on several complex environmental and clinical specimen types including urine, wound swabs, sputum, air, soil, water samples. The superiority of BioThreatSeq over traditional typing techniques include high sensitivity (e.g., a low limit of detection) and high specificity (e.g., discriminating among strains, detecting antimicrobial resistance, and profiling virulence signatures, etc.) in target agent detection. BioThreatSeq is also highly adaptable to new content, which allows for the flexibility to detect new biothreats agents and signatures. Thus, the assay methodology allows for the expansion of this tool to be used for several other BioThreat agents or applications.

Target Specific Amplicon

The Inventors used comparative genomics to identify a first genomic region that differentiates a target agent from its near neighbor relatives. In the first scenario, the first genomic region is present in all known target strains, and a lack of the first genomic region indicates an absence of the target agent in the sample. In the second scenario, the first genomic region not only is present in all known target strains, but is also absent in near-neighbor species. Thus, a presence of the first genomic region indicates a presence of the target agent in the sample.

In certain non-limiting embodiments, the presence or absence of the first genomic region in the nucleic acids of the

sample is determined by PCR using a first forward primer and a first reverse primer. The first forward primer and the first reverse primer amplify a Target Specific Amplicon, i.e., all strains of the threat agent, but not the near neighbors.

Differential Target Amplicons

The Inventors also used comparative genomics to identify a second genomic region that differentiates a target agent from its near neighbor relatives. The second genomic region is present in all strains of the near-neighbor relatives, but will not be in the target. Thus, a presence of the second genomic region indicates an absence of the target agent in the sample.

In certain non-limiting embodiments, the presence or absence of the second genomic region in the nucleic acids of the sample is determined by PCR using a second forward primer and a second reverse primer. The second forward primer and second reverse primer amplify Differential Target Amplicons, i.e., all strains of the near neighbors, but not the threat agent. Differential identification assays can be included in the multiplex assay to help nullify any false positive results. This optional step offers interpretive value in complex species.

The Inventors determined the exclusivity and inclusivity of the first forward and the first reverse primers in silico across all available threat agents (*Bacillus anthracis*, *Burkholderia pseudomallei*, *Burkholderia mallei*, *Francisella tularensis*, and *Yersinia pestis*) and near neighbor genomes. The in-silico validation included genomes from common contaminants such as humans. Because a large number of genomics sequences exist for both target and non-target organisms, the in-silico validation step eliminates any primers that are non-exclusive to the biothreat target. The assay primers were tested against the target and near-neighbor DNA templates to validate them under actual assay conditions.

Primer Design

The design of the first forward primer, the first reverse primer, the second forward primer, and the second reverse primer is consistent with a standard PCR method but is amendable to analysis using next-generation sequencing methods. This requirement includes the addition of “bar-codes” to allow for indexing of samples for combining into single DNA sequencing batches. The technical details are provided in the PCT Patent Application entitled “Systems And Methods for Universal Tail-Based Indexing Strategies for Amplicon Sequencing” (International Application Number: PCT/US2014/064890; International Publication Number: WO 2015/070187 A2), the contents of which are hereby incorporated in their entirety.

The Inventors determined the exclusivity and inclusivity of the second forward and the second reverse primers in silico across all available threat agents (*Bacillus anthracis*, *Burkholderia pseudomallei*, *Burkholderia mallei*, *Francisella tularensis*, and *Yersinia pestis*) and near neighbor genomes. The in-silico validation included genomes from common contaminants such as humans and soil DNA. Because a large number of genomics sequences exist for both target and non-target organisms, the in-silico validation step eliminates any primers that are non-exclusive to the near-neighbor species. The assay primers were tested against the target and near-neighbor DNA templates to validate them under actual assay conditions.

In the third scenario, the first genomic region is present in all known target strains and at least one near-neighbor species. In this case, producing an exclusive amplicon is not feasible and the combination of amplification and internal sequence is needed to distinguish target from near-neighbor.

bors. In the absence of exclusive-target amplification, the amplicon sequence could provide definitive identification of the target and non-target agents.

The Inventors has defined the phylogenetic structure of the first genomic region that includes both the target agent and its near neighbors and identified a variable internal sequence region which allows for: (1) differentiation of near neighbor from target species, (2) strain identification, (3) drug susceptibility identification, and/or (4) virulence prediction.

Performance of the Multiplex Assays

The Inventors have developed combined multi-agent amplicon sequencing assays for 2, 3, 4, 5, 6, or 7 biothreat agents and validated them under laboratory conditions. For the combined biothreat agent assays, important test parameters such as linearity, LOD, sensitivity, specificity, quantitative performance (absolute and relative), contaminant interference, performance with environmental samples (spikes), etc. have been determined.

Software

The Inventors have developed software that analyzes *B. anthracis* amplicon sequence data and provides actionable information (i.e., agent presence with confidence metrics, presence of virulence and antibiotic resistance factors, phylogenetic classification, etc.). The Inventors have also developed software that analyzes *B. anthracis* and other target agent (*F. tularensis*, *Y. pestis*, *B. mallei*, *B. pseudomallei*, *Brucella melitensis*, and *B. abortus*) and allow for on-site and remote reporting.

BTSeq comprises target agent and near neighbor (NN) species identification assays, antimicrobial resistance (AMR) assays, virulence gene assays, and uses TGen North's amplicon sequencing analysis pipeline (ASAP) to report results.

Use of the disclosed amplicon sequencing tool can be used to screen environmental samples for presence of target agent DNA, as well as war fighter and civilian patients for target agent carriage in the event of suspected exposure.

In some embodiments, the present invention relates to a method of detecting *Bacillus anthracis* in a sample, comprising detecting at least one *B. anthracis*-specific amplicon selected from the group consisting of: CP008853.1_5309, CP008853.1_5316, CP012725.1_3629, CP012725.1_5103, CP012725.1_5107, JSZQ01000034.1_220, JSZS01000036.1_5, LGCC01000010.1_232, and LGCC01000048.1_280 in the sample, wherein the presence of the *B. anthracis*-specific amplicon indicates the presence of *B. anthracis* in the sample, and an absence of the *B. anthracis*-specific amplicon indicates an absence of *B. anthracis* in the sample.

In other embodiments, the disclosed methods further comprise confirming the absence of *B. anthracis* by detecting at least one *B. anthracis* Near Neighbor-specific amplicon selected from the group consisting of: NN_LOMU01000090.1_49, NN_LOQC01000013.1_3, and ChimpKiller_9-159 in the sample, wherein detecting the *B. anthracis* Near Neighbor-specific amplicon confirms the absence of *B. anthracis*.

In yet other embodiments, the disclosed methods further comprise characterizing and/or subtyping *B. anthracis* by detecting at least one amplicon, single nucleotide polymorphism (SNP) or sequence variant (SV) selected from the group consisting of: ChimpKiller_91-320, ChimpKiller_481-698, plcR, pagA, pX01, pX01, gyrA, parC, gyrB, rpoB, AA_2502, AA_2503, Ba_AmesAnc_4669915, Ba_AmesAnc_4001578, Ba_AmesAnc_1069024,

Ba_AmesAnc_3668548, Ba_AmesAnc_371913, and Ba_AmesAnc_999035 in the sample.

In certain aspects, the disclosed methods further comprise characterizing and/or subtyping *B. anthracis* by detecting at least one amplicon, single nucleotide polymorphism (SNP) or sequence variant (SV) selected from the group consisting of: ChimpKiller_91-320, ChimpKiller_481-698, plcR, pagA, pX01, pX01, gyrA, parC, gyrB, rpoB, AA_2502, AA_2503, Ba_AmesAnc_4669915, Ba_AmesAnc_4001578, Ba_AmesAnc_1069024, Ba_AmesAnc_3668548, Ba_AmesAnc_371913, and Ba_AmesAnc_999035 in the sample.

In other aspects, the present invention relates to a method of detecting *Burkholderia pseudomallei* and/or *Burkholderia mallei* in a sample by detecting at least one *B. pseudomallei* or *B. mallei*-specific amplicon selected from the group consisting of: LWWC01000187.1_18, LWWB01000125.1_17183_17602, LXAY01000367.1_0_640, LWVY01000190.1_17226_17689, and LXAD01000059.1_24760_25075, wherein the presence of the *B. pseudomallei* or *B. mallei*-specific amplicon indicates the presence of *B. pseudomallei* and/or *B. mallei* in the sample, and an absence of the *B. pseudomallei* or *B. mallei*-specific amplicon indicates an absence of *B. pseudomallei* and *B. mallei* in the sample.

In some embodiments, the present invention provides a method of detecting *Burkholderia pseudomallei* and/or *Burkholderia mallei* in the sample by detecting at least one *B. pseudomallei* or *B. mallei*-specific amplicon selected from the group consisting of: LWWC01000187.1_18, LWWB01000125.1_17183_17602, LXAY01000367.1_0_640, LWVY01000190.1_17226_17689, and LXAD01000059.1_24760_25075, wherein the presence of the *B. pseudomallei* or *B. mallei*-specific amplicon indicates the presence of *B. pseudomallei* and/or *B. mallei* in the sample, and an absence of the *B. pseudomallei* or *B. mallei*-specific amplicon indicates an absence of *B. pseudomallei* and *B. mallei* in the sample.

In other embodiments, the present invention provides a method of detecting *B. pseudomallei* in a sample by detecting at least one *B. pseudomallei*-specific amplicon selected from the group consisting of: TTS1 BPSS1407, LXCC01000141.1_39296_39817, LXBY01000087.1_75760_76751, LXCD01000002.1_99652_100245, and LXCE01000123.1_34220_34747, wherein the presence of the *B. pseudomallei*-specific amplicon indicates the presence of *B. pseudomallei* in the sample, and an absence of the *B. pseudomallei*-specific amplicon indicates an absence of *B. pseudomallei* in the sample.

In yet other embodiments, the present invention provides a method of detecting *B. mallei* in the sample by detecting at least one *B. mallei*-specific amplicon selected from the group consisting of: Bm 11589 and Bm 11767, wherein the presence of the *B. mallei*-specific amplicon indicates the presence of *B. mallei* in the sample, and an absence of the *B. mallei*-specific amplicon indicates an absence of *B. mallei* in the sample.

In certain aspects, the disclosed methods further comprise characterizing and/or subtyping *B. pseudomallei* and/or *B. mallei* by detecting at least one amplicon, single nucleotide polymorphism (SNP) or sequence variant (SV) selected from the group consisting of: K9penA378-529, K9penA575-761, K9penA949-1172, pbp3-1, and pbp3-2 in the sample.

In other aspects, the disclosed methods further comprise confirming the absence of *B. pseudomallei* and *B. mallei* by detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific single nucleotide polymorphism (SNP) or sequence variant (SV) selected from the group consisting of: NC 006350 2289827, NC 006350 133027, NC 006350 2248145-2248193, and NC 006350 988041-988089 in the sample, wherein detecting the *B. pseudomallei* or *B. mallei* Near Neighbor-specific single nucleotide polymorphism (SNP) or sequence variant (SV) confirms the absence of *B. pseudomallei* and *B. mallei*.

In yet other aspects, the present invention provides a method of detecting *Francisella tularensis* in a sample by detecting at least one *F. tularensis*-specific amplicon selected from the group consisting of: *F. tularensis*_CP000915.1_1782, *F. tularensis*_CP000915.1-731, and *Ft_dup_CP000915.1_197*, wherein the presence of the *F. tularensis*-specific amplicon indicates that *F. tularensis* is present in the sample, and an absence of the *F. tularensis*-specific amplicon indicates that *F. tularensis* is absent in the sample.

In some aspects, the disclosed methods further comprise confirming the absence of *F. tularensis* by detecting at least one *F. tularensis* Near Neighbor-specific amplicon selected from the group consisting of: *F. tinoviciida*_CP009607.1, *F. philom*_CP009444.1_569, and *F. philom*_CP009444.1_285 in the sample, wherein detecting the *F. tularensis* Near Neighbor-specific amplicon confirms the absence of *F. tularensis*.

In other aspects, the disclosed methods further comprise confirming the absence of *F. tularensis* by detecting at least one *F. tularensis* Near Neighbor-specific SNP or SV selected from the group consisting of: *FtA1*, *FtA2*, *FtB*, *FtA*, *FtLVS_AM233362_1646546*, *FtLVS_AM233362_1643765*, and *FtLVS_AM233362_1562618* in the sample, wherein detecting the *F. tularensis* Near Neighbor-specific polymorphism confirms the absence of *F. tularensis*.

In yet other aspects, the present invention provides a method of detecting *Yersinia pestis* in the sample by detecting at least one *Y. pestis*-specific amplicon selected from the group consisting of: *Y. pestis*_LPQY01000176.1_7, *AGJT01000065.1_0_338*, and *FAUR01000053.1_96407_96884*, wherein the presence of the *Y. pestis*-specific amplicon indicates the presence of *Y. pestis* in the sample, and an absence of the *Y. pestis*-specific amplicon indicates an absence of *Y. pestis* in the sample.

In one embodiment, the disclosed methods further comprise confirming the absence of *Y. pestis* by detecting at least one *Y. pestis* Near Neighbor-specific SNP or SV selected from the group consisting of: *YpCO92_NC_003143_113190*, *YpCO92_NC_003143_161621*, *YpCO92_NC_003143_152213*, *YpCO92_NC_003143_129539*, *YpCO92_NC_003143_91203*, *YpCO92_NC_003143_121812*, and *Yp_AL590842.1_RX_SNP* in the sample, wherein detecting the *Y. pestis* Near Neighbor-specific SNP or SV confirms the absence of *Y. pestis*.

In another embodiment, the disclosed methods further comprise characterizing and/or subtyping *Y. pestis* by detecting at least one amplicon selected from the group consisting of: *YpPGM_AL031866.1_81*, *YpPGM_31-205*, *Yp-p1202_42780-43194*, *Yp-p1202_126386-126750*, and *Yp-p1202_156402-156711* in the sample.

In some embodiments, the presence or absence of *B. anthracis* in a sample is detected by identifying a specific mutation in the *PlcR* gene, a single base change at position 640, a nonsense mutation, which creates a dysfunctional protein. In other embodiments, the presence or absence of *B. anthracis* in a sample is detected by identifying the *pXO1* and/or *pXO2* plasmids.

PlcR is a global transcriptional regulator which controls most of the secreted virulence factors in *B. cereus* and *B. thuringiensis*. It is chromosomally encoded and is ubiquitous throughout the cell (Agaisse, H. et al. (June 1999). "PlcR is a pleiotropic regulator of extracellular virulence factor gene expression in *Bacillus thuringiensis*". Molecular Microbiology. 32 (5): 1043-53). In *B. anthracis*, however, the *plcR* gene contains a single base change at position 640, a nonsense mutation, which creates a dysfunctional protein. While 1% of the *B. cereus* group carries an inactivated *plcR* gene, none of them carries the specific mutation found only in *B. anthracis* (Slamti, L. et al. (June 2004). "Distinct mutations in *PlcR* explain why some strains of the *Bacillus cereus* group are nonhemolytic". Journal of Bacteriology. 186 (11): 3531-8).

The lack of *PlcR* in *B. anthracis* is a principle characteristic differentiating it from other members of the *B. cereus* group. While *B. cereus* and *B. thuringiensis* depend on the *plcR* gene for expression of their virulence factors, *B. anthracis* relies on the *pXO1* and *pXO2* plasmids for its virulence (Kolsto, A. et al. (October 2009). "What Sets *Bacillus anthracis* Apart from Other *Bacillus* Species?" Annual Review of Microbiology. 63 (1): 451-476). *Bacillus cereus* biovar anthracis, i.e. *B. cereus* with the two plasmids, is also capable of causing anthrax.

In various embodiments, the disclosed methods identify an antibiotic resistance gene selected from a beta-lactamase gene, such as *blaOXA*, encoding extended spectrum OM class D beta-lactamases, *blaCTX-M 82*, *blaCFX A4*, encoding extended spectrum class A serine beta-lactamases, and *AmpC*, encoding the extended spectrum cephalosporin-resistant class C beta-lactamases; a multidrug efflux transporter system gene such as *acrE*, encoding a component of the *AcrEF-TolC* multidrug efflux transporter system (Lau and Zgurskaya, 2005, J. Bacteriol. 187:7815); *baeR*, encoding a response regulator of the *MdtABC* multidrug efflux transporter system (Nagakubo et al., 2002, J. Bacteriol. 184:4161); *emrY*, encoding a component of the *EmrKY-TolC* multidrug efflux transporter system (Tanabe et al., 1997, J. Gen. Appl. Microbiol. 43:257); *mdtD*, encoding a component of the *MdtABC* multidrug efflux transporter system (Nagakubo et al., 2002, J. Bacteriol. 184:4161); and *mdtN*, encoding a multidrug resistance efflux pump from the major facilitator superfamily (Sulavik et al., 2001, Antimicrob. Agents Chemother. 45:1126); *pbp2*, encoding penicillin binding protein 2 (Bharat et al., 2015, Antimicrob. Agents Chemother. 59:5003); *pbp4*, encoding penicillin binding protein 4 (Sun et al., 2014, PLoS One 9:e97202); and aminoglycoside *strA* (Scholz et al., 1989, Gene 75:271) encodes an aminoglycoside phosphotransferase, and *Tetracycline_tet39* (Agerso and Guardabassi, 2005, J. Antimicrob. Chemother. 55:566) encodes a component of a tetracycline efflux pump.

Other antibiotic resistance genes are provided in the Antibiotic Resistance Genes Database (ARDB), see Nucl. Acids Res. (2009) 37 (suppl 1): D443-D447, the World Wide Web (www) at ardb.cbcb.umd.edu, Antimicrob. Agents Chemother. July 2013 vol. 57 no. 7 3348-3357, and the

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NCBI database (the World Wide Web (www) at ncbi.nlm.nih.gov), the entire contents of which are hereby incorporated by reference.

In various embodiments, the antibiotic resistance gene is one or more of the genes shown below:

Aminocoumarins:

Aminocoumarin-resistant DNA topoisomerases

Aminocoumarin-resistant GyrB, ParE, ParY

Aminoglycosides:

Aminoglycoside acetyltransferases

AAC(1), AAC(2), AAC(3), AAC(6')

Aminoglycoside nucleotidyltransferases

ANT(2"), ANT(3"), ANT(4), ANT(6), ANT(9)

Aminoglycoside phosphotransferases

APH(2"), APH(3"), APH(3'), APH(4), APH(6), APH(7"), APH(9)

16S rRNA methyltransferases

ArmA, RaitA, RntB, RniC, Sgrn

β -Lactams:

Class A β -lactamases

AER, BLA1, CTX-M, KPC, SHV, TEM, etc.

Class B (metallo-) β -lactamases

BlaB, CcrA, IMP, NDM, VIM, etc.

Class C β -lactamases

ACT, AmpC, CMY, LAT, PDC, etc.

Class D β -lactamases

OXA β -lactamase

mecA (methicillin-resistant PBP2)

Mutant porin proteins conferring antibiotic resistance

Antibiotic-resistant Omp36, OmpF, PIB (por)

Genes Modulating β -Lactam Resistance:

bla (blaI, blaR1) and mec (mecI, mecR1) operons

Chloramphenicol:

Chloramphenicol acetyltransferase (CAT)

Chloramphenicol phosphotransferase

Ethambutol:

Ethambutol-resistant arabinosyltransferase (FrbnB)

Mupirocin:

Mupirocin-resistant isoleucyl-tRNA synthetases

MupA, MupB

Peptide Antibiotics:

Integral membrane protein MprF

Phenicol:

Cfr 23S rRNA methyltransferase

Rifampin:

Rifampin ADP-ribosyltransferase (Arr)

Rifampin glycosyltransferase

Rifampin monooxygenase

Rifampin phosphotransferase

Rifampin resistance RNA polymerase-binding proteins

DnaA, RbpA

Rifampin-resistant beta-subunit of RNA polymerase

(RpoB)

Streptogramins:

Cfr 23S rRNA methyltransferase

Erm 23S rRNA methyltransferases

ErmA, ErmB, Erm(31), etc.

Streptogramin resistance ATP-binding cassette (ABC) efflux pumps

Lsa, MsrA, Vga, VgaB

Streptogramin Vgb lyase

Vat acetyltransferase

Fluoroquinolones:

Fluoroquinolone acetyltransferase

Fluoroquinolone-resistant DNA topoisomerases

Fluoroquinolone-resistant GyrA, GyrB, ParC

Quinolone resistance protein (Qnr)

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Fosfomycin:

Fosfomycin phosphotransferases

FomA, FomB, FosC

Fosfomycin thiol transferases

5 FosA, FosB, FosX

Glycopeptides:

VanA, VanB, VanD, VanR, VanS, etc.

Lincosamides:

Cfr 23S rRNA methyltransferase

10 Erm 23SrRNA methyltransferases

ErmA, ErmB, Erm(31), etc.

Lincosamide nucleotidyltransferase (Lin)

Linezolid:

15 Cfr 23S rRNA methyltransferase

Macrolides:

Cfr 23S rRNA methyltransferase

Erm 23S rRNA methyltransferases

ErmA, ErmB, Erm(31),

20 Macrolide esterases

EreA, EreB

Macrolide glycosyltransferases

GimA, Mgt, Ole

Macrolide phosphotransferases (MPH)

25 MPH(2')-I, MPH(2')-II

Macrolide resistance efflux pumps

MefA, MefE, Mel

Streptothricin:

Streptothricin acetyltransferase (sat)

30 Sulfonamides:

Sulfonamide-resistant dihydropteroate synthases

Sul1, Sul2, Sul3, sulfonamide-resistant FolP

Tetracyclines:

35 Mutant porin PIB (por) with reduced permeability

Tetracycline inactivation enzyme TetX

Tetracycline resistance major facilitator superfamily

(MFS) efflux pumps

TetA, TetB, TetC, Tet30, Tet31, etc.

40 Tetracycline resistance ribosomal protection proteins

TetM, TetO, TetQ, Tet32, Tet36, etc.

Efflux Pumps Conferring Antibiotic Resistance:

ABC antibiotic efflux pump

MacAR-TolC, MsbA, MsrA, VgaB, etc.

45 MFS antibiotic efflux pump

EmrD, EmrAB-TolC, NorB, GepA, etc.

Multidrug and toxic compound extrusion (MATE) transporter

MepA

50 Resistance-nodulation-cell division (RND) efflux pump

AdeABC, AcrD, MexAB-OprM, mtrCDE, etc.

Small multidrug resistance (SMR) antibiotic efflux pump

EmrE

Genes Modulating Antibiotic Efflux:

55 adeR, acrR, baeSR, mexR, phoPQ, mtrR

Multidrug Resistance:

plasmid pIP1202

In certain aspects, the disclosed methods the sample is obtained from a subject and the method further comprises administering at least one antibiotic to the subject.

In one aspect, the at least one antibiotic is a fluoroquinolone. Non-limiting fluoroquinolones for use as described herein include levofloxacin, ofloxacin, ciprofloxacin, enoxacin, gatifloxacin, gemifloxacin, grepafloxacin, besifloxacin, lomefloxacin, moxifloxacin, norfloxacin, ofloxacin, pefloxacin, sparfloxacin, garenoxacin, trovafloxacin, sitafloxacin, and DX-619.

In another aspect, the at least one antibiotic is an aminoglycoside such as amikacin, gentamycin, kanamycin, neomycin, netilmicin, paromomycin, streptomycin, or tobramycin.

In another aspect, the at least one antibiotic is a carbapenem such as ertapenem, imipenem, meropenem, or chloramphenicol.

In another aspect, the at least one antibiotic is a glycopeptide such as vancomycin.

In another aspect, the at least one antibiotic is a lincosamide such as clindamycin.

In another aspect, the at least one antibiotic is a macrolide/ketolide such as azithromycin, clarithromycin, dirithromycin, erythromycin, or telithromycin.

In another aspect, the at least one antibiotic is a cephalosporin such as (1st generation) cefadroxil, cefazolin, cephalixin, cephalothin, cephapirin, and cephadrine; or (2nd generation) cefaclor, cefamandole, cefonicid, cefotetan, cefoxitin, cefprozil, cefuroxime, and loracarbef, or (3rd generation) cefdinir, cefditoren, cefixime, cefoperazone, cefotaxime, cefpodoxime, ceftazidime, ceftibuten, ceftizoxime, and ceftriaxone, or (4th generation) cefepime.

In another aspect, the at least one antibiotic is a monobactam such as aztreonam.

In another aspect, the at least one antibiotic is a nitroimidazole such as metronidazole.

In another aspect, the at least one antibiotic is an oxazolidinone such as linezolid.

In another aspect, the at least one antibiotic is a penicillin such as amoxicillin, amoxicillin/clavulanate, ampicillin, ampicillin/sulbactam, bacampicillin, carbenicillin, cloxacillin, dicloxacillin, methicillin, mezlocillin, nafcillin, oxacillin, penicillin G, penicillin V, piperacillin, piperacillin/tazobactam, ticarcillin, or ticarcillin/clavulanate.

In another aspect, the at least one antibiotic is a streptogramin such as quinupristin/dalfopristin.

In another aspect, the at least one antibiotic is a tetracycline such as demeclocycline, doxycycline, minocycline, or tetracycline.

In another aspect, the at least one antibiotic is a β -lactam such as a penicillin, cephalosporin, carbapenem, or monobactam.

The at least one antibiotic may be a physiologically acceptable salt, prodrug, or combination of any one of the aforementioned antibiotics.

The following examples are given for purely illustrative and non-limiting purposes of the present invention.

EXAMPLES

Example 1. Detecting the Presence of Nucleic Acids from *Burkholderia*

This protocol describes procedures for: (1) PCR amplification of multiplexed *Burkholderia* targets; (2) Index extension PCR to prepare amplicons for MiSeq (ILLUMINA® sequencer); (3) SequelPrep™ Normalization Plate Kit (ThermoFisher); and (4) Agencourt AMPure XP bead cleanup for PCR purification (Beckman Coulter).

Universal-tailed gene-specific primers are pooled together in a "primer mix" in amounts relative to each other to help reduce PCR bias. Make the primer mixture by combining the primers and 1×TE (e.g., according to Table 8). Vortex and spin down each primer stock. The primer mixture can be stored at -20°C . Plan and arrange the layout of where the samples will go on a 96-well plate. If clinical samples are being processed, make sure to space the samples on the plate

accordingly. FIG. 11 can be used as an example clinical plate layout. Black wells are samples. Combine the following volumes of reagents as described in Table 10, except IPSC to create the *Burkholderia* Multiplex Master Mix. Reagents should be thawed and mixed before use. Vortexing PCR mastermix should be avoided to prevent damaging the enzymes in the mixture.

To prepare *Burkholderia* Multiplex, on experimental day, thaw out an aliquot of Internal Plasmid Sequencing Control (IPSC) at 10^6 copies per μL , dilute down to 10^3 copies per μL by three serial dilutions of 1 to 10. Make dilutions in tubes for better vortexing & spinning. For single reaction no-template control (NTC) reaction, add 12.5 μL Q5 2× HotStart, 5 μL 5M Betaine, and 4.5 μL Diluted Primer Mix. For single reaction, add 12.5 μL Q5 2× HotStart, 5 μL 5M Betaine, 1 μL IPSC 1000 copies per μL , and 4.5 μL Diluted Primer Mix. To prepare for Master Mix reaction, add 2475 μL Q5 2× HotStart, 990 μL 5M Betaine, 198 μL IPSC 1000 copies per μL , and 891 μL Diluted Primer.

Gently mix template DNA and spin down (Do not vortex genomic DNA). Add 2 μL of template DNA to its appropriate well, along with 2 μL H_2O for IPSC, and 3 μL H_2O for NTC. Seal plate with a thermocycler seal. Spin down the plate. Using a heated lid, put plate on thermocycler and run the following parameters:

Step	Repeats	Temperature ($^{\circ}\text{C}$)	Time
initial denaturation	1	98	5 min
denaturation	35	98	30 sec
annealing		68	15 sec
extension		72	20 sec
final extension	1	72	2 min
cooling	1	10	forever

Spin plates (*Burkholderia* target plate and *Bacillus*, *Yersinia*, *Francisella* target plate) down once the thermocycler finishes.

Prepare AMPure Beads. Prepare 10 mM Tris-HCl 0.05% Tween-20 in H_2O by adding 400 μL 10 mM Tris-HCl, 20 μL 0.05% Tween-20, and 39.580 mL Molecular Grade H_2O , and heat to 50°C . Equilibrate the bead to Room Temperature for 30 minutes. Add beads in a 1:1 ratio with reaction volume to each well (30 μL) and mix well by pipetting. Incubate the bead/reaction mixture for 5 minutes. Place the 96-well plate onto a magnetic stand, incubate for another 5 minutes. Aspirate supernatant out of wells without disturbing the beads. If beads were disturbed, let them incubate for another 2 minutes. Be sure to remove as much liquid as you can. Twice wash the beads by adding 80% EtOH (32 mL 100% Ethanol, and 8 mL Molecular Grade H_2O) to completely cover beads ($\sim 200 \mu\text{L}$) and incubate for 30 seconds. Aspirate. Fully remove liquids after the wash. Move plate off magnetic stand and allow beads to dry. Be sure to keep a close watch on the beads. If the beads start to crack, the DNA will be difficult to elute out. Move plate off the magnetic stand and add 32.54 heated 10 mM Tris-HCl 0.05% Tween-20 in H_2O to the wells, mix well. Incubate for 2 minutes. Move plate to magnetic stand and incubate for 2 minutes. Remove 304 of supernatant and transfer it to a new well, do not disturb or transfer any beads.

Burkholderia, *Bacillus*, *Francisella*, and *Yersinia* AmpSeq amplicons require two bead cleanups before Extension PCR. Repeat steps 12-21.

Detecting the Presence of Nucleic Acids from *Bacillus anthracis*, *Yersinia*, and *Francisella* UT-AmpSeq PCR and Bead Cleanup.

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1. PCR amplification of multiplexed *Bacillus*, *Yersinia*, and *Francisella* targets. Universal-tailed gene-specific primers are pooled together in a “primer mix” in amounts relative to each other to help reduce PCR bias. These amounts have been previously optimized. Please follow the Primer Mix parameters to create the needed mix for the multiplex currently in use.
 2. Index extension PCR to prepare amplicons for MiSeq (ILLUMINA® sequencer). Enter the total number of samples in the box below. Primer Mix Parameters, # of Samples
 3. SequelPrep™ Normalization Plate Kit (ThermoFisher).
 4. Agencourt AMPure XP bead cleanup for PCR purification (Beckman Coulter).
- Protocol
- SOP for *Burkholderia* UT-AmpSeq PCR and Bead Cleanup
- This SOP describes procedures for the following:
1. PCR amplification of multiplexed *Burkholderia* targets
 2. Index extension PCR to prepare amplicons for MiSeq (ILLUMINA® sequencer)
 3. SequelPrep™ Normalization Plate Kit (ThermoFisher)
 4. Agencourt AMPure XP bead cleanup for PCR purification (Beckman Coulter)
- Steps and Procedures
1. Universal-tailed gene-specific primers are pooled together in a “primer mix” in amounts relative to each other to help reduce PCR bias. These amounts have been previously optimized.
 - Please follow the Primer Mix parameters to create the needed mix for the multiplex currently in use
 2. Enter the total number of samples in the box below. Primer Mix Parameters
 - # of Samples
 - 180
 - ensure that all values in column K “How much starting primer conc. to add in mix stock” are above 2.0u1 and not highlighted in red
 - 3a. Make the primer mixture by combining the following primers.
 - 3b. Vortex and spin down each primer stock.
 - 3c. Using the “Start (uM)” concentration primer stock of each primer, add the volume from “Amount to add (uL)” into a 1.7 mL microcentrifuge tube, unless “Total (uL)” at bottom of table is above 1200, then split volume evenly across necessary tubes. Vortex to mix and spin.
 - (Refer to Table 8)
 - 4a. This mixture can be stored at -20° C. for future use. To use mixture, let thaw, vortex and spin down.
 - 4b. If using mixture previously made, write down the initials of the person who made it and when: Initials _____ Date _____
 5. Plan and arrange the layout of where your samples will go on a 96-well plate. If you are processing clinical samples make sure to space your samples on the plate accordingly. 6. Use the Plate Maps sheet for convenience and record keeping.
 - (Refer to FIG. 11).
 - 6a. Reagents should be thawed and mixed before use. Avoid vortexing PCR mastermix as this can damage the enzymes in the mixture.
 - 6b. Combine the following volumes of reagents as described in the following table except IPSC to create the *Burkholderia* Multiplex Master Mix

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<i>Burkholderia</i> Multiplex Reagent	uL for single reaction	uL addition in Master Mix
Q5 2x HotStart	12.5	2475
Betaine 5M	5	990
IPSC 1000 copies/uL	1	198
Diluted Primer Mix	4.5	891
		4554

- 6c. Add 22 uL of *Burkholderia* Multiplex Master Mix without IPSC to any NTC reactions you are processing
- 7a. Thaw out an aliquot of Internal Plasmid Sequencing Control (IPSC) at 10⁶ copies per uL. Dilute this down to 10³ copies/uL by three serial dilutions of 1 to 10. Make dilutions in tubes for better vortexing & spinning. Make this fresh the day of.
- 7b. Add volume with the # of NTCs subtracted of 10³ copies/uL IPSC to master mix. For example, if you had 3 NTCs that you had aliquoted master mix for, and the above table indicated 9 uL of IPSC be added, you would add 6 instead.
- 7d. Mix well and spin down
- 7e. Add 23 uL of Master Mix to each appropriate well on your plate
- 8a. Gently mix template DNA and spin down (Do not vortex genomic DNA)
- 8b. Add 2 uL of template DNA to its appropriate well, along with 2 uL H₂O for IPSC, and 3 uL H₂O for NTC
- 8c. Seal plate with a thermocycler seal
- 8d. Spin down plate
9. Using a heated lid, put plate on thermocycler and run the following parameters

Step:	Reps:	Temp:	Time:
initial denaturation	1	98	5 min
denaturation	35	98	30 sec
annealing		68	15 sec
extension		72	20 sec
final extension	1	72	2 min
cooling	1	10	forever

10. During this time, take out AMPure Beads to equilibrate them to Room Temperature for 30 minutes and heat some 10 mM Tris-HCl 0.05% Tween-20 in H₂O to 50 C.

10 mM Tris-HCl 0.05% Tween-20 in H2O Example Formula	
10 mM Tris-HCl	400 uL
0.05% Tween-20	20 uL
Molecular Grade H2O	39.580 mL
80% Ethanol in H2O	
100% Ethanol	32 mL
MBG H2O	8 mL

11. Spin plates (*Burkholderia* target plate and *Bacillus*, *Yersinia*, *Francisella* target plate) down once the thermocycler finishes
12. Combine 15 uL of *Burkholderia* target reaction with 15 uL of *Bacillus*, *Yersinia*, *Francisella* target reaction
13. Add beads in a 1:1 ratio with reaction volume to each well (30 uL) and mix well by pipette
14. Incubate the bead/reaction mixture for 5 minutes

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15. Place 96-well plate onto a magnetic stand, incubate for another 5 minutes
16. Aspirate supernatant out of wells without disturbing the beads. If beads ARE disturbed, let them incubate for another 2 minutes. Be sure to remove as much liquid as you can
- 17a. Add 80% EtOH to completely cover beads (~200 uL) and incubate for 30 seconds. Aspirate.
- 17b. Repeat 16a and remove as much liquid as you can (two washes total), following with a 20 uL pipette to ensure full removal
18. Move plate off magnetic stand and allow beads to dry. Be sure to keep a close watch on the beads. If the beads start to crack, the DNA will be harder to elute out.
19. Move plate off the magnetic stand and add 32.5 uL heated 10 mM Tris-HCl 0.05% Tween-20 in H₂O to the wells, mix well
20. Incubate for 2 minutes
21. Move plate to magnetic stand and incubate for 2 minutes
22. Remove 30 uL of supernatant and transfer it to a new well, do not disturb or transfer any beads
22. *Burkholderia*, *Bacillus*, *Francisella*, and *Yersinia* AmpSeq amplicons require two bead cleanups before Extension PCR. Repeat steps 12-21
23. Store amplicons at -20 C
- Index Extension PCR
24. Thaw, gently mix, and spin down the following reagents in the following amounts for the Index Extension of the Target Amplicons
- * Amounts are in respect to number of samples entered in Step 2

Index Extension Master Mix Reagent	uL	Lot#
2x Kapa Hifi	2475	
Betaine 5M	990	
Molecular Grade H2O	693	

25. Combine the above volumes together, mix gently, and spin down.
- 26a. Each reaction will require a unique pair of index primers (UT1 and UT2), prepare a chart of what indexes will be used and where
- 26b. Thaw, vortex, and spin down the stock 10 uM aliquots of each index that will be used for this run
- 26c. If some tubes appear empty, create a new 10 uM aliquot of that index. Dilute in TE.
27. Once all indexes are accounted for, add 21 uL of Index Extension Master Mix to each appropriate well in a 96-well plate
28. Add 1 uL of each 10 uM index to its appropriate well
- 29a. After all UT1 and UT2 indexes have been added to their wells add 2 uL of CLEANED AMPLICONS
- 29b. The following should now be in each reaction well

12.5 uL	2x Kapa Hifi
3.5 uL	H2O
5 uL	5M Betaine
2 uL	DNA
1 uL	10 uM UT1
1 uL	10 uM UT2
25 uL	

22

30. Seal the plate with a thermocycler seal
31. Spin down plate
- 32a. Using a heated lid, put plate on thermocycler and run the following parameters

Step:	Reps:	Temp:	Time:
initial denaturation	1	98	2 min
denaturation	8	98	30 sec
annealing		60	20 sec
extension		72	30 sec
final extension	1	72	2 min
cooling	1	10	forever

- 32b. After the PCR has completed, spin down plate
- 33a. Samples will be cleaned and normalized using the Invitrogen SequalPrep system
- 33b. In a new plate, add equal amounts of illext DNA template and SequalPrep Normalization Binding Buffer
- 33c. Mix completely by pipette mixing several times, take care not to etch the sides of the well with the pipette tip
- 33d. Incubate the plate for 1 hour at room temperature to allow binding of DNA to the plate surface (longer than 1 hr is acceptable but will not increase binding or final elution concentration, can be overnight)
- 33e. Aspirate the liquid from the wells
- 33f. Add 50 uL Sequal Prep Normalization Wash Buffer, mix by pipetting up and down twice
- 33g. Completely aspirate the buffer, a small amount of residual Wash Buffer (1-3 uL) is typical
- 33h. Add 20 uL SequalPrep Normalization Elution Buffer to each well of the plate, mix by pipette
- 33i. Incubate at room temperature for 5 minutes
- 33j. Transfer samples to a new plate
- 34a. Samples should all be normalized now so pool them together in equal volumes
- 34b. The final DNA concentration will be fairly low, so perform an AMPure XP bead cleanup on the pool at a 1:1 ratio of pool to beads (be sure to note total volume of pooled samples)
- 34c. However, when eluting the DNA off the beads with heated Tris-Tween use 1/10 the initial pool volume used
35. Store DNA at -20 C
- SOP for *Bacillus*, *Yersinia*, and *Francisella* UT-AmpSeq PCR and Bead Cleanup
- This SOP describes procedures for the following:
 1. PCR amplification of multiplexed *Bacillus*, *Yersinia*, and *Francisella* targets
 2. Index extension PCR to prepare amplicons for MiSeq (ILLUMINA® sequencer)
 3. SequalPrep™ Normalization Plate Kit (ThermoFisher)
 4. Agencourt AMPure XP bead cleanup for PCR purification (Beckman Coulter)

Steps and Procedures

1. Universal-tailed gene-specific primers are pooled together in a "primer mix" in amounts relative to each other to help reduce PCR bias. These amounts have been previously optimized.

Please Follow the Primer Mix Parameters to Create the Needed Mix for the Multiplex Currently in use

2. Enter the total number of samples in the box below.

Primer Mix Parameters

of Samples

180

- ensure that all values in column K "How much starting primer conc. to add in mix stock" are above 2.0u1 and not highlighted in red

23

3a. Make the primer mixture by combining the following primers.

3b. Vortex and spin down each primer stock.

3c. Using the "Start (uM)" concentration primer stock of each primer, add the volume from "Amount to add (uL)" into a 1.7 mL microcentrifuge tube, unless "Total (uL)" at bottom of table is above 1200, then split volume evenly across necessary tubes. Vortex to mix and spin.

(Refer to Table 10)

4a. This mixture can be stored at -20°C . for future use. To use mixture, let thaw, vortex and spin down.

4b. If using mixture previously made, write down the initials of the person who made it and when: Initials _____ Date _____

5. Plan and arrange the layout of where your samples will go on a 96-well plate. If you are processing clinical samples make sure to space your samples on the plate accordingly.

6. Use the Plate Maps sheet for convenience and record keeping.

(Refer to FIG. 11).

6a. Reagents should be thawed and mixed before use. Avoid vortexing PCR mastermix as this can damage the enzymes in the mixture.

6b. Combine the following volumes of reagents as described in the following table except IPSC to create the *Bacillus*, *Francisella*, and *Yersinia* Multiplex Master Mix

<i>Bacillus</i> , <i>Francisella</i> , <i>Yersinia</i> Multiplex Reagent	uL for single reaction	uL addition in Master Mix
Q5 2x HotStart	12.5	2475
H ₂ O	5	990
IPSC 1000 copies/uL	1	198
Diluted Primer Mix	4.5	891
		4554

6c. Add 22 uL of *Burkholderia* Multiplex Master Mix without IPSC to any NTC reactions you are processing

7a. Thaw out an aliquot of Internal Plasmid Sequencing Control (IPSC) at 10^6 copies per uL. Dilute this down to 10^2 copies/uL by three serial dilutions of 1 to 10. Make dilutions in tubes for better vortexing & spinning. Make this fresh the day of.

7b. Add volume with the # of NTCs subtracted of 10^2 copies/uL IPSC to master mix. For example, if you had 3 NTCs that you had aliquoted master mix for, and the above table indicated 9 uL of IPSC be added, you would add 6 instead.

7d. Mix well and spin down

7e. Add 23 uL of Master Mix to each appropriate well on your plate

8a. Gently mix template DNA and spin down (Do not vortex genomic DNA)

8b. Add 2 uL of template DNA to its appropriate well, along with 2 uL H₂O for IPSC, and 3 uL H₂O for NTC

8c. Seal plate with a thermocycler seal

8d. Spin down plate

9. Using a heated lid, put plate on thermocycler and run the following parameters

24

Step:	Reps:	Temp:	Time:
initial denaturation	1	98	5 min
denaturation	35	98	30 sec
annealing		55	15 sec
extension		72	20 sec
final extension	1	72	2 min
cooling	1	10	forever

10. During this time, take out AMPure Beads to equilibrate them to Room Temperature for 30 minutes and heat some 10 mM Tris-HCl 0.05% Tween-20 in H₂O to 50 C

10 mM Tris-HCl 0.05% Tween-20 in H ₂ O Example Formula	
10 mM Tris-HCl	400 uL
0.05% Tween-20	20 uL
Molecular Grade H ₂ O	39.580 mL
80% Ethanol in H ₂ O	
100% Ethanol	32 mL
MBG H ₂ O	8 mL

11. Spin plates (*Burkholderia* target plate and *Bacillus*, *Yersinia*, *Francisella* target plate) down once the thermocycler finishes

12. Combine 15 uL of *Burkholderia* target reaction with 15 uL of *Bacillus*, *Yersinia*, *Francisella* target reaction

13. Add beads in a 1:1 ratio with reaction volume to each well (30 uL) and mix well by pipette

14. Incubate the bead/reaction mixture for 5 minutes

15. Place 96-well plate onto a magnetic stand, incubate for another 5 minutes

16. Aspirate supernatant out of wells without disturbing the beads. If beads ARE disturbed, let them incubate for another 2 minutes. Be sure to remove as much liquid as you can

17a. Add 80% EtOH to completely cover beads (~200 uL) and incubate for 30 seconds.

Aspirate.

17b. Repeat 16a and remove as much liquid as you can (two washes total), following with a 20 uL pipette to ensure full removal

18. Move plate off magnetic stand and allow beads to dry. Be sure to keep a close watch on the beads. If the beads start to crack, the DNA will be harder to elute out.

19. Move plate off the magnetic stand and add 32.5 uL heated 10 mM Tris-HCl 0.05% Tween-20 in H₂O to the wells, mix well

20. Incubate for 2 minutes

21. Move plate to magnetic stand and incubate for 2 minutes

22. Remove 30 uL of supernatant and transfer it to a new well, do not disturb or transfer any beads

22. *Burkholderia*, *Bacillus*, *Francisella*, and *Yersinia* AmpSeq amplicons require two bead cleanups before Extension PCR. Repeat steps 12-21

23. Store amplicons at -20°C

Index Extension PCR

24. Thaw, gently mix, and spin down the following reagents in the following amounts for the Index Extension of the Target Amplicons

*Amounts are in respect to number of samples entered in Step 2

25

Index Extension Master Mix Reagent	uL
2x Kapa Hifi	2475
Betaine 5M	990
Molecular Grade H2O	693
	4158
Lot#	

25. Combine the above volumes together, mix gently, and spin down.

26a. Each reaction will require a unique pair of index primers (UT1 and UT2), prepare a chart of what indexes will be used and where

26b. Thaw, vortex, and spin down the stock 10 uM aliquots of each index that will be used for this run

26c. If some tubes appear empty, create a new 10 uM aliquot of that index. Dilute in TE.

27. Once all indexes are accounted for, add 21 uL of Index Extension Master Mix to each appropriate well in a 96-well plate

28. Add 1 uL of each 10 uM index to its appropriate well

29a. After all UT1 and UT2 indexes have been added to their wells add 2 uL of CLEANED AMPLICONS

29b. The following should now be in each reaction well

12.5 uL	2x Kapa Hifi
3.5 uL	H2O
5 uL	5M Betaine
2 uL	DNA
1 uL	10 uM UT1
1 uL	10 uM UT2
25 uL	

30. Seal the plate with a thermocycler seal

31. Spin down plate

32. Using a heated lid, put plate on thermocycler and run the following parameters

Step:	Reps:	Temp:	Time:
initial denaturation	1	98	2 min
denaturation	8	98	30 sec
annealing		60	20 sec
extension		72	30 sec
final extension	1	72	2 min
cooling	1	10	forever

32b. After the PCR has completed, spin down plate

33a. Samples will be cleaned and normalized using the Invitrogen SequalPrep system

33b. In a new plate, add equal amounts of illex DNA template and SequalPrep Normalization Binding Buffer

33c. Mix completely by pipette mixing several times, take care not to etch the sides of the well with the pipette tip

33d. Incubate the plate for 1 hour at room temperature to allow binding of DNA to the plate surface (longer than 1 hr is acceptable but will not increase binding or final elution concentration, can be overnight)

33e. Aspirate the liquid from the wells

33f. Add 50 uL Sequal Prep Normalization Wash Buffer, mix by pipetting up and down twice

33g. Completely aspirate the buffer, a small amount of residual Wash Buffer (1-3 uL) is typical

26

33h. Add 20 uL SequalPrep Normalization Elution Buffer to each well of the plate, mix by pipette

33i. Incubate at room temperature for 5 minutes

33j. Transfer samples to a new plate

34a. Samples should all be normalized now so pool them together in equal volumes

34b. The final DNA concentration will be fairly low, so perform an AMPure XP bead cleanup on the pool at a 1:1 ratio of pool to beads (be sure to note total volume of pooled samples)

34c. However, when eluting the DNA off the beads with heated Tris-Tween use $\frac{1}{10}$ the initial pool volume used

35. Store DNA at -20 C

Burkholderia pseudomallei, *Burkholderia mallei*, *Bacillus anthracis*, *Yersinia pestis* and *Francisella tularensis* are all Tier 1 select agents, posing a potentially severe threat to public health.¹

Current surveillance methods rely upon single locus PCR techniques that allow for only presence/absence of SA results.

Has been known to lead to false positives, especially due to the complexity of environmental samples including huge numbers of microorganisms, many of which can be highly similar target pathogens.

Constantly working to develop technologies that significantly improve the identification of biological contaminants in varying sample types

Both sequencing costs and sequencing time are decreasing.²

Targeted amplicon sequencing can provide the necessary information at a fraction of the cost of WGS.

Targeted amplicon sequencing can also provide a more manageable data set for researchers with less background in bioinformatics with an appropriate processing tool.

Summary

With the cost of sequencing decreasing and the ability to combine higher and higher numbers of samples on a single run, amplicon sequencing provides a cost effective alternative to individual PCR methods.

A screening panel of target organism strains as well as near neighbor strains allowed for differentiation with 100% sensitivity for all target agents and 91-100% specificity.

Future Directions

Limit of detection testing across a subset of target organisms in a pristine sample.

Limit of detection testing across a subset of target organisms in samples with environmental and human backgrounds.

Testing on other sequencing platforms.

REFERENCES

- ¹ Select Agents and Toxins Regulations. 42 C.F.R. Part 73.3 HHS Select Agents and Toxins
- ² Goodwin S, McPherson J D, McCombie W R. Coming of age: ten years of next-generation sequencing technologies. Nat Rev Genet. 2016

TABLE 1

BTseq Loci Summary				
	<i>B. anthracis</i>	<i>B. pseudo-mallei/ mallei</i>	<i>F. tularensis</i>	<i>Y. pestis</i>
Target Species Loci PA	9	12*	3	3
NN Species Loci PA	4	0	3	0
NN Species Loci SNP/SV	7	9	7	7
AMR Loci	6	4	0	3
Virulence/Plasmid Loci	4	0	0	2
Total	30	25	13	15

*Five loci amplify both Bp and Bm

TABLE 2

SNPs for detecting the presence of nucleic acids from <i>Bacillus anthracis</i> .							
LocusID	Ba_AmesAnc						
	1069024::91	1712462::85	3668548::88	371913::77	388555::108	3981642::97	4001578::113
Reference	T	C	C	T	A	A	C
Ba_A0098_S45_L001	T	C	C	T	A	A	C
Ba_A0330_S46_L001	T	C	C	T	A	A	C
Ba_A0490_S47_L001	T	C	C	T	A	A	C
Ba_A0605_S48_L001	T	C	C	T	A	A	C
Ba_A0615_S49_L001	T	C	C	T	A	A	C
Ba_A0706_S50_L001	T	C	C	T	A	A	C
Ba_A071_S52_L001	T	C	C	T	A	A	C
Ba_A072_S58_L001	T	C	C	T	A	A	C
Ba_A073_S59_L001	T	C	C	T	A	A	C
Ba_A074_S60_L001	T	C	C	T	A	A	C
Ba_A0767_S51_L001	T	C	C	T	A	A	C
Ba_A0847_S53_L001	T	C	C	T	A	A	C
Ba_A1085_S54_L001	T	C	C	T	A	A	C
Ba_A2010_S55_L001	T	C	C	T	A	A	C
Ba_A2105_S56_L001	T	C	C	T	A	A	C
Ba_A2175_S57_L001	T	C	C	T	A	A	C
Ba_NN295171-grainy_S63_L001	A	X	X	X	X	X	T
Ba_NN295171-smooth_S70_L001	X	X	X	X	X	X	X
Ba_NN295618_S71_L001	X	X	X	X	X	X	X
Ba_NNAI-hakem-grainy_S77_L001	X	X	X	X	X	X	X
Ba_NNAI-hakem-smooth_S64_L001	A	X	X	C	G	G	T
Ba_NNATCC10792_S62_L001	A	X	X	C	G	G	T
Ba_NNATCC13402_S76_L001	X	X	X	X	X	X	X
Ba_NNATCC31293_S74_L001	X	X	X	X	X	X	T
Ba_NNBacillus-cereus_S69_L001	X	X	X	X	X	X	X
Ba_NNBacillus-megaterium_S75_L001	A	X	X	X	X	X	T
Ba_NNFRI33_S72_L001	X	X	X	X	X	X	X
Ba_NNFRI35_S73_L001	A	X	X	X	G	G	T
Ba_NNHD-1011_S93_L001	A	X	T	C	G	G	T
Ba_NNHD-1012_S94_L001	A	X	X	C	X	X	T
Ba_NNHD-1015_S61_L001	A	X	X	X	X	G	T
Ba_NNHD-288_S81_L001	A	X	X	C	G	G	T
Ba_NNHD-34_S78_L001	X	X	X	X	X	X	T
Ba_NNHD-44-grainy_S79_L001	X	X	X	C	X	G	T
Ba_NNHD-44-smooth_S66_L001	X	X	X	X	X	X	X
Ba_NNHD-47_S80_L001	A	X	X	C	G	X	T
Ba_NNHD-526-grainy_S82_L001	A	X	X	T	X	G	T

TABLE 2-continued

SNPs for detecting the presence of nucleic acids from <i>Bacillus anthracis</i> .							
Ba_NNHD-526-smooth_S67_L001	A	T	X	C	G	G	T
Ba_NNHD-557_S83_L001	X	X	X	X	X	X	T
Ba_NNHD-571-grainy_S84_L001	A	X	T	C	G	G	T
Ba_NNHD-571-smooth_S65_L001	A	X	T	C	G	G	T
Ba_NNHD-621_S85_L001	X	X	X	X	X	X	T
Ba_NNHD-681_S86_L001	A	X	X	X	G	X	T
Ba_NNHD-682_S87_L001	X	X	X	X	X	X	X
Ba_NNHD-711_S88_L001	X	X	X	X	X	X	X
Ba_NNHD-754_S89_L001	A	X	X	X	X	X	T
Ba_NNHD-789_S90_L001	X	X	X	X	X	X	X
Ba_NNHD-930_S91_L001	A	X	X	X	G	X	T
Ba_NNHD-974-grainy_S92_L001	A	X	X	X	G	G	T
Ba_NNHD-974-smooth_S68_L001	A	X	X	X	G	G	T
Ba_NNNTC_Ba_1_S95_L001	X	X	X	X	X	X	X
Ba_NNNTC_Ba_2_S96_L001	X	X	X	X	X	X	X
F1057_S97_L001	X	X	X	X	X	X	X
F1062_S98_L001	X	X	X	X	X	X	X
NTC_Ft1_S99_L001	X	X	X	X	X	X	X
NTC_Ft2_S100_L001	X	X	X	X	X	X	X
NTC_Yp1_55_S104_L001	X	X	X	X	X	X	X
NTC_Yp2_55_S105_L001	X	X	X	X	X	X	C
NTC_Yp3_60_S109_L001	X	X	X	X	X	X	C
NTC_Yp4_60_S110_L001	X	X	X	X	X	X	C
NTC_Yp5_65_S114_L001	X	X	X	X	X	X	C
NTC_Yp6_65_S115_L001	X	X	X	X	X	X	C
Yp1763_55_S101_L001	X	X	X	X	X	X	X
Yp1763_60_S106_L001	X	X	X	X	X	X	X
Yp1763_65_S111_L001	X	X	X	X	X	X	X
Yp2051_55_S102_L001	X	X	X	X	X	X	X
Yp2051_60_S107_L001	X	X	X	X	X	X	X
Yp2051_65_S112_L001	X	X	X	X	X	X	X
Yp2126_55_S103_L001	X	X	X	X	X	X	X
Yp2126_60_S108_L001	X	X	X	X	X	X	X
Yp2126_65_S113_L001	X	X	X	X	X	X	X

LocusID	Ba_AmesAnc					plcR::147
	4087624::116	4669915::40	734209::105	999035::79		
Reference	G	T	T	G		A
Ba_A0098_S45_L001	G	T	T	G		A
Ba_A0330_S46_L001	G	T	T	G		A
Ba_A0490_S47_L001	G	T	T	G		A
Ba_A0605_S48_L001	G	T	T	G		A
Ba_A0615_S49_L001	G	T	T	G		A
Ba_A0706_S50_L001	G	T	T	G		A
Ba_A071_S52_L001	G	T	T	G		A
Ba_A072_S58_L001	G	T	T	G		A
Ba_A073_S59_L001	G	T	T	G		A
Ba_A074_S60_L001	G	T	T	G		A
Ba_A0767_S51_L001	G	T	T	G		A
Ba_A0847_S53_L001	G	T	T	G		A
Ba_A1085_S54_L001	G	T	T	G		A
Ba_A2010_S55_L001	G	T	T	G		A
Ba_A2105_S56_L001	G	T	T	G		A
Ba_A2175_S57_L001	G	T	T	G		A
Ba_NN295171-grainy_S63_L001	X	X	X	A		X
Ba_NN295171-smooth_S70_L001	X	X	X	X		X
Ba_NN295618_S71_L001	X	X	X	X		X
Ba_NNAI-hakem-grainy_S77_L001	X	X	X	X		X
Ba_NNAI-hakem-smooth_S64_L001	T	C	C	A		X

TABLE 2-continued

SNPs for detecting the presence of nucleic acids from <i>Bacillus anthracis</i> .					
Ba_NNATCC10792_S62_L001	T	C	C	A	C
Ba_NNATCC13402_S76_L001	X	X	X	X	X
Ba_NNATCC31293_S74_L001	X	X	X	X	X
Ba_NNBacillus-cereus_S69_L001	X	X	X	X	X
Ba_NNBacillus-megaterium_S75_L001	X	X	X	X	X
Ba_NNFRI33_S72_L001	X	X	X	X	X
Ba_NNFRI35_S73_L001	T	C	X	A	C
Ba_NNHD-1011_S93_L001	T	C	C	A	C
Ba_NNHD-1012_S94_L001	T	C	X	A	C
Ba_NNHD-1015_S61_L001	X	C	X	A	X
Ba_NNHD-288_S81_L001	X	C	C	A	X
Ba_NNHD-34_S78_L001	X	X	X	X	X
Ba_NNHD-44-grainy_S79_L001	X	C	X	X	C
Ba_NNHD-44-smooth_S66_L001	X	X	X	X	X
Ba_NNHD-47_S80_L001	X	C	X	A	X
Ba_NNHD-526-grainy_S82_L001	X	C	X	X	A
Ba_NNHD-526-smooth_S67_L001	T	C	X	A	X
Ba_NNHD-557_S83_L001	X	X	X	X	X
Ba_NNHD-571-grainy_S84_L001	T	C	C	A	C
Ba_NNHD-571-smooth_S65_L001	T	C	C	A	C
Ba_NNHD-621_S85_L001	X	X	X	X	X
Ba_NNHD-681_S86_L001	X	C	X	A	X
Ba_NNHD-682_S87_L001	X	X	X	X	X
Ba_NNHD-711_S88_L001	X	X	X	X	X
Ba_NNHD-754_S89_L001	X	C	X	A	X
Ba_NNHD-789_S90_L001	X	X	X	X	X
Ba_NNHD-930_S91_L001	X	C	X	A	X
Ba_NNHD-974-grainy_S92_L001	X	C	X	X	X
Ba_NNHD-974-smooth_S68_L001	X	C	X	X	X
Ba_NNNTC_Ba_1_S95_L001	X	X	X	X	X
Ba_NNNTC_Ba_2_S96_L001	X	X	X	X	X
F1057_S97_L001	X	X	X	X	X
F1062_S98_L001	X	X	X	X	X
NTC_Ft1_S99_L001	X	X	X	X	X
NTC_Ft2_S100_L001	X	X	X	X	X
NTC_Yp1_55_S104_L001	X	X	X	X	X
NTC_Yp2_55_S105_L001	X	X	X	X	A
NTC_Yp3_60_S109_L001	X	X	X	X	X
NTC_Yp4_60_S110_L001	X	X	X	X	A
NTC_Yp5_65_S114_L001	X	X	X	X	X
NTC_Yp6_65_S115_L001	X	X	X	X	A
Yp1763_55_S101_L001	X	X	X	X	X
Yp1763_60_S106_L00	X	X	X	X	X
Yp1763_65_S111_L001	X	X	X	X	X
Yp2051_55_S102_L001	X	X	X	X	X
Yp2051_60_S107_L001	X	X	X	X	X
Yp2051_65_S112_L001	X	X	X	X	X
Yp2126_55_S103_L001	X	X	X	X	X
Yp2126_60_S108_L001	X	X	X	X	X
Yp2126_65_S113_L001	X	X	X	X	X

TABLE 3

Primers for detecting the presence of nucleic acids from <i>Bacillus anthracis</i> .						
Assay name	Assay Type	Target species/gene		primer name	sequence (5' -> 3')	SEQ ID NO
CP008853.1_5309	PA	<i>B. anthracis</i>	F	Ba-specific-3F_UT1	ACGTCAGGTGATTATTGGAC	1
			R	Ba-specific-3R_UT2	CAACAATTATATCCGCCATT	2
CP008853.1_5316	PA	<i>B. anthracis</i>	F	Ba-specific-5F_UT1	GAAGATGTACGCTCGATAGG	3
			R	Ba-specific-5R_UT2	GAAATTCTTTTGGCCATCAC	4
CP012725.1_3629	PA	<i>B. anthracis</i>	F	Ba-specific-6F_UT1	CACAATTGAATGAAATGCT	5
			R	Ba-specific-6R_UT2	CACGAAACCTGTTTACCTTT	6
CP012725.1_5103	PA	<i>B. anthracis</i>	F	Ba-specific-8F_UT1	GATATTCGACGAGCTTTCTG	7
			R	Ba-specific-8R_UT2	TATTCATCGTCATCCTCTC	8
CP012725.1_5107	PA	<i>B. anthracis</i>	F	Ba-specific-9F_UT1	TATTGAACGCATTGAATCAG	9
			R	Ba-specific-9R_UT2	TATTGGTAAGCAAACCGTCT	10
JSZQ01000034.1_220	PA	<i>B. anthracis</i>	F	Ba-specific-11F_UT1	GGTTCAGGACAAAATGTAGC	11
			R	Ba-specific-11R_UT2	TAACTTCTGAAGCGAAAACC	12
JSZS01000036.1_5	PA	<i>B. anthracis</i>	F	Ba-specific-12F_UT1	GCGAATTTTAGACGACAATC	13
			R	Ba-specific-12R_UT2	TAACCGTGTCTAATTCGTTT	14
LGCC01000010.1_232	PA	<i>B. anthracis</i>	F	Ba-specific-14F_UT1	ATTAATAAGGCGACTGGTGA	15
			R	Ba-specific-14R_UT2	TTACCCATCCAGAATGAGAC	16
LGCC01000048.1_280	PA	<i>B. anthracis</i>	F	Ba-specific-16F_UT1	ACAATTCCTAAAAGGCGACA	17
			R	Ba-specific-16R_UT2	TGTAGCGTCTCCGATATTTT	18
NN_LOMU01000090.1_49	PA	near neighbor species	F	Ba-specific-20F_UT1	CATGGGGCTTTTCTATTATGT	19
			R	Ba-specific-20R_UT2	TTCGTTCTTTTCATAAGTTTCCT	20
NN_LOQC01000013.1_3	PA	near neighbor species	F	Ba-specific-22F_UT1	TTGGAGTTTGTGTTTGTCTTTT	21
			R	Ba-specific-22Rv2_UT2	GTAACAATTAATCCACGTCTCT	22
ChimpKiller_9-159	PA	<i>B. cereus</i> spp. <i>anthracis</i>	F	ChimpKiller_9F	TTATCGTCCATTCTTTCGTC	23
			R	ChimpKiller_159R	AAACCTAATGAAACGGGATT	24
ChimpKiller_91-320	SV	<i>B. cereus</i> spp. <i>anthracis</i>	F	ChimpKiller_91F	TATGAAAGGAGCCGTAAC	25
			R	ChimpKiller_320R	TGAATATGAAGCGGAAACT	26
ChimpKiller_481-698	SV	<i>B. cereus</i> spp. <i>anthracis</i>	F	ChimpKiller_481F	TCGAACATACCTCCATTTCT	27
			R	ChimpKiller_698R	AAAGATAGCTTTGCACTTGG	28
plcR	PA	Virulence locus plcR	F	Ba-specific-1F_UT1	TTTTTCGTAAGCATCTTCAA	29
			R	Ba-specific-1R_UT2	TTTGATGTGAAGGTGAGACA	30
pagA	PA	Virulence locus pagA	F	801F_pagAv3_UT1	GGTTACAGGACGGATTGATA	31
			R	1042R_pagAv3_UT2	TCCCACCAATATCAAAGAAC	32
pX01	PA	Virulence plasmid	F	pX01_113F_UT1	TGAGCCTACCTAGTGATTGG	33
			R	pX01-315Rv2_UT2	TTGGATAAATTCCACAAATTCC TC	34
pX02	PA	Virulence plasmid pX02	F	pX02_101F_UT1	CGCCAGCGTATTATATAGGT	35
			R	pX02_269R_UT2	GCTAATTCTGGGTGTGTGTT	36
gyrA	SNP	Drug resistance SNP gyrA	F	gyrA_28Fv2_UT1	TCGGTAAGTATCACCTCA	37
			R	gyrA_182Ry2_UT2	TGCTTCTGTATAACGCATT	38
parC	SNP	Drug resistance SNP parC	F	parC_1F_UT1	CAGTCGGTAACGTTATTGGT	39
			R	parC_197R_UT2	TAACTCAGATGCAATTGGTG	40
gyrB	SNP	Drug resistance SNP gyrB	F	gyrB_8F_UT1	ATTGTAGAGGGTGACTCTGC	41
			R	gyrB_194R_UT2	TATCAAAATCTCCGCAAT	42
rpoB	SNP	Drug resistance SNP rpoB	F	rpoB_29F_UT1	TTCTTCGGAAGTTCTCAGTT	43
			R	rpoB_196R_UT2	CGGACACATACGACCATAG	44
AA_2502	SNP	Drug resistance SNP	F	AA_2502_UT1	AAGTTTGAGGTGTGGAAATG	45
			R	AA_2502_UT2	TCGAAATGAGTTCCAATTTT	46
AA_2503	SNP	Drug resistance SNP	F	AA_2503v2_UT1	CAAACTAATAGGGGAGGGTG	47
			R	AA_2503_UT2	CCGAGAACCTACCTCGTTA	48

TABLE 3-continued

Primers for detecting the presence of nucleic acids from <i>Bacillus anthracis</i> .						
Assay name	Assay Type	Target species/gene		primer name	sequence (5' -> 3')	SEQ ID NO
Ba_AmesAnc_4669915	SV	near neighbor species	F	Ba&NN32_F	AGGAGATGAGAGTTTTGCAC	49
			R	Ba&NN32_R	ACCCCCATAATTACCATGA	50
Ba_AmesAnc_4001578	SV	near neighbor species	F	Ba&NN33_F	CGTTGCGTAAGTATGTGCTA	51
			R	Ba&NN33_R	AGGTGGCGTAATTAACGTAG	52
Ba_AmesAnc_1069024	SV	near neighbor species	F	Ba&NN37_F	CGAAAAGTTGTCGACCTAAT	53
			R	Ba&NN37_R	ACTGCGTTCACGAAGAATAG	54
Ba_AmesAnc_3668548	SV	near neighbor species	F	Ba&NN38_F	TCTCTTGATTCAACGTTTCC	55
			R	Ba&NN38_R	GATGCAAAACCAATTCACTT	56
Ba_AmesAnc_371913	SV	near neighbor species	F	Ba&NN40_F	GTGAAACATCGCTTTTTAGG	57
			R	Ba&NN40_R	TCCGCAATGATATACTTCAA	58
Ba_AmesAnc_999035	SV	near neighbor species	F	Ba&NN41_F	ATACGGTGAAAATGAAGCAG	59
			R	Ba&NN41_R	CGTCTTTGGTAATCGTTCA	60

TABLE 4

Primers for detecting the presence of nucleic acids from <i>Burkholderia</i> .					
Assay name	Assay Type	Target species/gene	primer name	sequence (5' -> 3')	SEQ ID NO:
TTS1_BPSS1407	PA	TTS1	F BpAmpSeq_1_F	TCGTCGTACCGGGAT	61
			R BpAmpSeq_1_R	GGTC GGCCTTTGCCCGCATA CTCG	62
LXCC01000141.1_392 PA 96_39817	PA	<i>B. pseudomallei</i>	F BpAmpSeq_3_F	TCGCAWGAAGTGCGT	63
			R BpAmpSeq_3_R	TGCCG GCCGCTTGCGAAGCGA TGAT	64
LXBY01000087.1_757 PA 60_76751	PA	<i>B. pseudomallei</i>	F BpAmpSeq_4_F	CGCGCTTGCCCACTA	65
			R BpAmpSeq_4_R	CCAG GCGCAACGGTGCGAG ACAAT	66
LXCD01000002.1_996 PA 52_100245	PA	<i>B. pseudomallei</i>	F BpAmpSeq_5_F	AATCCATGCATGTCGY	67
			R BpAmpSeq_5_R	GCCC GCGATCGCTCAACGGG CTTC	68
LXCE01000123.1_342 PA 20_34747	PA	<i>B. pseudomallei</i>	F BpAmpSeq_6_F	TCGCATTGCAAYACGC	69
			R BpAmpSeq_6_R	TCCC AGTGCGCAAACCTGGC GAGG	70
BpCEN586498	PA	<i>pseudomallei</i>	F BpCEN586498_F102	CACCGAAAGATTTCAG	71
			R BpCEN586498_R388	TTCCGCCTCATTCA GGCCGTCGATGGTTTC GTCGGTTTTC	72
BpCEN617822	PA	<i>pseudomallei</i>	F BpCEN617822_F43	TGCATTGAGCACGGCA	73
			R BpCEN617822_R260	CGCAGATTC GAAAAATTTATCGGAT CGAGCACCATGGTTG	74
BpCEN972235	PA	<i>pseudomallei</i>	F BpCEN972235_F107	ATACGCGGCGCGGCTC	75
			R BpCEN972235_R305	ATTTTCG GCGTCGCGCTCGTCGA TACGGTCA	76
BpCEN70178	PA	<i>pseudomallei</i>	F BpCEN70178-2F	TGCGCAGCGAGTGGTT	77
			R BpCEN70178-182R	CAGGTTGTC CGACGATACGGATAC GGCACGGAAGC	78

TABLE 4-continued

Primers for detecting the presence of nucleic acids from <i>Burkholderia</i> .					
Assay name	Assay Type	Target species/ gene	primer name	sequence (5' -> 3')	SEQ ID NO:
BpCEN508364	PA	<i>pseudomallei</i>	F UT1-BpCEN508364_F37	CCGCGCCGGCCGCAG ACC	79
			R UT2-BpCEN508364_R184	CGGGCGTGCCGGACTC CTCGTC	80
LWWC01000187.1_18	PA	<i>B. pseudomallei</i> <i>mallei</i>	F BpAmpSeq_8_F	CCTTTGCGGCAAGCGT CGAA	81
			R BpAmpSeq_8_R	GAGCCAACGCACATG GACGG	82
LWWB01000125.1_17 183_17602	PA	<i>B. pseudomallei</i> <i>mallei</i>	F BpAmpSeq_10_F	CCAGTCGGGCCGGGA AAAAC	83
			R BpAmpSeq_10_R	GGCGGCAAAAGCGTC GATGA	84
LXAY01000367.1_0_6 40	PA	<i>B. pseudomallei</i> <i>mallei</i>	F BpAmpSeq_11_F	GCCGGAACCGTCGAG CATTG	85
			R BpAmpSeq_11_R	TGGATTGCGACTGCCTC CGCT	86
LWVY01000190.1_17 226_17689	PA	<i>B. pseudomallei</i> <i>mallei</i>	F BpAmpSeq_12_F	TCGATATCCGCCGTCT CGCC	87
			R BpAmpSeq_1_R	ATGTGTCGGTGGGCTT CGGT	88
LXAD01000059.1_247 60_25075	PA	<i>B. pseudomallei</i> <i>mallei</i>	F BpAmpSeq_13_F	GAAAGGCGATGTGCC GAGCG	89
			R BpAmpSeq_13_R	TTCGGAGAAGCGCCA AACGC	90
BpmCEN322640	PA	<i>pseudomallei/mallei</i>	F BpCEN322640_F2	CGCGGACAGCATCGAT TACGTGAATC	91
			R BpCEN32264_R2	CCGCCGAATCCGATGC TCAATTTT	92
BpmCEN1761486	PA	<i>pseudomallei/mallei</i>	F BpmCEN1761486_F1	GACCTGCAGCAGGTAT TCGACATTATCGTTC	93
			R BpmCEN1761486_R1	AGCTTCGCATACAGCA CTTCCGCCAG	94
BpmCEN1235988	PA	<i>pseudomallei/mallei</i>	F BpmCEN1235988_F1	GCGCTGCCCGTTTCAC CACTGG	95
			R BpmCEN1235988_R1	CGTGACGCCGTCGGGA AAGATCATC	96
BpmCEN1565214	PA	<i>pseudomallei/mallei</i>	F BpmCEN1565214_F1	CTGACCGAACGATGGC TGGAGATACATGC	97
			R BpmCEN1565214_R1	CAAAATGGGAAGCGAG CTCCCTTCCGA	98
BpmCEN276339-1	PA	<i>pseudomallei/mallei</i>	F BpmCEN276339-1_F1	CGGACGCCCTGTGCCCC GAAACCTAT	99
			R BpmCEN276339-1_R1	CGCGAGCACGCCGAG CGACAT	100
BpmCEN276339-2	PA	<i>pseudomallei/mallei</i>	F BpmCEN276339-2_F1	CGTCGACGCCCGGGC TTTCTG	101
			R BpmCEN276339-2_R1	CGCCGCGCACCGGTTT CAATC	102
BpmCEN894337	PA	<i>pseudomallei/mallei</i>	F BpmCEN894337_F_1	CGAAAATAATTTTCGG CCGGCGCAC	103
			R BpmCEN894337_R1	CGACAGGCATCGGGC GACTACTACCG	104
BpmCEN1722622	PA	<i>pseudomallei/mallei</i>	F UT2-Bpm_CEN1722622-f1	CAACGGGCGAGTTTGC AACGGAATC	105
			R UT1-Bpm_CEN1722622-1-1	GCCCGCTTGGCTTCGT CCTTGTC	106
BpmCEN357268	PA	<i>pseudomallei/mallei</i>	F UT1-Bpm_CEN357268-f1	CGGCATGCGCGGCCG AATC	107
			R UT2-Bpm_CEN357268-r1	ATCGCGCCCTGCAGCG AGCAC	108

TABLE 4-continued

Primers for detecting the presence of nucleic acids from <i>Burkholderia</i> .					
Assay name	Assay Type	Target species/ gene	primer name	sequence (5' -> 3')	SEQ ID NO:
NC_006350_2289827	SV	<i>B. pseudomallei</i> complex SNP	F BpAmpSeq_16_F	GCCAGCGCATCCACCA	109
			R BpAmpSeq_16_R	ACAT AGAGGAAGAAGGGCG AGGCG	110
NC_006350_133027	SV	<i>B. cepacia</i> complex SNPs	F BpAmpSeq_18_F	CGCGCARYTCGTCGTC	111
			R BpAmpSeq_18_R	CTCG CGAACCTSGTGCMGGT RCAG	112
NC_006350_2248145- 2248193	SV	<i>B. pseudomallei</i> complex SNP	F BpAmpSeq_19_F	CACGTTGCCSGRAAR	113
			R BpAmpSeq_19_R	TACG CCGTCGACAAGATCGC GCTS	114
NC_006350_988041- 988089	SV	<i>B. pseudomallei</i> complex SNP	F BpAmpSeq_20_F	CAGAACGCGCTRTYCC	115
			R BpAmpSeq_20_R	ACG TGCCGCGTGATCCATT GCAT	116
Bm_11589	PA	<i>B. mallei</i>	F BpAmpSeq_21_F	AGGGGGTGGTTTCCTG	117
			R BpAmpSeq_21_R	AGTGGCGTGAC AGCGGTGTCACGGGT GGAAAGGATG	118
Bm_11767	PA	<i>B. mallei</i>	F BpAmpSeq_22_F	ACGGGCGCTTCACGAT	119
			R BpAmpSeq_22_R	CTCGGTGTTT GCGCGGCAGTTCGATC AGGCATTG	120
Bm_11589	PA	<i>B. mallei</i>	F Bm-11589-f1_UT1	GACGGCGGGCTTTGGG	121
			R Bm-11589-r1_UT2	GAGTCC GCTCGCGGCAGCGGT GTCG	122
Bm_11767	PA	<i>B. mallei</i>	F Bm-11767-f1_UT1	GACGGCCCCGGGCGG	123
			R Bin-11767-r1_UT2	CTTTAC CGCGGCAGTTCGATCA GGCATTGAG	124
Bm_11767	PA	<i>B. mallei</i>	F Bm-11767-f1_UT1	GACGGCCCCGGGCGG	125
			R Bm-11767-r2_UT2	CTTTAC CGAGGGGCGAAATTC CCCTTATAGATCAGTT G	126
K9penA378-529	SNP	penA	F BpAmpSeq_26_F	CGGTCGCCACAAATTC	127
			R BpAmpSeq_26_R	GCACGCACTC AGCGAGCGGCGCAAC GGAGAATGATT	128
K9penA575-761	SNP	penA	F BpAmpSeq_27_F	GCTGCGCGCCAAGC	129
			R BpAmpSeq_27_R	GAAAAACG CGCGAGGACCGCAGC GCAAAGC	130
K9penA949-1172	SNP	penA	F BpAmpSeq_28_F	GGCCGCGAGACCGTCAC	131
			R BpAmpSeq_28_R	CGCGTATG GTCGCCCCGTCTTGTTG CCGAGCATC	132
penA_-78promoter	SNP	penA	F K9penA281fUT1	GCCCCGTCAATCCGATG	133
			R K9penA565rUT2	CMGTATCTGG GCGCCGATCARTGGGG TGGAATG	134
penA_C69Y_S72F	SNP	penA	F K9penA696fUT1	CATCGCGGCGACGAG	135
			R K9penA848rUT2	CGTTTCC CTCGGTGATCGGCGAA TAGCGGATGAGA	136

TABLE 4-continued

Primers for detecting the presence of nucleic acids from <i>Burkholderia</i> .					
Assay name	Assay Type	Target species/ gene	primer name	sequence (5' -> 3')	SEQ ID NO:
penA_P167S	SNP	penA	F K9penA881fUT1	GCTGTGCGCGGCGACG	137
			R K9penA1258rUT2	CTTCAGTA CCGATGTCGTTCGCCG TTCCGTAGTC	138
PBP3-170f-505r	PA	penA	F PBP3-170f3UT1	ATCCGCCGTCGCCGCC	139
			R PBP3-505r3UT2	AGCAATAG GGGTTGCGCCAGATTT CGTAGGTGGTGAG	140
pbp3-1	PA	pbp3	F K9pbp336fUT1	TCGCCGTTTCACGCC	141
			R K9PBP3331rUT2	CGCAAC GCGCCGAACGCGAGG AACACGA	142
pbp3-2	PA	pbp3	F K9pbp31292fUT1	GCTCGCGAAGCTCGCG	143
			R K9PBP31527rUT2	CTGAACC GGATCGTGCCGTCGCC CGCATAC	144
V15G_R20	SV	folM pteridine reductase	F 1026pter371fU	ACAAGCCCGGYGTCGTCG	145
			R 1026pter636rUT2	AGATGGTGAC CGCGTCGGCCGAAYG GTCGTAGT	146
bpeT HTH	SV	bpeT HTH region	F bpeT_-76fUT1	AATCGTCGGCTGCGTC	147
			R bpeT_596rUT2	GCCTTCA CGGGTAGCGTGAGTG GAATTCGCAGAG	148
bpeT substrate binding	SV	bpeT substrate binding region	F bpeT_695fUT1	CCTCGAAGGCTTCGGG	149
			R bpeT_1014rUT2	CTGATCCAG GACTAACCGCTTACGC CACCCACTCGTTC	150
bpeS HTH	SV	bpeS HTH region	F bpeS_-83fUT1	AAAGCGAATAGTCGC	151
			R bpeS_230rUT2	GAAGCGGCTTGA GCGATCTCGGTGATGA TCTTGATGCAGTG	152
bpeS substrate binding	SV	bpeS substrate binding region	F bpeS_648fUT1	AACGGCGGCGTGACC	153
			R bpeS_977rUT2	GTCAACG CGCTACGCGGCCACCT GCCC	154
bimA	PA	bimA	F Bpvir_bimA_407F	CGGAGCTTCAGAACA	155
			R Bpvir_bimA_654R	ACCCGCGTGTAAC CCTTCGGACCTTTTCC CGCAACTGGC	156
cheD	PA	cheD	F Bpvir_cheD_29F	AAATCGGCCGCGAGGC	157
			R Bpvir_cheD_297R	GGTACG CGCGCGCAGCCGGCAT TTG	158
fhaB1long	PA	fhaB1long	F Bpvir_fhaB1long_8410F	CCCTTCGGTCCCCACC	159
			R Bpvir_fhaB1long_8599R	AGAAAAATTCG AGCCGTACAGGCAAT GCAGCCATCTATG	160
fhaB1short	PA	fhaB1short	F Bpvir_fhaB1short_63F	GCGCCGCGTGTTCTGTG	161
			R Bpvir_fhaB1short_316R	ACCTTGTC CGCTGATCGGCGCATC GGACAC	162
fhaB2	PA	fhaB2	F Bpvir_fhaB2_1812F	ATCGTGATATCGCCGG	163
			R Bpvir_fhaB2_2100R	TTCTGGTTGTG CACGTTTGGCGGCAGT GCAAGGTGTAG	164

TABLE 4-continued

Primers for detecting the presence of nucleic acids from <i>Burkholderia</i> .					
Assay name	Assay Type	Target species/ gene	primer name	sequence (5' -> 3')	SEQ ID NO:
fhaB3	PA	fhaB3	F Bpvir_fhaB3_3966F	TCTGCTGATCGGCCTT	165
			R Bpvir_fhaB3_4324R	CGCCAGATAYAC GCGGATGAACAATTTT CTGTCTGAGCGACTATT AC	166
LPSA	PA	LPSA	F Bpvir_LPSA_1087F	GCAGGGCGCCTTGATA	167
			R Bpvir_LPSA_1407R	TCCGCTATGAG CGGCGCAAGGTTCTCC TGCCACATC	168
LPSb1	PA	LPSb1	F Bpvir_LP_Sb1_65F	GTGTGATCGACKGCGT	169
			R Bpvir_LPSb1_256R	CCTCCCTGAG CAAGCCGCTGATACCC GTGTCGCTG	170
LPSb2	PA	LPSb2	F Bpvir_LPSb2_88F	GCGCTTCTCGGTGGGT	171
			R Bpvir_LPSb2_400R	ACGAAAAACAGC CGAGTCGGCCAAGATC ATTCAGGACCAG	172
wcbj	PA	wcbj	F Bpvir_wcbj_252F	CACCTTGACACTGATC	173
			R Bpvir_wcbj_508R	CGCGCGGTAG CTTCCTTCGCACAACC GAGCAAATACTGAGT AAATC	174
ylf	PA	ylf	F Bpvir_ylf_865F	GATCTTGCGACCGATG	175
			R Bpvir_ylf_1153R	CTCAGCGTGTG TGGCGCGGGCCAAGG ATATCAGTTC	176
thai_small_15666	PA	<i>thailandensis</i> (small clade)	F thai_small_15666_3F_UT1	GCCTTACGCCTTCGGG	177
			R thai_small_15666_342R_UT2	ATCG GAATGCGCTCACCAGA TGCT	178
thai_small_28301	PA	<i>thailandensis</i> (small clade)	F thai_small_28301_11F_UT1	AGCAAGCCATCCGCGT	179
			R thai_small_28301_297R_UT2	CATC CAGGATGCCACCGTTG GTGA	180
thai_large_48054	PA	<i>thailandensis</i> (large clade)	F thai_large_48054_286F	GCCACAGGCATGGTG	181
			R thai_large_48054_534R	AGCAA CGGCATTCCCTCAATC ACGAA	182
thai_all_110625	PA	<i>thailandensis</i>	F thai_all_110625_212F	CTGCGTCCCAAAACCGA	183
			R thai_all_110625_512R	CGA CCGTCGATGCCACGAA TGAA	184
humpty_7099	PA	<i>humptydooensis</i>	F humpty_7099_510F	CCCCAAAAATCCCGCT	185
			R humpty_7099_762R	CTGG CGGCACAAAGCCGGT GAAAG	186
humpty_45647	PA	<i>humptydooensis</i>	F HUMPTY_45647_173F_UT1	TGCCGTTTCAGTTGGGC	187
			R HUMPTY_45647_390R_UT2	CTTT TGCCGCTTCCAACGTC TTCA	188
humpty_7093	PA	<i>humptydooensis</i>	F HUMPTY_7093_48F_UT1	GGGCGGGCCAATCTTT	189
			R HUMPTY_7093_381R_UT2	TCTG TCCGCGATGTGACCAA ACGA	190
humpty_38764	PA	<i>humptydooensis</i>	F humpty_38764_44F	TCGGAGATTCGACGG	191
			R humpty_38764_390R	ACCA CCGCATATCGCCCTGA CACA	192

TABLE 4-continued

Primers for detecting the presence of nucleic acids from <i>Burkholderia</i> .					
Assay name	Assay Type	Target species/ gene	primer name	sequence (5' -> 3')	SEQ ID NO:
okla_24632	PA	<i>oklahomensis</i>	F OK_24632_724F_UT1	GGCACCGACGTGCAA AAAGC	193
			R OK_24632_996R_UT2	GGCCGATCTCGGCACT ACGA	194
okla_5812	PA	<i>oklahomensis</i>	F okla_5812_382F	GCGGGGTACGGGCTA ACCAA	195
			R okla_5812_700R	TCCGTACGCTCGCCAC AACA	196
okla_3784	PA	<i>oklahomensis</i>	F okla_3784_454F	GCAAAGGCGCCAGGA AACAA	197
			R okla_3784_742R	ACCGCCCGATTGACC AAGT	198
okla_like_18345	PA	<i>oklahomensis</i> -like	F OK_like_18345_56F_UT1	TCCAGGCGGTTCTCCG ATTG	199
			R OK_like_18345_372R_UT2	GTTGCCGATGTCGAGG CACA	200
okla_like_18342	PA	<i>oklahomensis</i> -like	F OK_like_18342_361F_UT1	TCTTCGGCGAGCGTCT ACGG	201
			R OK_like_18342_685R_UT2	CGCGTCGGACGAGTGT CGTA	202
MSMB175_14005	PA	MSMB175 group	F MSMB175_14005_422F	GGCTCACACGGCTGGG TCAT	203
			R MSMB175_14005_746R	ACGGCGTTTTGGACCA CGAG	204
MSMB175_1868	PA	MSMB175 group	F MSMB175_1868_197F_UT1	CCGCCTACTGGTGGCA GGTG	205
			R MSMB175_1868_480R_UT2	GCCAGTCCCGGGAAG GAGTG	206
MSMB175_8900	PA	MSMB175 group	F MSMB175_8900_29F_UT1	GCTCATCTGCCAGGC CAGT	207
			R MSMB175_8900_345R_UT2	GATACCCACCGCCGGA ACCT	208
MSMB175_9798	PA	MSMB175 group	F MSMB175_9798_517F_UT1	AGCGGCGGATTATGG GCACT	209
			R MSMB175_9798_782R_UT2	ACGCTGGGGCTGTTTT GCAG	210
MSMB264_34074	PA	MSMB264 group	F MSMB264_34074_554F	CGCCCTTCGAGCTTGC TTCC	211
			R MSMB264_34074_778R	CCGCAACAGGTGGCTT CTGAC	212
MSMB264_4163	PA	MSMB264 group	F MSMB264_4163_252F	CGTTGCCCCGCCCAC GTAG	213
			R MSMB264_4163_596R	CCGTGTGGCGGTCCT CCAT	214
vietnam_61292	PA	<i>vietnamiensis</i>	F vietnam_61292_183F	TGGGCTCATCCTCGCA AAGC	215
			R vietnam_61292_527R	ACGCGCTCGGTGGAA AACAG	216
vietnam_98057	PA	<i>vietnamiensis</i>	F vietnam_98057_111F	TCACACCATGGGCTCC GAGA	217
			R vietnam_98057_460R	CGGGCGGGTAGACGA GTTCC	218
vietnam_226017	PA	<i>vietnamiensis</i>	F vietnam_226017_33F_UT1	ACCACGAGTGTGTGCG GCATT	219
			R vietnam_226017_285R_UT2	GCGCTCGATGGTTCCC GAAG	220
ubon_small_102920	PA	<i>ubonensis</i> (small clade)	F ubon_small_102920_511F	CTTGCCCTCCAGGCGC ACAT	221
			R ubon_small_102920_802R	TGCCAAGCGGAAGCTC CTTG	222

TABLE 4-continued

Primers for detecting the presence of nucleic acids from <i>Burkholderia</i> .					
Assay name	AssayType	Target species/ gene	primer name	sequence (5' -> 3')	SEQ ID NO:
ubon_small_111449	PA	ubonensis (small clade)	F ubon_small_111449_167F	UTGCCGTGTCCGCATGAT	223
			R ubon_small_111449_431R_U	CGCTCCAGTGC GTTGT	224
			T2	CGAG	
ubon_large_1438777	PA	ubonensis (large clade)	F ubon_large_41F_1438777	BHCACTGTTCCGCATCGGT	225
			R ubon_large_240R_1438777	BCTYGCCGTGTCCGTCA	226
			H_RP	CGACAAG	
ubon_all_1328624	PA	ubonensis	F ubon_all_1328624_220F	GGCGCCTTCTGGTGGT	227
			R ubon_all_1328624_563R	TGGCTTTGCGACCAGT	228
				CGTG	
cepacia_1208120	PA	cepacia-complex	F cepacia_1208120_1F	ATGGCAARGATTCTKG	229
			R cepacia_1208120_311R	TTCACGATCCAGCCCT	230
				T	

TABLE 5

Primers for detecting the presence of nucleic acids from <i>Yersinia</i> .					
Assay name	AssayType	Target species/ gene	primer name	sequence (5' -> 3')	SEQ ID NO:
Ypestis_	PA	<i>Y. pestis</i>	F Yp&NN1_F	AACAAGCTAAAACGAACAA	231
LPQY01000176.1_7			R Yp&NN1_R	ATAGCCTCAACTGCTTTTGT	232
AGJT01000065.1_0_338	PA	<i>Y. pestis</i>	F Yp&NN11_F	CAGTACCGACAAAACCTTC	233
			R Yp&NN11_R	TTTACTACTCTGAAAACGAG	234
FAUR01000053.1_96407_96884	PA	<i>Y. pestis</i>	F Yp&NN12_F	GCACTACAAATTTAAATCCC	235
			R Yp&NN12_R	GTCGATTATCAACCTCTATG	236
Wagner_Yp_pla_Forward	PA	<i>Y. pestis</i>	F Yp&NN2_F	GAAAGGAGTGCGGGTAATAGG	237
			R Yp&NN2_R	GGCCTGCAAGTCCAATATATGG	238
YpPGM_8-158	PA	Virulence locus PGM	F YpPGM_8F	TTAATATCCCGGCACTCATA	239
			R YpPGM_158R	TCCTTAACTGAATAAGTGCTCA	240
YpPGM_31-205	PA	Virulence locus PGM	F YpPGM_31Fv2	TTTAATGAACGGTGCCTAG	241
			R YpPGM_205Rv2	GTCTGCGTTTCTCCAGTAT	242
Yp-p1202_42780-43194	PA	Drug Resistance plasmid p1P1202	F Yp-p1202_42780F-UT1	TCTGGCCTGCTAAATAAAAACG	243
			R Yp-p1202_43194R-UT2	AACC CAGGCCTCAGCATTTTATTATG	244
				GTGAT	
Yp-p1202_126386-126750	PA	Drug Resistance plasmid p1P1202	F Yp-p1202_126386F-UT1	GGGGCGGATACCTTCACCTATG	245
			R Yp-p1202_126750R-UT2	CTGGGGTTCTAGTCTGGACGAGA	246
				T	
Yp-p1202_156402-156711	PA	Drug Resistance plasmid p1P1202	F Yp-p1202_156402F2-UT1	ACCATCCGGCGCTAAATCGTC	247
			R Yp-p1202_156711R-UT2	GAAATGCGCCTGGTAAGCAGA	248
				GT	
YpCO92_NC_003143_113190	SV	near neighbor species	F Yp&NN4_F	ACTCGGGATACTCCATACTG	249
			R Yp&NN4_R	CGAAAGCAGTGGTCAATC	250
YpCO92_NC_003143_161621	SV	near neighbor species	F Yp&NN5_F	CATGCGCTTTACGTTATATG	251
			R Yp&NN5_R	GCGTTCTGCACTCTGTCT	252
YpCO92_NC_003143_152213	SV	near neighbor species	F Yp&NN6_F	AGCGACTTCCGTGATAAAG	253
			R Yp&NN6_R	ACTCAGGATACCGTGTGGT	254

TABLE 5-continued

Primers for detecting the presence of nucleic acids from <i>Yersinia</i> .					
Assay name	Assay Type	Target species/ gene	primer name	sequence (5'→3')	SEQ ID NO:
YpCO92_NC_003143_129539	SV	near neighbor species	F Yp&NN7_F R Yp&NN7_R	TTCACGATAATCCCCTAATG TTCTGTGCTCTGGCTGATA	255 256
YpCO92_NC_003143_91203	SV	near neighbor species	F Yp&NN8_F R Yp&NN8_R	ATTATCTGTGCCCTTCTTT GGAGTGGATGCCACTAAAC	257 258
YpCO92_NC_003143_121812	SV	near neighbor species	F Yp&NN9_F R Yp&NN9_R	CCTCACACAACAATTCACCTG TTTTTCCGACAAATTTAAGG	259 260
Yp_AL590842.1_RX_SNP	SV	near neighbor species	F Yp&NN10_F R Yp&NN10_R	AGCATGAAGGTTGCTAAAAG GGTGACTTCAAACCGTTAG	261 262
Yentero_FR729477.2_1623	PA	<i>Y. enterocoliticus</i>	F Yp&NN3_F R Yp&NN3_R	GATGCTTCTGCTATCAGSTT GTGTGRCTTTGAASTCTTGT	263 264

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TABLE 6

Primers for detecting the presence of nucleic acids from <i>Francisella</i> .					
Assay name	Assay Type	Target species/ gene	primer name	sequence (5'→3')	SEQ ID NO:
Ftularensis_CP000915.1_1782	PA	<i>F. tularensis</i>	F Ft&NN2_F R Ft&NN2_R	GAAGTGGCTC265 ATGTTAGAGG AGCGAGCCTA266 TATGTAACCA	25 30
Ftularensis_CP000915.1_731	PA	<i>F. tularensis</i>	F Ft&NN3_F R Ft&NN3_R	TTTAATGTCC267 GTCAACCTCT ACGAGTTTGT268 GAGTCGCTAT	35
Ft_dup_CP000915.1_197	PA	<i>F. tularensis</i>	F Ft&NN8_F R Ft&NN8_R	TGTTACGTAC269 AGGCTGTCAA ATCATATCCC270 GTAGCACAAAG	40
FtA1	SNP	FtA1 Clade	F 9F_FtA1_UT1 R 246R_FtA1_UT2	CATAACCCAT271 CGCAATATCT AAATTATCTG272 TAGCGGCAAA	45
FtA2	SNP	FtA2 Clade	F 34F_FtA2_UT1 R 169R_FtA2_UT2	GTGTCCAACG273 AAACCATAAT TTTGGTTGAT274 TCTGTCACTG	50
FtB	SNP	FtB Clade	F 28F_FtB_UT1 R 173R_FtB_UT2	AAGCTTAAC275 GGTGATTGGA CGCCTAACAT276 CTTATCTGCT	55
FtA	SNP	FtA Clade	F 14F_FtA_UT1 R 207R_FtA_UT2	GGGTGATGCA277 GTAGAGAAAA TACCAGATGA278 ACGAATAGCC	60
FtLVS_AM233362_1646546	SV	near neighbor species	F Ft&NN9_F R Ft&NN9_R	ATCAAGCTCA279 TCTTCAAAGC AACCATGTTC280 AGATCCAAAA	65

TABLE 6-continued

Primers for detecting the presence of nucleic acids from <i>Francisella</i> .					
Assay name	Assay Type	Target species/ gene	primer name	sequence (5'→3')	SEQ ID NO:
FtLVS_AM233362_1643765	SV	near neighbor species	F Ft&NN10_F R Ft&NN10_R	TACCTCTGCC281 AAAAATTCAT GGCATACTCA282 AGGTAGTGGT	25 30
FtLVS_AM233362_1562618	SV	near neighbor species	F Ft&NN11_F R Ft&NN11_R	TCTTTGGTAG283 CTTGCTGACT CAGACGACAC284 TTGCTTATT	35
Ftnovicida_CP009607.1	PA	<i>F. tularensis</i> spp. <i>Novicida</i>	F Ft&NN1_F R Ft&NN1_R	GGTAGGATAA285 CTACCAAG GTCATGAGTT286 TTACCAATAC TC	40
Fphilom_CP009444.1_569	PA	<i>F. philomiragia</i>	F Ft&NN6_F R Ft&NN6_R	CTTATGCAGC287 AAGAGGAAC ATACACCGGG288 ATAGGTTTCT	45
Fphilom_CP009444.1_285	PA	<i>F. philomiragia</i>	F Ft&NN7_F R Ft&NN7_Rv2	CTGATGGAAG289 AGAGTTTCGAG GTAGATATAA290 TCAGCGCCAC	50
Fnoatunensis_CP003402.1_1749	PA	<i>F. noatunensis</i>	F Ft&NN4_F R Ft&NN4_R	CGGTAAGAAT291 ACGACCAGAG AGAGGATTTTC292 TTCCTCCTTG	55
Fnoatunensis_CP003402.1_424	PA	<i>F. noatunensis</i>	F Ft&NN5_F R Ft&NN5_R	AATTCTACAA293 GCACCTGGAA TCCTATTAAA294 AGCGCCATAG	60

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TABLE 7

Primers for Sequence Control.						
Assay name	Target species/gene	For-ward primer name	Forward sequence ID (5' -> 3')	Re-verse primer name	Reverse sequence ID (5' -> 3')	SEQ NO
IPSC-1	se-quencing control	UT1-IPSC-f1	GGGCGGAC 295	UT2-IPSC-r1	GCCGGGAT 298	5
			GAAAACCC		GCCTTACC	10
			TTGAGCAC		TAGACGCA	
			AG		ATGA	
IPSC-1	se-quencing control	UT1-IPSC-flv2	GCTCGGC 296	UT2-IPSC-r1	GCCGGGAT 299	15
			ACCCTTGA		GCCTTACC	
					TAGACGCA	
					ATGA	

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TABLE 7-continued

Primers for Sequence Control.						
Assay name	Target species/gene	For-ward primer name	Forward sequence ID (5' -> 3')	Re-verse primer name	Reverse sequence ID (5' -> 3')	SEQ NO
IPSC-2	se-quencing control	UT1-IPSC-f2	GCGGCAGC 297	UT2-IPSC-r2	CGAGTTCC 300	
			CGTTGAGG		GTCCGGTT	
			CAAAAGTG		AAGCGTGA	
			ATAC		CAGTC	
Forward sequence w/UT: ACCCAACTGAATGGAGC (SEQ ID NO: 301) at 5' of the forward sequence, e.g., UT1-IPSC-f1: ACCCAACTGAATGGAGCGGGGGGACGAAACCCCTTGAGCACAG (SEQ ID NO: 302)						
Reverse sequence w/UT: ACGCACTTGACTTGCTCTC (SEQ ID NO: 303) at 5' of the reverse sequence, e.g., UT2-IPSC-r1: ACGCACTTGACTTGCTCTCGCGGGATGCCTTACCTAGACGCAATGA (SEQ ID NO: 304)						

TABLE 8

Preparation of Primer Mixture for detecting the presence of nucleic acids from <i>Burkholderia</i> .								
Assay name	Primer	Target	Start (uM)	Desired final uM in rxn.	uM in mix stock (final uM x 5.56)	Volume needed in Mix stock (uL)	starting primer conc. to add in mix stock	Recalculated final conc. of primer in a single rxn.
TTS1_BPSS1407	BpAmpSeq_1_F	TTS1	100	0.10	0.56	0.03	5.0	0.10
TTS1_BPSS1407	BpAmpSeq_1_R	TTS1	100	0.10	0.56	0.03	5.0	0.10
LXCC01000141.1_39296_39817	BpAmpSeq_3_F	<i>pseudomallei</i>	100	0.20	1.11	0.05	9.9	0.20
LXCC01000141.1_39296_39817	BpAmpSeq_3_R	<i>pseudomallei</i>	100	0.20	1.11	0.05	9.9	0.20
LXBY01000087.1_75760_76751	BpAmpSeq_4_F	<i>pseudomallei</i>	100	0.20	1.11	0.05	9.9	0.20
LXBY01000087.1_75760_76751	BpAmpSeq_4_R	<i>pseudomallei</i>	100	0.20	1.11	0.05	9.9	0.20
LXCD01000002.1_99652_100245	BpAmpSeq_5_F	<i>pseudomallei</i>	100	0.40	2.22	0.10	19.8	0.40
LXCD01000002.1_99652_100245	BpAmpSeq_5_R	<i>pseudomallei</i>	100	0.40	2.22	0.10	19.8	0.40
LXCE01000123.1_34220_34747	BpAmpSeq_6_F	<i>pseudomallei</i>	100	0.40	2.22	0.10	19.8	0.40
LXCE01000123.1_34220_34747	BpAmpSeq_6_R	<i>pseudomallei</i>	100	0.40	2.22	0.10	19.8	0.40
LWWC01000187.1_18	BpAmpSeq_8_F	<i>pseudomallei mallei</i>	100	0.30	1.67	0.08	14.9	0.30
LWWC01000187.1_18	BpAmpSeq_8_R	<i>pseudomallei mallei</i>	100	0.30	1.67	0.08	14.9	0.30
LWWB01000125.1_17183_17602	BpAmpSeq_10_F	<i>pseudomallei mallei</i>	100	0.20	1.11	0.05	9.9	0.20
LWWB01000125.1_17183_17602	BpAmpSeq_10_R	<i>pseudomallei mallei</i>	100	0.20	1.11	0.05	9.9	0.20
LXAY1000367.1_0_640	BpAmpSeq_11_F	<i>pseudomallei mallei</i>	100	0.10	0.56	0.03	5.0	0.10
LXAY1000367.1_0_640	BpAmpSeq_11_R	<i>pseudomallei mallei</i>	100	0.10	0.56	0.03	5.0	0.10
LWVY01000190.1_17226_17689	BpAmpSeq_12_F	<i>pseudomallei mallei</i>	100	0.40	2.22	0.10	19.8	0.40
LWVY01000190.1_17226_17689	BpAmpSeq_12_R	<i>pseudomallei mallei</i>	100	0.40	2.22	0.10	19.8	0.40
LXAD01000059.1_24760_25075	BpAmpSeq_13_F	<i>pseudomallei mallei</i>	100	0.30	1.67	0.08	14.9	0.30
LXAD01000059.1_24760_25075	BpAmpSeq_13_R	<i>pseudomallei mallei</i>	100	0.30	1.67	0.08	14.9	0.30
NC_006350_2289827	BpAmpSeq_16_F	<i>pseudomallei</i> complex SNP	100	0.40	2.22	0.10	19.8	0.40
NC_006350_2289827	BpAmpSeq_16_R	<i>pseudomallei</i> complex SNP	100	0.40	2.22	0.10	19.8	0.40
NC_006350_133027	BpAmpSeq_18_F	<i>cepacia</i> complex SNPs	100	0.20	1.11	0.05	9.9	0.20
NC_006350_133027	BpAmpSeq_18_R	<i>cepacia</i> complex SNPs	100	0.20	1.11	0.05	9.9	0.20
NC_006350_2248145-2248193	BpAmpSeq_19_F	Bpc MSS	100	0.40	2.22	0.10	19.8	0.40
NC_006350_2248145-2248193	BpAmpSeq_19_R	Bpc MSS	100	0.40	2.22	0.10	19.8	0.40
NC_006350_988041-988089	BpAmpSeq_20_F	Bpc MSS	100	0.20	1.11	0.05	9.9	0.20
NC_006350_988041-988089	BpAmpSeq_20_R	Bpc MSS	100	0.20	1.11	0.05	9.9	0.20
Bm_11589	BpAmpSeq_21_F	<i>mallei</i>	100	0.40	2.22	0.10	19.8	0.40
Bm_11589	BpAmpSeq_21_R	<i>mallei</i>	100	0.40	2.22	0.10	19.8	0.40
Bm_11767	BpAmpSeq_22_F	<i>mallei</i>	100	0.40	2.22	0.10	19.8	0.40
Bm_11767	BpAmpSeq_22_R	<i>mallei</i>	100	0.40	2.22	0.10	19.8	0.40
PBP3-170-505	BpAmpSeq_24_F	pbp3	100	0.20	1.11	0.05	9.9	0.20
PBP3-170-505	BpAmpSeq_24_R	pbp3	100	0.20	1.11	0.05	9.9	0.20
K9penA378-529	BpAmpSeq_26_F	penA	100	0.10	0.56	0.03	5.0	0.10
K9penA378-529	BpAmpSeq_26_R	penA	100	0.10	0.56	0.03	5.0	0.10
K9penA575-761	BpAmpSeq_27_F	penA	100	0.10	0.56	0.03	5.0	0.10
K9penA575-761	BpAmpSeq_27_R	penA	100	0.10	0.56	0.03	5.0	0.10
K9penA949-1172	BpAmpSeq_28_F	penA	100	0.10	0.56	0.03	5.0	0.10
K9penA949-1172	BpAmpSeq_28_R	penA	100	0.10	0.56	0.03	5.0	0.10
IPSC	IPSC-flv2	IPSC	20	0.05	0.28	0.06	12.4	0.05
IPSC	IPSC-r1	IPSC	20	0.05	0.28	0.06	12.4	0.05

Volume of primer mix in a single rxn: 4.5 uL.

Desired volume of primer mix stock: 891 uL.

TABLE 10

Preparation of Primer Mixture for detecting the presence of nucleic acids from SOP for <i>Bacillus</i> , <i>Yersinia</i> , and <i>Francisella</i> UT-AmpSeq PCR and Bead Cleanup									
Primer name	Assay name	Start (uM)	Volume of primer mix in a single rxn (uL)	Desired final uM in rxn.	uM in mix stock (final uM × 5.56)	Volume needed in Mix stock (uL)	Desired volume of Primer mix stock (uL)	How much starting primer conc. to add in mix stock	Re- calculated final conc. Of primer in a single rxn.
Ba-specific-1F_UT1	plcR	100	4.5	0.1	0.56	0.03	891	5	0
Ba-specific-1R_UT2	plcR	100		0.1	0.56	0.03		5	0
Ba-specific-3F_UT1	CP008853.1_5309	500		0.2	1.11	0.01		2	0
Ba-specific-3R_UT2	CP008853.1_5309	500		0.2	1.11	0.01		2	0
Ba-specific-5F_UT1	CP008853.1_5316	500		0.2	1.11	0.01		2	0
Ba-specific-5R_UT2	CP008853.1_5316	500		0.2	1.11	0.01		2	0
Ba-specific-6F_UT1	CP012725.1_3629	500		1.6	8.9	0.08		15.9	0
Ba-specific-6R_UT2	CP012725.1_3629	500		1.6	8.9	0.08		15.9	0
Ba-specific-8F_UT1	CP012725.1_5103	100		0.1	0.56	0.03		5	0
Ba-specific-8R_UT2	CP012725.1_5103	100		0.1	0.56	0.03		5	0
Ba-specific-9F_UT1	CP012725.1_5107	100		0.1	0.56	0.03		5	0
Ba-specific-9R_UT2	CP012725.1_5107	100		0.1	0.56	0.03		5	0
Ba-specific-11F_UT1	JSZQ01000034.1_220	100		0.1	0.56	0.03		5	0
Ba-specific-11R_UT2	JSZQ01000034.1_220	100		0.1	0.56	0.03		5	0
Ba-specific-12F_UT1	JSZS01000036.15	100		0.1	0.56	0.03		5	0
Ba-specific-12R_UT2	JSZS01000036.15	100		0.1	0.56	0.03		5	0
Ba-specific-14F_UT1	LGCC01000010.1_232	500		0.4	2.22	0.02		4	0
Ba-specific-14R_UT2	LGCC01000010.1_232	500		0.4	2.22	0.02		4	0
Ba-specific-16F_UT1	LGCC01000048.1_280	100		0.1	0.56	0.03		5	0
Ba-specific-16R_UT2	LGCC01000048.1_280	100		0.1	0.56	0.03		5	0
Ba-specific-20F_UT1	NN_LOMU01000090.1_49	500		1	5.56	0.05		9.9	0
Ba-specific-20R_UT2	NN_LOMU01000090.1_49	500		1	5.56	0.05		9.9	0
Ba-specific-22F_UT1	NN_LOQC01000013.1_3	500		0.2	1.11	0.01		2	0
Ba-specific-22Rv2_UT2	NN_LOQC01000013.1_3	500		0.2	1.11	0.01		2	0
pX01_113F_UT1	pX01	500		0.8	4.45	0.04		7.9	0
pX01-315Rv2_UT2	pX01	500		0.8	4.45	0.04		7.9	0
pX02_101F_UT1	pX02	100		0.1	0.56	0.03		5	0
pX02_269R_UT2	pX02	100		0.1	0.56	0.03		5	0
gyrA_28Fv2_UT1	gyrA	50		0.05	0.28	0.03		5	0
gyrA_182Rv2_UT2	gyrA	50		0.05	0.28	0.03		5	0
parC_1F_UT1	parC	100		0.05	0.28	0.01		2.5	0
parC_197R_UT2	parC	100		0.05	0.28	0.01		2.5	0
gyrB_8F_UT1	gyrB	100		0.1	0.56	0.03		5	0
gyrB_194R_UT2	gyrB	100		0.1	0.56	0.03		5	0
801F_pagAv3_UT1	pagAv3	100		0.1	0.56	0.03		5	0
1042R_pagAv3_UT2	pagAv3	100		0.1	0.56	0.03		5	0
rpoB_29F_UT1	rpoB	50		0.05	0.28	0.03		5	0
rpoB_196R_UT2	rpoB	50		0.05	0.28	0.03		5	0
AA_2502_UT1	AA_2502	500		0.8	4.45	0.04		7.9	0
AA_2502_UT2	AA_2502	500		0.8	4.45	0.04		7.9	0
AA_2503v2_UT1	AA_2503	500		0.8	4.45	0.04		7.9	0
AA_2503_UT2	AA_2503	500		0.8	4.45	0.04		7.9	0
Ba&NN32_F	Ba_AmesAnc_4669915	100		0.1	0.56	0.03		5	0
Ba&NN32_R	Ba_AmesAnc_4669915	100		0.1	0.56	0.03		5	0
Ba&NN33_F	Ba_AmesAnc_4001578	100		0.05	0.28	0.01		2.5	0
Ba&NN33_R	Ba_AmesAnc_4001578	100		0.05	0.28	0.01		2.5	0
Ba&NN37_F	Ba_AmesAnc_1069024	500		0.2	1.11	0.01		2	0
Ba&NN37_R	Ba_AmesAnc_1069024	500		0.2	1.11	0.01		2	0
Ba&NN38_F	Ba_AmesAnc_3668548	500		0.2	1.11	0.01		2	0
Ba&NN38_R	Ba_AmesAnc_3668548	500		0.2	1.11	0.01		2	0
Ba&NN40_F	Ba_AmesAnc_371913	500		0.2	1.11	0.01		2	0
Ba&NN40_R	Ba_AmesAnc_371913	500		0.2	1.11	0.01		2	0
Ba&NN41_F	Ba_AmesAnc_999035	100		0.05	0.28	0.01		2.5	0
Ba&NN41_R	Ba_AmesAnc_999035	100		0.05	0.28	0.01		2.5	0
ChimpKiller_9F	ChimpKiller_9-159	100		0.1	0.56	0.03		5	0
ChimpKiller_159R	ChimpKiller_9-159	100		0.1	0.56	0.03		5	0
ChimpKiller_91F	ChimpKiller_91-320	500		0.8	4.45	0.04		7.9	0
ChimpKiller_320R	ChimpKiller_91-320	500		0.8	4.45	0.04		7.9	0
ChimpKiller_481F	ChimpKiller_481-698	500		0.8	4.45	0.04		7.9	0
ChimpKiller_698R	ChimpKiller_481-698	500		0.8	4.45	0.04		7.9	0
Yp&NN1_F	Ypestis_LPQY01000176.1_7	500		0.2	1.11	0.01		2	0
Yp&NN1_R	Ypestis_LPQY01000176.1_7	500		0.2	1.11	0.01		2	0
Yp&NN2_F	Wagner_Yp_pla_Forward	100		0.1	0.56	0.03		5	0
Yp&NN2_R	Wagner_Yp_pla_Forward	100		0.1	0.56	0.03		5	0
Yp&NN3_F	Yentero_FR729477.2_1623	500		0.2	1.11	0.01		2	0
Yp&NN3_R	Yentero_FR729477.2_1623	500		0.2	1.11	0.01		2	0
Yp&NN4_F	YpCO92_NC_003143_113190	500		0.4	2.22	0.02		4	0
Yp&NN4_R	YpCO92_NC_003143_113190	500		0.4	2.22	0.02		4	0
Yp&NN5_F	YpCO92_NC_003143_161621	100		0.05	0.28	0.01		2.5	0
Yp&NN5_R	YpCO92_NC_003143_161621	100		0.05	0.28	0.01		2.5	0

TABLE 10-continued

Preparation of Primer Mixture for detecting the presence of nucleic acids from SOP for <i>Bacillus</i> , <i>Yersinia</i> , and <i>Francisella</i> UT-AmpSeq PCR and Bead Cleanup									
Primer name	Assay name	Start (uM)	Volume of primer mix in a single rxn (uL)	Desired final uM in rxn.	uM in mix stock (final uM × 5.56)	Volume needed in Mix stock (uL)	Desired volume of Primer mix stock (uL)	How much starting primer conc. to add in mix stock	Re- calculated final conc. Of primer in a single rxn.
Yp&NN6_F	YpCO92_NC_003143_152213	100		0.05	0.28	0.01		2.5	0
Yp&NN6_R	YpCO92_NC_003143_152213	100		0.05	0.28	0.01		2.5	0
Yp&NN7_F	YpCO92_NC_003143_129539	100		0.1	0.56	0.03		5	0
Yp&NN7_R	YpCO92_NC_003143_129539	100		0.1	0.56	0.03		5	0
Yp&NN8_F	YpCO92_NC_003143_91203	100		0.05	0.28	0.01		2.5	0
Yp&NN8_R	YpCO92_NC_003143_91203	100		0.05	0.28	0.01		2.5	0
Yp&NN9_F	YpCO92_NC_003143_121812	100		0.1	0.56	0.03		5	0
Yp&NN9_R	YpCO92_NC_003143_121812	100		0.1	0.56	0.03		5	0
Yp&NN10_F	Yp_AL590842.1_RX_SNP	50		0.05	0.28	0.03		5	0
Yp&NN10_R	Yp_AL590842.1_RX_SNP	50		0.05	0.28	0.03		5	0
Yp&NN11_F	AGJT01000065.1_0_338	100		0.1	0.56	0.03		5	0
Yp&NN11_R	AGJT01000065.1_0_338	100		0.1	0.56	0.03		5	0
Yp&NN12_F	FAUR01000053.1_96407_96884	100		0.1	0.56	0.03		5	0
Yp&NN12_R	FAUR01000053.1_96407_96884	100		0.1	0.56	0.03		5	0
YpPGM_8F	YpPGM_8-158	500		0.8	4.45	0.04		7.9	0
YpPGM_158R	YpPGM_8-158	500		0.8	4.45	0.04		7.9	0
YpPGM_31Fv2	YpPGM_31-205	50		0.05	0.28	0.03		5	0
YpPGM_205Rv2	YpPGM_31-205	50		0.05	0.28	0.03		5	0
Yp-p1202_42780F-UT1	Yp-p1202_42780-43194	500		0.2	1.11	0.01		2	0
Yp-p1202_43194R-UT2	Yp-p1202_42780-43194	500		0.2	1.11	0.01		2	0
Yp-p1202_126386F-UT1	Yp-p1202_126386-126750	500		0.2	1.11	0.01		2	0
Yp-p1202_126750R-UT2	Yp-p1202_126386-126750	500		0.2	1.11	0.01		2	0
Yp-p1202_156402F2-UT1	Yp-p1202_156402-156711	500		0.2	1.11	0.01		2	0
Yp-p1202_156711R-UT2	Yp-p1202_156402-156711	500		0.2	1.11	0.01		2	0
Ft&NN1_F	<i>Fmovicida</i> _CP009607.1	500		0.2	1.11	0.01		2	0
Ft&NN1_R	<i>Fmovicida</i> _CP009607.1	500		0.2	1.11	0.01		2	0
Ft&NN2_F	<i>Ftularensis</i> _CP000915.1_1782	100		0.1	0.56	0.03		5	0
Ft&NN2_R	<i>Ftularensis</i> _CP000915.1_1782	100		0.1	0.56	0.03		5	0
Ft&NN3_F	<i>Ftularensis</i> _CP000915.1_731	500		1.6	8.9	0.08		15.9	0
Ft&NN3_R	<i>Ftularensis</i> _CP000915.1_731	500		1.6	8.9	0.08		15.9	0
Ft&NN4_F	<i>Fnoatumensis</i> _CP003402.1_1749	500		0.2	1.11	0.01		2	0
Ft&NN4_R	<i>Fnoatumensis</i> _CP003402.1_1749	500		0.2	1.11	0.01		2	0
Ft&NN5_F	<i>Fnoatumensis</i> _CP003402.1_424	500		0.2	1.11	0.01		2	0
Ft&NN5_R	<i>Fnoatumensis</i> _CP003402.1_424	500		0.2	1.11	0.01		2	0
Ft&NN6_F	<i>Fphilom</i> _CP009444.1_569	500		0.2	1.11	0.01		2	0
Ft&NN6_R	<i>Fphilom</i> _CP009444.1_569	500		0.2	1.11	0.01		2	0
Ft&NN7_F	<i>Fphilom</i> _CP009444.1_285	500		0.2	1.11	0.01		2	0
Ft&NN7_Rv2	<i>Fphilom</i> _CP009444.1_285	500		0.2	1.11	0.01		2	0
Ft&NN8_F	Ft_dup_CP000915.1_197	100		0.05	0.28	0.01		2.5	0
Ft&NN8_R	Ft_dup_CP000915.1_197	100		0.05	0.28	0.01		2.5	0
Ft&NN9_F	FtLVS_AM233362_1646546	100		0.05	0.28	0.01		2.5	0
Ft&NN9_R	FtLVS_AM233362_1646546	100		0.05	0.28	0.01		2.5	0
Ft&NN10_F	FtLVS_AM233362_1643765	100		0.1	0.56	0.03		5	0
Ft&NN10_R	FtLVS_AM233362_1643765	100		0.1	0.56	0.03		5	0
Ft&NN11_F	FtLVS_AM233362_1562618	100		0.1	0.56	0.03		5	0
Ft&NN11_R	FtLVS_AM233362_1562618	100		0.1	0.56	0.03		5	0
9F_FtA1_UT1	FtA1	100		0.1	0.56	0.03		5	0
246R_FtA1_UT2	FtA1	100		0.1	0.56	0.03		5	0
34F_FtA2_UT1	FtA2	100		0.1	0.56	0.03		5	0
169R_FtA2_UT2	FtA2	100		0.1	0.56	0.03		5	0
28F_FtB_UT1	FtB	100		0.1	0.56	0.03		5	0
173R_FtB_UT2	FtB	100		0.1	0.56	0.03		5	0
14F_FtA_UT1	FtA	100		0.1	0.56	0.03		5	0
207R_FtA_UT2	FtA	100		0.1	0.56	0.03		5	0
IPSC-f2	IPSC	20		0.03	0.14	0.03		6.2	0
IPSC-r2	IPSC	20		0.03	0.14	0.03		6.2	0

580.6 uL total volume of primers

310.4 uL of 1x TE to add to bring up to desired volume of primer mix stock

891.0 Total (uL)

TABLE 11

Burkholderia primers and primers with Universal Tail (UT) . The UT sequence is underlined.					
Assay name	Target species/ gene		SEQ ID NO:		SEQ ID NO:
		Forward primer name	Forward sequence (5' ->3')	Forward sequence w/UT	
TTS1_ BPSS1 407	TTS1	BpAmpSeq_1 _F	TCGTCGTCACCGGGATGGTC	61 <u>ACCCAACTGAATGGAGCTCGTCG</u> TCACCGGGATGGTC	307
		Reverse primer name	Reverse sequence (5' ->3')	Reverse sequence w/UT	
		BpAmpSeq_1 _R	GGCCTTTGCCCGCATACTCG	62 <u>ACGCACTTGACTTGTCTTCGGCC</u> TTTGCCCGCATACTCG	308
		Forward primer name	Forward sequence (5' ->3')	Forward sequence w/UT	
LXCCO 100014 1.1_392 96_398	pseudomali	BpAmpSeq_3 _F	TCGCAWGAAGTGCGTTGCC	63 <u>ACCCAACTGAATGGAGCTCGCA</u> WGAAGTGCGTTGCCG	309
		Reverse primer name	Reverse sequence (5' ->3')	Reverse sequence w/UT	
17		BpAmpSeq_3 _R	GCCGCTTGCGAAGCGATGAT	64 <u>ACGCACTTGACTTGTCTTCGCCG</u> CTTGCGAAGCGATGAT	310
		Forward primer name	Forward sequence (5' ->3')	Forward sequence w/UT	
LXBY0 100008 7.1_757 60_767	pseudomali	BpAmpSeq_4 _F	CGCGCTTGCCCACTACCAG	65 <u>ACCCAACTGAATGGAGCCGCGCT</u> TGCCCACTACCAG	311
		Reverse primer name	Reverse sequence (5' ->3')	Reverse sequence w/UT	
51		BpAmpSeq_4 _R	GCGCAACGGTGCGAGACAA T	66 <u>ACGCACTTGACTTGTCTTCGCGC</u> AACGGTGCGAGACAAT	312
		Forward primer name	Forward sequence (5' ->3')	Forward sequence w/UT	
LXCD0 100000 2.1_996 52_100	pseudomali	BpAmpSeq_5 _F	AATCCATGCATGTCGYGCCC	67 <u>ACCCAACTGAATGGAGCAATCCA</u> TGCATGTCGYGCCC	313
		Reverse primer name	Reverse sequence (5' ->3')	Reverse sequence w/UT	
245		BpAmpSeq_5 _R	GCGATCGCTCAACGGGCTTC	68 <u>ACGCACTTGACTTGTCTTCGCGA</u> TCGCTCAACGGGCTTC	314
		Forward primer name	Forward sequence (5' ->3')	Forward sequence w/UT	
LXCE0 100012 3.1_342 20_347	pseudomali	BpAmpSeq_6 _F	TCGCATTGTCAYACGCTCCC	69 <u>ACCCAACTGAATGGAGCTCGCAT</u> TTGCAYACGCTCCC	315
		Reverse primer name	Reverse sequence (5' ->3')	Reverse sequence w/UT	
47		BpAmpSeq_6 _R	AGTGCGCAAACCTGGCGAG G	70 <u>ACGCACTTGACTTGTCTTCAGTG</u> CGCAAACCTGGCGAGG	316
		Forward primer name	Forward sequence (5' ->3')	Forward sequence w/UT	
LWWC 010001 87.1 18	pseudomali mallei	BpAmpSeq_8 _F	CCTTTGCGGCAAGCGTCGAA	81 <u>ACCCAACTGAATGGAGCCCTTTG</u> CGGCAAGCGTCGAA	317
		Reverse primer name	Reverse sequence (5' ->3')	Reverse sequence w/UT	
		BpAmpSeq_8 _R	GAGCCAACGCACATGGACG G	82 <u>ACGCACTTGACTTGTCTTCGAGC</u> CAACGCACATGGACGG	318

TABLE 11-continued

Burkholderia primers and primers with Universal Tail (UT) . The UT sequence is underlined.					
Assay name	Target species/ gene		SEQ ID NO:		SEQ ID NO:
		Forward primer name	Forward sequence (5'->3')	Forward sequence w/UT	
LWWB	<i>pseudomal- lei</i>	BpAmpSeq_1	CCAGTCGGGCCGGGAAAAA	83 ACCCAACTGAATGGAGCCAGTC	319
010001	<i>mallei</i>	0_F	C	GGGCCGGGAAAAAC	
25.1_17		Reverse	Reverse sequence		
183_17		primer name	(5'->3')	Reverse sequence w/UT	
602		BpAmpSeq_1	GGCGGCAAAAGCGTCGATG	84 ACGCACTTGACTTGCTTCCGGCG	320
		0_R	A	GCAAAAGCGTCGATGA	
		Forward primer name	Forward sequence (5'->3')	Forward sequence w/UT	
LXAY0	<i>pseudomal- lei</i>	BpAmpSeq_1	GCCGGAACCGTCGAGCATT	85 ACCCAACTGAATGGAGCGCCGG	321
100036	<i>mallei</i>	1_F	G	AACCGTCGAGCATTG	
7.1_0_6		Reverse	Reverse sequence		
40		primer name	(5'->3')	Reverse sequence w/UT	
		BpAmpSeq_1	TGGATTCGACTGCCTCCGCT	86 ACGCACTTGACTTGCTTCTTGA	322
		1_R		TTCGACTGCCTCCGCT	
		Forward primer name	Forward sequence (5'->3')	Forward sequence w/UT	
LWVY	<i>pseudomal- lei</i>	BpAmpSeq_1	TCGATATCCGCCGTCTCGCC	87 ACCCAACTGAATGGAGCTCGATA	323
010001	<i>mallei</i>	2_F		TCCGCCGTCTCGCC	
90.1_17		Reverse	Reverse sequence		
226_17		primer name	(5'->3')	Reverse sequence w/UT	
689		BpAmpSeq_1	ATGTGTCGGTGGGCTTCGGT	88 ACGCACTTGACTTGCTTTCATGT	324
		2_R		GTCGGTGGGCTTCGGT	
		Forward primer name	Forward sequence (5'->3')	Forward sequence w/UT	
LXAD0	<i>pseudomal- lei</i>	BpAmpSeq_1	GAAAGGCGATGTGCCGAGC	89 ACCCAACTGAATGGAGCGAAAG	325
100005	<i>mallei</i>	3_F	G	GCGATGTGCCGAGCG	
9.1_247		Reverse	Reverse sequence		
60_250		primer name	(5'->3')	Reverse sequence w/UT	
75		BpAmpSeq_1	TTCGGAGAAGCGCCAAACG	90 ACGCACTTGACTTGCTTCTTCGG	326
		3_R	C	AGAAGCGCCAAACGC	
		Forward primer name	Forward sequence (5'->3')	Forward sequence w/UT	
NC_00	<i>pseudomal- lei</i>	BpAmpSeq_1	GCCAGCGCATCCACCAACAT	109 ACCCAACTGAATGGAGCGCCAGC	327
6350_2	complex SNP	6_F		GCATCCACCAACAT	
289827		Reverse	Reverse sequence		
		primer name	(5'->3')	Reverse sequence w/UT	
		BpAmpSeq_1	AGAGGAAGAAGGGCGAGGC	110 ACGCACTTGACTTGCTTTCAGAG	328
		6_R	G	GAAGAAGGGCGAGGCG	
		Forward primer name	Forward sequence (5'->3')	Forward sequence w/UT	
NC_00	<i>cepacia</i>	BpAmpSeq_1	CGCGCARYTCGTCGTCTCG	111 ACCCAACTGAATGGAGCCGCGCA	329
6350_1	complex	8_F		RYTCGTCTGTCTCG	
33027	SNPs	Reverse	Reverse sequence		
		primer name	(5'->3')	Reverse sequence w/UT	
		BpAmpSeq_1	CGAACCTSGTGCMGGTRCA	112 ACGCACTTGACTTGCTTCCGAA	330
		8_R	G	CCTSGTGCMGGTRCAG	

TABLE 11-continued

Burkholderia primers and primers with Universal Tail (UT). The UT sequence is underlined.				
Assay name	Target species/ gene		SEQ ID NO:	SEQ ID NO:
		<u>Forward primer name</u>	<u>Forward sequence (5'→3')</u>	<u>Forward sequence w/UT</u>
NC_00 6350_2 248145- 224819	Bpc MSS	BpAmpSeq_1 9_F	CACGTTGCCSGGRAARTACG113	ACCCAACTGAATGGAGCCACGTT 331 GCCSGGRAARTACG
		<u>Reverse primer name</u>	<u>Reverse sequence (5'→3')</u>	<u>Reverse sequence w/UT</u>
3		BpAmpSeq_1 9_R	CCGTCGACAAGATCGCGCTS114	ACGCACTTGACTTGCTTCCCGTC 332 GACAAGATCGCGCTS
		<u>Forward primer name</u>	<u>Forward sequence (5'→3')</u>	<u>Forward sequence w/UT</u>
NC_00 6350_9 88041- 988089	Bpc MSS	BpAmpSeq_2 0_F	CAGAACGCGCTRTYCCACG 115	ACCCAACTGAATGGAGCCAGAA 333 CGCGCTRTYCCACG
		<u>Reverse primer name</u>	<u>Reverse sequence (5'→3')</u>	<u>Reverse sequence w/UT</u>
		BpAmpSeq_2 0_R	TGCCGCGTGATCCATTGCAT116	ACGCACTTGACTTGCTTCTGCC 334 GCGTGATCCATTGCAT
		<u>Forward primer name</u>	<u>Forward sequence (5'→3')</u>	<u>Forward sequence w/UT</u>
Bm_11 589	mallei	BpAmpSeq_2 1_F	AGGGGGTGGTTTCTGAGTG117	ACCCAACTGAATGGAGCAGGGG 335 GTGGTTTCTGAGTGGCGTGAC
		<u>Reverse primer name</u>	<u>Reverse sequence (5'→3')</u>	<u>Reverse sequence w/UT</u>
		BpAmpSeq_2 1_R	AGCGGTGTCGACGGTGGA 118	ACGCACTTGACTTGCTTCAGCG 336 GTGTCGACGGGTGGAAGGATG
		<u>Forward primer name</u>	<u>Forward sequence (5'→3')</u>	<u>Forward sequence w/UT</u>
Bm_11 767	mallei	BpAmpSeq_2 2_F	ACGGGCGCTTCACGATCTCG119	ACCCAACTGAATGGAGCACGGG 337 CGCTTCACGATCTCGGTGTTC
		<u>Reverse primer name</u>	<u>Reverse sequence (5'→3')</u>	<u>Reverse sequence w/UT</u>
		BpAmpSeq_2 2_R	GCGCGGCAGTTCGATCAGG 120	ACGCACTTGACTTGCTTCGCGC 338 GGCAGTTCGATCAGGCATTG
		<u>Forward primer name</u>	<u>Forward sequence (5'→3')</u>	<u>Forward sequence w/UT</u>
PBP3- 170- 505	pbp3	BpAmpSeq_2 4_F	ATCCGCGCTCCCGCCAGCA305	ACCCAACTGAATGGAGCATCCGC 339 CGTCCCGCCAGCAATAG
		<u>Reverse primer name</u>	<u>Reverse sequence (5'→3')</u>	<u>Reverse sequence w/UT</u>
		BpAmpSeq_2 4_R	GGGTTGCGCCAGATTTCGTA306	ACGCACTTGACTTGCTTCGGGT 340 TCGCCAGATTTCGTAGGTGGTG AG
		<u>Forward primer name</u>	<u>Forward sequence (5'→3')</u>	<u>Forward sequence w/UT</u>
K9pen A378- 529	penA	BpAmpSeq_2 6_F	CGGTGCGCCACAAATTCGCAC127	ACCCAACTGAATGGAGCCGGTCG 341 CCACAAATTCGCACGCACTC
		<u>Reverse primer name</u>	<u>Reverse sequence (5'→3')</u>	<u>Reverse sequence w/UT</u>
		BpAmpSeq_2 6_R	AGCGAGCGGCGCAACGGAG 128	ACGCACTTGACTTGCTTCAGCG 342 AGCGGCGCAACGGAGAATGATT

TABLE 12-continued

Bacillus primers and primers with Universal Tail (UT). The UT sequence is underlined.					
				SEQ ID NO:	SEQ ID NO:
Assay name	Target species/ gene	Forward primer name	Forward sequence (5' -> 3')	Forward sequence w/UT	
CP(K) 88 53.1_ 5316		Ba- specific- 5FUT1	GAAGATGTACGCTCGATAG G	3 <u>ACCCAACTGAATGGAGCGAAGA</u> TGTACGCTCGATAGG	351
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		Ba- specific- 5RUT2	GAAATTCTTTTGGCCATCAC	4 <u>ACGCACTTGACTTGTCTTCGAAA</u> TTCTTTTGGCCATCAC	352
Assay name	Target species/ gene	Forward primer name	Forward sequence (5' -> 3')	Forward sequence w/UT	
CP0127 25.1_ 3629		Ba- specific- 6F_UT1	CACAATTGAATGAAAATGCT	5 <u>ACCCAACTGAATGGAGCCACAAT</u> TGAATGAAAATGCT	353
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		Ba- specific- 6RUT2	CACGAAACCTGTTTACCTTT	6 <u>ACGCACTTGACTTGTCTTCCACG</u> AAACCTGTTTACCTTT	354
Assay name	Target species/ gene	Forward primer name	Forward sequence (5' -> 3')	Forward sequence w/UT	
CP0127 25.1_ 5103		Ba- specific- 8F_UT1	GATATTCGACGAGCTTTCTG	7 <u>ACCCAACTGAATGGAGCGATATT</u> CGACGAGCTTTCTG	355
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		Ba- specific- 8R_UT2	TATTCATCGTCATCCTCCTC	8 <u>ACGCACTTGACTTGTCTTCTATT</u> CATCGTCATCCTCCTC	356
Assay name	Target species/ gene	Forward primer name	Forward sequence (5' -> 3')	Forward sequence w/UT	
CP0127 25.1_ 5107		Ba- specific- 9F_UT1	TATTGAACGCATTGAATCAG	9 <u>ACCCAACTGAATGGAGCTATTGA</u> ACGCATTGAATCAG	357
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		Ba- specific- 9RUT2	TATTGGTAAGCAAACCGTCT	10 <u>ACGCACTTGACTTGTCTTCTATTG</u> GTAAGCAAACCGTCT	358

TABLE 12-continued

<i>Bacillus</i> primers and primers with Universal Tail (UT). The UT sequence is underlined.					
Assay name	Target species/ gene	Forward primer name	Forward sequence (5' -> 3')	SEQ ID NO:	SEQ ID NO:
JSZQ01 000034.1_220		Ba-specific-11F_UT1	GGTTCAGGACAAAATGTAGC	11	ACCCAACTGAATGGAGCGGTTCA GGACAAAATGTAGC
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT
		Ba-specific-11R_UT2	TAACTTCTGAAGCGAAAACC	12	ACGCACTTGACTTGTCTTCTAACT TCTGAAGCGAAAACC
JSZS010 00036.1_5		Ba-specific-12F_UT1	GCGAATTTTAGACGACAATC	13	ACCCAACTGAATGGAGCGCGAAT TTTAGACGACAATC
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse Sequence w/UT
		Ba-specific-12RUT2	TAACCGTGCTTAATTCGTTT	14	ACGCACTTGACTTGTCTTCT AACCGTGCTTAATTCGTTT
LGCC0 1000010.1_232		Ba-specific-14F_UT1	ATTAATAAGGCGACTGGTGA	15	ACCCAACTGAATGGAGCATT AATAAGGCGACTGGTGA
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse Sequence w/UT
		Ba-specific-14RUT2	TTACCCATCCAGAATGAGAC	16	ACGCACTTGACTTGTCTTCT TACCCATCCAGAATGAGAC
LGCC0 1000048.1_280		Ba-specific-16F_UT1	ACAATTCCTAAAAGGCGACA	17	ACCCAACTGAATGGAGCACA ATTCTrAAAAGGCGACA
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse Sequence w/UT
		Ba-specific-16R_UT2	TGTAGCGTCTCCGATATTTT	18	ACGCACTTGACTTGTCTTCT GTAGCGTCTCCGATATTTT
NNLO MU010000 90.1_49		Ba-specific-20FUT1	CATGGGGCTTCTATTATGT	19	ACCCAACTGAATGGAGCCATGGG GCTTCTATTATGT
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/ UT

TABLE 12-continued

Bacillus primers and primers with Universal Tail (UT). The UT sequence is underlined.						
				SEQ ID NO:	SEQ ID NO:	
		Ba-specific-20R_UT2	TTCGTTCTTTCATAAGTTTCC T	20	ACGCACTTGACTTGCTTCTTCGT TCTTTCATAAGIF1CCT	368
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
NN_LO QC 0100 0013.1_3		Ba-specific-22F_UT1	TTGGAGTITGTTITGCTTTT	21	ACCCAACTGAATGGAGCTTGGAG TTTGTTTTGCTTTT	369
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		Ba-specific-22Rv2_UT2	GTAACAATTAATCCACGTCC T	22	ACGCACTTGACTTGCTTCGTAA CAATTAATCCACGTCTCT	370
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
pX01	pX01	pX01_113F_UT1	TGAGCCTACCTAGTGATTGG	33	ACCCAACTGAATGGAGCTGAGCC TACCTAGTGATTGG	371
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		pX01-315Rv2_UT2	TTGGATAAATCCACAAATT CCTC	34	ACGCACTTGACTTGCTTCTTGG ATAAATTCACAAATTCCTC	372
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
pX02	pX02	pX02_101F_UT1	CGCCAGCGTATTATATAGGT	35	ACCCAACTGAATGGAGCGCCAG CGTATTATATAGGT	373
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		pX02_269R_UT2	GCTAATCTGGGTTGTGTTT	36	ACGCACTTGACTTGCTTCTGCTA ATTCTGGGTTGTGTTT	374
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
gyrA	gyrA	gyrA_28Fv2_UT1	TCGGTAAGTATCACCTCA	37	ACCCAACTGAATGGAGCTCGGTA AGTATCACCTCA	375
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		gyrA_182Rv2_UT2	TGCTTCTGTATAACGCATT	38	ACGCACTTGACTTGCTTCTGCTT CTGTATAACGCATT	376
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
parC	parC	parC_1F_UT1	CAGTCGGTAACGTTATTGGT	39	ACCCAACTGAATGGAGCCAGTCG GTAACGTTATTGGT	377
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		parC_197R_UT2	TAACTCAGATGCAATTGGTG	40	ACGCACTTGACTTGCTTCTAACT CAGATGCAATTGGTG	378

TABLE 12-continued

<i>Bacillus</i> primers and primers with Universal Tail (UT). The UT sequence is underlined.						
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')	SEQ ID NO:	Forward sequence w/UT	SEQ ID NO:
gyrB	gyrB	gyrB_8F_UT1	ATTGTAGAGGGTGACTCTGC	41	ACCCAACTGAATGGAGCATTTGTA <u>GAGGGTGACTCTGC</u>	379
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		gyrB_194R_UT2	TATCAAAATCTCCGCCAAT	42	ACGCACTTGACTTGTCTTCTATCA <u>AAATCTCCGCCAAT</u>	380
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
pagA	pagA	801F_pagAv3_UT1	GGTTACAGGACGGATTGAT A	31	ACCCAACTGAATGGAGCGGTAC <u>AGGACGGATTGATA</u>	381
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		1042R_pagAv3_UT2	TCCCACCAATATCAAAGAAC	32	ACGCACTTGACTTGTCTTCTCCCA <u>CCAATATCAAAGAAC</u>	382
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
rpoB	rpoB	rpoB_29F_UT1	TTCTTCGGAAGTTCTCAGTT	43	ACCCAACTGAATGGAGCTTCTTC <u>GGAAGTTCTCAGTT</u>	383
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		rpoB_196R_UT2	CGGACACATACGACCATAG	44	ACGCACTTGACTTGTCTTCCGGA <u>CACATACGACCATAG</u>	384
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
AA_2502		AA_2502_UT1	AAGTTTGAGGTGTGGAAT G	45	ACCCAACTGAATGGAGCAAGTTT <u>GAGGTGTGGAATG</u>	385
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		AA_2502_UT2	TCGAAATGAGTTCCAATTTT	46	ACGCACTTGACTTGTCTTCTCGA <u>AATGAGTTCCAATTTT</u>	386
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
AA_2503		AA_2503v2_UT1	CAAACTAATAGGGGAGGG TG	47	ACCCAACTGAATGGAGCCAAAA <u>CTAATAGGGGAGGGTG</u>	387
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		AA_2503_UT2	CCGAGAACCTACCTCGTTA	48	ACGCACTTGACTTGTCTTCCGA <u>GAACCTACCTCGTTA</u>	388
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
Ba_AmesAnc_4669915		Ba&NN32_F	AGGAGATGAGAGTTTTGCA C	49	ACCCAACTGAATGGAGCAGGAG <u>ATGAGAGTTTTGCAC</u>	389
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		Ba&NN32_R	ACCCCATTAATTACCATGA	50	ACGCACTTGACTTGTCTTACCC <u>CCATAATTACCATGA</u>	390

TABLE 12-continued

<i>Bacillus</i> primers and primers with Universal Tail (UT). The UT sequence is underlined.					
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')	SEQ ID NO:	SEQ ID NO:
Ba_AmesAnc_4001578		Ba&NN33_F	CGTTGCGTAAGTATGTGCTA	51	ACCCAAGTGAATGGAGCCGTTGC GTAAGTATGTGCTA
		Reverse primer name	Reverse sequence (5' -> 3')		
		Ba&NN33_R	AGGTGGCGTAATTAACGTA G	52	ACGCACTTGACTTGTCTTCAGGT GGCGTAATTAACGTAG
		Reverse primer name	Reverse sequence (5' -> 3')		
Ba_AmesAnc_1069024		Ba&NN37_F	CGAAAAGTTGTCGACCTAAT	53	ACCCAAGTGAATGGAGCCGAAA AGTTGTCGACCTAAT
		Reverse primer name	Reverse sequence (5' -> 3')		
		Ba&NN37_R	ACTGCGTTCACGAAGAATA G	54	ACGCACTTGACTTGTCTTCCTG CGTTCACGAAGAATAG
		Reverse primer name	Reverse sequence (5' -> 3')		
Ba_AmesAnc_3668548		Ba&NN38_F	TCTCTTGATTCAACGTTTCC	55	ACCCAAGTGAATGGAGCTCTCTT GATTCAACGTTTCC
		Reverse primer name	Reverse sequence (5' -> 3')		
		Ba&NN38_R	GATGCAAAACCAATTCACCTT	56	ACGCACTTGACTTGTCTTCGATG CAAAACCAATTCACCTT
		Reverse primer name	Reverse sequence (5' -> 3')		
Ba_AmesAnc_371913		Ba&NN40_F	GTGAAACATCGCTTTTATAGG	57	ACCCAAGTGAATGGAGCGTGAA ACATCGCTTTTATAGG
		Reverse primer name	Reverse sequence (5' -> 3')		
		Ba&NN40_R	TCCGCAATGATATACTTCAA	58	ACGCACTTGACTTGTCTTCTCCGC AATGATATACTTCAA
		Reverse primer name	Reverse sequence (5' -> 3')		
BaA_mesAnc_999035		Ba&NN41_F	ATACGGTGAAAATGAAGCA G	59	ACCCAAGTGAATGGAGCATACGG TGAAAATGAAGCAG
		Reverse primer name	Reverse sequence (5' -> 3')		
		Ba&NN41_R	CGTCTTTGGTAATCGTTCA	60	ACGCACTTGACTTGTCTTCCGTCT TTGGTAATCGTTCA
		Reverse primer name	Reverse sequence (5' -> 3')		
ChimpKiller_9-159		ChimpKiller_9F	TTATCGTCCATTCTTTCGTC	23	ACCCAAGTGAATGGAGCTTATCG TCCATTCTTTCGTC
		Reverse primer name	Reverse sequence (5' -> 3')		
		ChimpKiller_159R	AAACCTAATGAAACGGGAT T	24	ACGCACTTGACTTGTCTTCAAAC CTAATGAAACGGGATT
		Reverse primer name	Reverse sequence (5' -> 3')		

TABLE 12-continued

<i>Bacillus</i> primers and primers with Universal Tail (UT). The UT sequence is underlined.						
				SEQ ID NO:		SEQ ID NO:
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
ChimpKiller_ 91-320		ChimpKiller_ 91F	TATGAAAGGAGCCGTAAAA C	25	<u>ACCCAACTGAATGGAGCT</u> TATGAA AGGAGCCGTAAAAC	403
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		ChimpKiller_ 320R	TGAATATGAAGCGGAAAAC T	26	<u>ACGCACTTGACTTGCTTCT</u> TGAA TATGAAGCGGAAAAC	404
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
ChimpKiller_ 481-698		ChimpKiller_ 481F	TCGAACATACCTCCATTCT	27	<u>ACCCAACTGAATGGAGCT</u> CGAAC ATACCTCCATTCT	405
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		ChimpKiller_ 698R	AAAGATAGCTTTGCACTTGG	28	<u>ACGCACTTGACTTGCTTCA</u> AAG ATAGCTTTGCACTTGG	406

TABLE 13

Yersinia primers and primers with Universal Tail (UT). The UT sequence is underlined.						
				SEQ ID NO:		SEQ ID NO:
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
Ypestis_LPQY01000176.1_7		Yp&NN1_F	AACAAGCTAAAACCGAACA A	231	ACCCAACTGAATGGAGCAACAA GCTAAAACCGAACAA	407
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		Yp&NN1_R	ATAGCCTCAACTGCTTTTGT	232	ACGCACTTGACTTGCTTTCATAG CCTCAACTGCTTTTGT	408
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
Wagner_Yp_pla_Forward		Yp&NN2_F	GAAAGGAGTGCGGGTAATA GGTT	237	ACCCAACTGAATGGAGCGAAAG GAGTGCGGGTAATAGGTT	409
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		Yp&NN2_R	GGCCTGCAAGTCCAATATA TGG	238	ACGCACTTGACTTGCTTCGGCC TGCAAGTCCAATATATGG	410
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')		Forward sequence w/UT	
YenteroFR729477.2_623		Yp&NN3_F	GATGCTTCTGCTATCAGSTT	263	ACCCAACTGAATGGAGCGATGCT TCTGCTATCAGSTT	411
		Reverse primer name	Reverse sequence (5' -> 3')		Reverse sequence w/UT	
		Yp&NN3_R	GTGTGRCTTTGAATCCTTGT	264	ACGCACTTGACTTGCTTCTCGTGT GRCTTTGAATCCTTGT	412

TABLE 13-continued

<i>Yersinia</i> primers and primers with Universal Tail (UT). The UT sequence is underlined.					
			SEQ ID NO:	SEQ ID NO:	
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence (5' -> 3')	Forward sequence w/UT	
YPC092_NC_00 3143_113190		Yp&NN4_F	ACTCGGATACTCCATACT G	249 ACCCAACTGAATGGAGCACTCGG GATACTCCATACTG	413
		Reverse primer name (5' -> 3')	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		Yp&NN4_R	CGAAAGCAGTGGTCAATC	250 ACGCACTTGACTTGTCTTCCGAA AGCAGTGGTCAATC	414
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence (5' -> 3')	Forward sequence w/UT	
YPC092_NC_00 3143_161621		Yp&NN5_F	CATGCGCTTTACGTTATATG	251 ACCCAACTGAATGGAGCCATGCG CTTTACGTTATATG	415
		Reverse primer name (5' -> 3')	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		Yp&NN5_R	GCGTTCTGCACTCTGTCT	252 ACGCACTTGACTTGTCTTCGCGTT CTGCACTCTGTCT	416
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence (5' -> 3')	Forward sequence w/UT	
YPC092_NC_00 3143_152213		Yp&NN6_F	AGCGACTTCCGTGATAAAG	253 ACCC' AACTGAATGGAGCAGCGA CTTCCGTGATAAAG	417
		Reverse primer name (5' -> 3')	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		Yp&NN6_R	ACTCAGGATACCGTGTGGT	254 ACGCACTTGACTTGTCTTCACTC AGGATACCGTGTGGT	418
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence (5' -> 3')	Forward sequence w/UT	
YPC092_NC_00 3143_129539		Yp&NN7_F	TTACAGATAATCCCCTAAT G	255 ACCCAACTGAATGGAGCTTCAG ATAATCCCCTAATG	419
		Reverse primer name (5' -> 3')	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		Yp&NN7_R	TTCTGTGCTCTGGCTGATA	256 ACGCACTTGACTTGTCTTCTTCTG TGCTCTGGCTGATA	420
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence (5' -> 3')	Forward sequence w/UT	
YPC092_NC_00 3143_91203		Yp&NN8_F	ATTATCTGTGCCCTTCTTT	257 ACCCAACTGAATGGAGCATTATC TGTGCCCTTCTTT	421
		Reverse primer name (5' -> 3')	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		Yp&NN8_R	GGAGTGGATGCCACTAAAC	258 ACGCACTTGACTTGTCTTCGGAG TGGATGCCACTAAAC	422
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence (5' -> 3')	Forward sequence w/UT	
YpC092_NC_00 3143_121812		Yp&NN9_F	CCTCACACAACAATTCAC G	259 ACCCAACTGAATGGAGCCCTCAC ACAACAATTCAC	423
		Reverse primer name (5' -> 3')	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		Yp&NN9_R	TTTTTCCGACAAATTTAAG G	260 ACGCACTTGACTTGTCTTCTTTT CCGACAAATTTAAGG	424

TABLE 13-continued

<i>Yersinia</i> primers and primers with Universal Tail (UT). The UT sequence is underlined.					
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence	SEQ ID NO:	SEQ ID NO:
Yp_AL590842.1_RX_SNP		Yp&NN10_F	AGCATGAAGGTTGCTAAAA G	261	ACCCAACTGAATGGAGCAGCATG AAGGTTGCTAAAAAG
		Reverse primer name (5' -> 3')	Reverse sequence		Reverse sequence w/UT
		Yp&NN10_R	GGTGACTTCAAACCGTTA G	262	ACGCACTTGACTTGCTCTCGGTG ACTTCAAACCGTTAG
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence	Forward sequence w/UT	
AGJT01000065.1_0_338		Yp&NN11_F	CAGTACCGACAAAACCTTC	233	ACCCAACTGAATGGAGCCAGTAC CGACAAAACCTTC
		Reverse primer name (5' -> 3')	Reverse sequence		Reverse sequence w/UT
		Yp&NN11_R	TTTACTACTCTGAAAACGA G	234	ACGCACTTGACTTGCTCTTCTTAC TACTCTGAAAACGAG
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence	Forward sequence w/UT	
FAURO1000053.1_9640_796884		Yp&NN12_F	GCAC TACAAATTTAAATCC C	235	ACCCAACTGAATGGAGCGCACTA CAAATTTAAATCCC
		Reverse primer name (5' -> 3')	Reverse sequence		Reverse sequence w/UT
		Yp&NN12_R	GTCGATTATCAACCTCTAT G	236	ACGCACTTGACTTGCTCTCGTCG ATTATCAACCTCTATG
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence	Forward sequence w/UT	
YpPGM_8-158	PGM	YpPGM_8F	TTAATATCCCGCACTCAT A	239	ACCCAACTGAATGGAGCTTAATA TCCCGCACTCATA
		Reverse primer name (5' -> 3')	Reverse sequence		Reverse sequence w/UT
		YpPGM_158R	TCCTTAAGTGAATAAGTGC TCA	240	ACGCACTTGACTTGCTCTTCCTT AACTGAATAAGTGCTCA
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence	Forward sequence w/UT	
YpPGM_31-205	PGM	YpPGM_31F v2	TTTAATGAACGGTGCCTAG	241	ACCCAACTGAATGGAGCTTTAAT GAACGGTGCCTAG
		Reverse primer name (5' -> 3')	Reverse sequence		Reverse sequence w/UT
		YpPGM_205Rv2	GTCTGCGTTTCTCCAGTAT	242	ACGCACTTGACTTGCTCTCGTCTG CGTTTCTCCAGTAT
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence	Forward sequence w/UT	
Yp-p1202_42780-43194	p1P1202	Yp-P1202_42780 F-UT1	TCTGGCCTGCTAAATAAAA ACGAACC	243	ACCCAACTGAAIGGAGCTCTGGC CTGCTAAATAAAAACGAACC

TABLE 13-continued

<i>Yersinia</i> primers and primers with Universal Tail (UT). The UT sequence is underlined.				
			SEQ ID NO:	SEQ ID NO:
Assay name	Target species/gene	Reverse primer name (5' -> 3')	Reverse sequence w/UT	
		Yp-pl202_43194 R-UT2	CAGGCCTCAGCATTTTATT ATGGTGAT	244 AC ^{CG} CACTTGACTTGTCTTCCAGG CCTCAGCATTTTATTATGGTGAT 436
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence w/UT	
Yp-pl202_1 26386- 126750	plP1202	Yp-P1202_12638 6F-UT1	GGGGCGGATACCTTCACCT ATG	245 ACCCAACTGAATGGAGCGGGGC GGATACCTTCACCTATG 437
		Reverse primer name (5' -> 3')	Reverse sequence w/UT	
		Yp-pl202_126750R-UT2	CTGGGGTTCAGTCTGGACG AGAT	246 AC ^{CG} CACTTGACTTGTCTTCCTGG GGTTCAGTCTGGACGAGAT 438
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence w/UT	
Yp-pl202_156402- 156711	plP1202	Yp-pl202_15640 2F2-UT1	ACCATCCGGCGCTAAATCG TC	247 ACCCAACTGAATGGAGCACCATC CGGCGCTAAATCGTC 439
		Reverse primer name (5' -> 3')	Reverse sequence w/UT	
		Yp-pl202_15671 1R-UT2	GAAATGCGCCTGGTAAGCA GAGT	248 AC ^{CG} CACTTGACTTGTCTTCGAAA TGCGCCTGGTAAGCAGAGT 440

TABLE 14

<i>Francisella</i> primers and primers with Universal Tail (UT). The UT sequence is underlined.				
			SEQ ID NO:	SEQ ID NO:
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence w/UT	
Ftnovicida_CP009607.1		Ft&NN1_F	GGTAGGATAACTACCAAG	285 ACCCAACTGAATGGAGCGGTAG GATAACTACCAAG 441
		Reverse primer name (5' -> 3')	Reverse sequence w/UT	
		Ft&NN1_R	GTCATGAGTTTTACCAATA CTC	286 AC ^{CG} CACTTGACTTGTCTTCGTCAT GAGTTTACCAATACTC 442
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence w/UT	
Ftularensis_CP00915.1_1782		Ft&NN2_F	GAAGTGGCTCATGTTAGAG G	265 ACCCAACTGAATGGAGCGAAGT GGCTCATGTTAGAGG 443
		Reverse primer name (5' -> 3')	Reverse sequence w/UT	
		Ft&NN2_R	AGCGAGCCTATATGTAACC A	266 AC ^{CG} CACTTGACTTGTCTTCAGCG AGCCTATATGTAACCA 444

TABLE 14-continued

Francisella primers and primers with Universal Tail (UT). The UT sequence is underlined.				
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence	Forward sequence w/UT
Ftularensis_CP000915.1_731		Ft&NN3_F	TTTAATGTCCGTCACCTCT	267 ACCCAACTGAATGGAGCTTTAAT GTCCGTCACCTCT
		Reverse primer name (5' -> 3')	Reverse sequence	Reverse sequence w/UT
		Ft&NN3_R	ACGAGTTTGTGAGTCGCTA T	268 ACGCACTTGACTTGCTTTCACGA GTTTGTGAGTCGCTAT
		Reverse primer name (5' -> 3')	Reverse sequence	Reverse sequence w/UT
Fhoatunensis_CP003402.1_1749		Ft&NN4_F	CGGTAAGAATACGACCAGA G	291 ACCCAACTGAATGGAGCCGGTAA GAATACGACCAGAG
		Reverse primer name (5' -> 3')	Reverse sequence	Reverse sequence w/UT
		Ft&NN4_R	AGAGGATTCTTCCTCCTTG	292 ACGCACTTGACTTGCTTTCAGAG GATTCTTCCTCCTTG
		Reverse primer name (5' -> 3')	Reverse sequence	Reverse sequence w/UT
Fhoatunensis_CP003402.1_424		Ft&NN5_F	AATTCTACAAGCACCTGGA A	293 ACCCAACTGAATGGAGCAATTCT ACAAGCACCTGGAA
		Reverse primer name (5' -> 3')	Reverse sequence	Reverse sequence w/UT
		Ft&NN5R	TCCTATTAAGCGCCATA G	294 ACGCACTTGACTTGCTTCTCCTA TTAAAGCGCCATAG
		Reverse primer name (5' -> 3')	Reverse sequence	Reverse sequence w/UT
Fphilom_CP009444.1_569		Ft&NN6F	CTTATGCAGCAAGAGGAAC T	287 ACCCAACTGAATGGAGCCTTATG CAGCAAGAGGAAC
		Reverse primer name (5' -> 3')	Reverse sequence	Reverse sequence w/UT
		Ft&NN6_R	ATACACCGGATAGGTTTC T	288 ACGCACTTGACTTGCTTTCATAC ACCGGATAGGTTTCT
		Reverse primer name (5' -> 3')	Reverse sequence	Reverse sequence w/UT
Fphilom_CP009444.1_285		Ft&NN7_F	CTGATGGAAGAGAGTTCGA G	289 ACCCAACTGAATGGAGCCTGATG GAAGAGAGTTTCGAG
		Reverse primer name (5' -> 3')	Reverse sequence	Reverse sequence w/UT
		Ft&NN7_Rv2	GTAGATATAATCAGCGCCA C	290 ACGCACTTGACTTGCTTTCGTAG ATATAATCAGCGCCAC
		Reverse primer name (5' -> 3')	Reverse sequence	Reverse sequence w/UT
Ft_dup_CP000915.1_197		Ft&NN8_F	TGTTACGTACAGGCTGTCA A	269 ACCCAACTGAATGGAGCTGTTAC GTACAGGCTGTCAA
		Reverse primer name (5' -> 3')	Reverse sequence	Reverse sequence w/UT
		Ft&NN8_R	ATCATATCCCGTAGCACAA G	270 ACGCACTTGACTTGCTTTCATCAT ATCCCGTAGCACAAAG
		Reverse primer name (5' -> 3')	Reverse sequence	Reverse sequence w/UT

TABLE 14-continued

Francisella primers and primers with Universal Tail (UT). The UT sequence is underlined.					
			SEQ ID NO:	SEQ ID NO:	
Assay name	Target species/gene	Forward primer name	Forward sequence (5' -> 3')	Forward sequence w/UT	
FtLVS_AM2333 62_1646546		Ft&NN9_F	ATCAAGCTCATCTTCAAAG C	279 ACCCAACTGAATGGAGCATCAAG CTCATCTTCAAAGC	457
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		Ft&NN9_R	AACCATGTTCAGATCCAAA A	280 ACGCACTTGACTTGTCTTCAACC ATGTTTCAGATCCAAA	458
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
FtLVS_AM2333 62_1643765		Ft&NN10_F	TACCTCTGCCAAAAATTCA T	281 ACCCAACTGAATGGAGCTACCTC TGCCAAAAATTCAT	459
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		Ft&NN10_R	GGCATACTCAAGGTAGTGG T	282 ACGCACTTGACTTGTCTTCGGCA TACTCAAGGTAGTGGT	460
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
FtLVS_AM2333 62_1562618		Ft&NN11_F	TCTTTGGTAGCTTGCTGACT	283 ACCCAACTGAATGGAGCTCTTTG GTAGCTTGCTGACT	461
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		Ft&NN11_R	CAGACGACACTTGGCTTAT T	284 ACGCACTTGACTTGTCTTCCAGA CGACACTTGGCTTATT	462
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
FtA1	FtA1	9F_FtA1_UT 1	CATAACCCATCGCAATATC T	271 ACCCAACTGAATGGAGCCATAAC CCATCGCAATATCT	463
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		246R_FtA1_ UT2	AAATTATCTGTAGCGGCAA A	272 ACGCACTTGACTTGTCTTCAAAT TATCTGTAGCGGCAA	464
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
FtA2	FtA2	34F_FtA2_U T1	GTGTCCAACGAAACCATAA T	273 ACCCAACTGAATGGAGCGTGTCC AACGAAACCATAAT	465
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		169R_FtA2_ UT2	TTTGGTTGATTCTGTCAGTG	274 ACGCACTTGACTTGTCTTCTTTGG TTGATTCTGTCAGTG	466
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
FtB	FtB	28F_FtB_UT 1	AAGCTTAACGGTGATTGG A	275 ACCCAACTGAATGGAGCAAGCTT AACTGGTGATTGGA	467
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	
		173R_FtB_U T2	CGCCTAACATCTTATCTGCT	276 ACGCACTTGACTTGTCTTCCGCCT AACATCTTATCTGCT	468
		Reverse primer name	Reverse sequence (5' -> 3')	Reverse sequence w/UT	

TABLE 14-continued

Francisella primers and primers with Universal Tail (UT). The UT sequence is underlined.					
			SEQ ID NO:		SEQ ID NO:
Assay name	Target species/gene	Forward primer name (5' -> 3')	Forward sequence	Forward sequence w/UT	
FtA	FtA	14F_FtA_UT 1	GGGTGATGCAGTAGAGAAA A	277 ACCCAACTGAATGGAGCGGGTG ATGCAGTAGAGAAA	469
		Reverse primer name (5' -> 3')	Reverse sequence	Reverse sequence w/UT	
		207R_FtA_U T2	TACCAGATGAACGAATAGC C	278 ACGCACTTGACTTGTCTTCTACC AGATGAACGAATAGCC	470

All headings are for the convenience of the reader and should not be used to limit the meaning of the text that follows the heading, unless so specified.

Unless defined otherwise, all technical and scientific terms herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although any methods and materials, similar or equivalent to those described herein, can be used in the practice or testing of the present invention, the preferred methods and materials are described herein. All publications, patents, and patent publications cited are incorporated by reference herein in their entirety for all purposes.

The publications discussed herein are provided solely for their disclosure prior to the filing date of the present appli-

cation. Nothing herein is to be construed as an admission that the present invention is not entitled to antedate such publication by virtue of prior invention.

While the invention has been described in connection with specific embodiments thereof, it will be understood that it is capable of further modifications and this application is intended to cover any variations, uses, or adaptations of the invention following, in general, the principles of the invention and including such departures from the present disclosure as come within known or customary practice within the art to which the invention pertains and as may be applied to the essential features hereinbefore set forth and as follows in the scope of the appended claims.

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cgggcgtgcc ggactcctcg tc 22

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gagccaacgc acatggacgg 20

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<212> TYPE: DNA

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24

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31

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26

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22

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25

<210> SEQ ID NO 97
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29

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26

<210> SEQ ID NO 99
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25

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<212> TYPE: DNA
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21

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22

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cgaaaataat ttctggccgg cgcac 25

<210> SEQ ID NO 104
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<212> TYPE: DNA
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cgacaggcat cgggcgacta ctaccag 27

<210> SEQ ID NO 105
<211> LENGTH: 25
<212> TYPE: DNA
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caacgggcga gtttgcaacg gaatc 25

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<212> TYPE: DNA
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gccggcttgg cttcgtcctt gtc 23

<210> SEQ ID NO 107
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<212> TYPE: DNA
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cggcatgcgc ggccgaatc 19

<210> SEQ ID NO 108
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<212> TYPE: DNA
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<400> SEQUENCE: 108

atcgcgccct gcagcgagca c 21

<210> SEQ ID NO 109
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<213> ORGANISM: Artificial Sequence
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<400> SEQUENCE: 109

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<210> SEQ ID NO 110
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<212> TYPE: DNA
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<400> SEQUENCE: 110

agaggaagaa gggcgaggcg 20

<210> SEQ ID NO 111
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<400> SEQUENCE: 111

cgcgcarytc gtcgtcctcg 20

<210> SEQ ID NO 112
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cgaacctsgt gcmggtrcag 20

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<400> SEQUENCE: 113

cacgttgccs ggraartacg 20

<210> SEQ ID NO 114
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<212> TYPE: DNA
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<400> SEQUENCE: 114

ccgtcgacaa gatcgcgcts 20

<210> SEQ ID NO 115
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<212> TYPE: DNA
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<212> TYPE: DNA
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<400> SEQUENCE: 116

tgccgcgtga tccattgcat

20

<210> SEQ ID NO 117
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<212> TYPE: DNA
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<400> SEQUENCE: 117

aggggggtggt ttctgagtg gcgtgac

27

<210> SEQ ID NO 118
<211> LENGTH: 26
<212> TYPE: DNA
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<400> SEQUENCE: 118

agcgggtgtcg acgggtggaa aggatg

26

<210> SEQ ID NO 119
<211> LENGTH: 26
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 119

acgggcgctt cagcatctcg gtgttc

26

<210> SEQ ID NO 120
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 120

gcgcggcagt tcgacaggc atttg

25

<210> SEQ ID NO 121
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 121

gacggcgggc tttggggagt cc

22

<210> SEQ ID NO 122
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<212> TYPE: DNA
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<223> OTHER INFORMATION: Primer

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<210> SEQ ID NO 123

<211> LENGTH: 21

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<211> LENGTH: 32

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 126

cgaggggcga aattcccctt atagatcagt tg

32

<210> SEQ ID NO 127

<211> LENGTH: 26

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 127

cggtcgccac aaattcgcac gcactc

26

<210> SEQ ID NO 128

<211> LENGTH: 26

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 128

agcgagcggc gcaacggaga atgatt

26

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 129

gctgcgcggc caagcgaaaa acg 23

<210> SEQ ID NO 130
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 130

cgcgaggacc gcagcgcaaa gc 22

<210> SEQ ID NO 131
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 131

ggccgcagac cgtcaccgcg tatg 24

<210> SEQ ID NO 132
<211> LENGTH: 25
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 132

gtcgcccgtc ttgttgccga gcatc 25

<210> SEQ ID NO 133
<211> LENGTH: 26
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 133

gcccgtcaat ccgatgcmgt atctgg 26

<210> SEQ ID NO 134
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 134

gcgccgatca rtggggtgga aatg 24

<210> SEQ ID NO 135
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 135

catcgcggcg acgagcggtt cc

22

<210> SEQ ID NO 136

<211> LENGTH: 28

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<400> SEQUENCE: 136

ctcgggtgatc ggcgaatagc ggatgaga

28

<210> SEQ ID NO 137

<211> LENGTH: 24

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer

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<210> SEQ ID NO 138

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<212> TYPE: DNA

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<220> FEATURE:

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<400> SEQUENCE: 138

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<220> FEATURE:

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atccgccgtc ccgccagca atag

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<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<211> LENGTH: 22

<212> TYPE: DNA

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<400> SEQUENCE: 141

tcgccgtttc acgcccgca ac

22

<210> SEQ ID NO 142

<211> LENGTH: 22

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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 142

gcgccgaacg cgaggaacac ga 22

<210> SEQ ID NO 143
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 143

gctcgcggaag ctcgcgctga acc 23

<210> SEQ ID NO 144
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 144

ggatcggtgcc gtgcccgcga tac 23

<210> SEQ ID NO 145
<211> LENGTH: 25
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 145

acaagcccgg ygtcgagatg gtgac 25

<210> SEQ ID NO 146
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 146

cgcgtcggcc gaayggctgt agt 23

<210> SEQ ID NO 147
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 147

aatcgtcggc tgcgtcgct tca 23

<210> SEQ ID NO 148
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 148

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cgggtagcgt gagtggaatt cgcagag 27

<210> SEQ ID NO 149
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 149

cctcgaaggc ttcgggctga tccag 25

<210> SEQ ID NO 150
<211> LENGTH: 29
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 150

gactaaccgc ttacgccacc cactcgttc 29

<210> SEQ ID NO 151
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 151

aaagcgaata gtcggaagc ggcttga 27

<210> SEQ ID NO 152
<211> LENGTH: 29
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 152

gcgatctcgg tgatgatctt gatgcagtg 29

<210> SEQ ID NO 153
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 153

aacggcggcg tgaccgtcaa cg 22

<210> SEQ ID NO 154
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 154

cgctacgcgg ccacctgccc 20

<210> SEQ ID NO 155
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 155

cggagcttca gaacaaccg cgtgtaac 28

<210> SEQ ID NO 156
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 156

ccttcggacc tttcccgca actggc 26

<210> SEQ ID NO 157
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 157

aattcggccg gcaggcggta cg 22

<210> SEQ ID NO 158
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 158

cgcgcgagc cggcatttg 19

<210> SEQ ID NO 159
<211> LENGTH: 27
<212> TYPE: DNA
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cccttcggtc cccaccagaa aaattcg 27

<210> SEQ ID NO 160
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<212> TYPE: DNA
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<400> SEQUENCE: 160

agccgtacag gccaatgcag ccatctatg 29

<210> SEQ ID NO 161
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 161

gcgcgcgtg ttcgtagcct tgtc 24

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<210> SEQ ID NO 162
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 162

cgctgacgga cgcatcggaac ac 22

<210> SEQ ID NO 163
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 163

atcgatgatat cgccggttcc tgggtgtg 28

<210> SEQ ID NO 164
<211> LENGTH: 27
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 164

cacgtttggc ggcagtgcaa ggtgtag 27

<210> SEQ ID NO 165
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 165

tctgctgac gcccttcgcc agatayac 28

<210> SEQ ID NO 166
<211> LENGTH: 34
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 166

gcggatgaac aatttcctgt cgagcgacta ttac 34

<210> SEQ ID NO 167
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 167

gcagggcgcc ttgatccg ctatgag 27

<210> SEQ ID NO 168
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 168

cggcgcaagg ttctcctgcc acatc

25

<210> SEQ ID NO 169

<211> LENGTH: 26

<212> TYPE: DNA

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<211> LENGTH: 25

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 170

caagccgctg ataccctgtg cgctg

25

<210> SEQ ID NO 171

<211> LENGTH: 28

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 171

gcgcttctcg gtgggtacga aaaacagc

28

<210> SEQ ID NO 172

<211> LENGTH: 28

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 172

cgagtcggcc aagatcattc aggaccag

28

<210> SEQ ID NO 173

<211> LENGTH: 26

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer

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<210> SEQ ID NO 174

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<212> TYPE: DNA

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<400> SEQUENCE: 174

cttccttcgc acaaccgagc aaatactgag taaatc

36

<210> SEQ ID NO 175

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aacaagctaa aaccgaacaa 20

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22

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28

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28

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atccgccgtc ccgccagca atag 24

<210> SEQ ID NO 306
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<223> OTHER INFORMATION: Primer

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<212> TYPE: DNA

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<223> OTHER INFORMATION: Primer

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<211> LENGTH: 37

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 455

acccaactga atggagctgt tacgtacagg ctgtcaa 37

<210> SEQ ID NO 456

<211> LENGTH: 39

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 456

acgcacttga cttgtcttca tcatatcccg tagcacaag 39

<210> SEQ ID NO 457

<211> LENGTH: 37

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 457

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<210> SEQ ID NO 458

<211> LENGTH: 39

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 458

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<210> SEQ ID NO 459
<211> LENGTH: 37
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 459

acccaactga atggagctac ctctgccaaa aattcat 37

<210> SEQ ID NO 460
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 460

acgcacttga cttgtcttcg gcatactcaa ggtagtggt 39

<210> SEQ ID NO 461
<211> LENGTH: 37
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 461

acccaactga atggagctct ttggtagctt gctgact 37

<210> SEQ ID NO 462
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 462

acgcacttga cttgtcttcc agacgacact tggcttatt 39

<210> SEQ ID NO 463
<211> LENGTH: 37
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 463

acccaactga atggagccat aacccatcgc aatatct 37

<210> SEQ ID NO 464
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<210> SEQ ID NO 465
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 465

acccaactga atggagcgtg tccaacgaaa ccataat 37

<210> SEQ ID NO 466
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 466

acgcacttga cttgtcttct ttggttgatt ctgtcagtg 39

<210> SEQ ID NO 467
<211> LENGTH: 37
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 467

acccaactga atggagcaag cttaactggt gattgga 37

<210> SEQ ID NO 468
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 468

acgcacttga cttgtcttcc gcctaacatc ttatctgct 39

<210> SEQ ID NO 469
<211> LENGTH: 37
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 469

acccaactga atggagcggg tgatgcagta gagaaaa 37

<210> SEQ ID NO 470
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Primer

<400> SEQUENCE: 470

acgcacttga cttgtcttct accagatgaa cgaatagcc 39

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What is claimed is:

1. A method of detecting *Bacillus anthracis* in a sample, comprising:

generating one or more amplicons from the sample using at least one primer pair comprising a forward and reverse primer in at least one amplification reaction; sequencing the one or more amplicons using next-generation sequencing; and

detecting at least one *B. anthracis*-specific amplicon in the sequenced amplicons, wherein each forward and reverse primer comprise a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 1 and SEQ ID NO: 2;

SEQ ID NO: 3 and SEQ ID NO: 4;

SEQ ID NO: 5 and SEQ ID NO: 6;

SEQ ID NO: 7 and SEQ ID NO: 8;

SEQ ID NO: 9 and SEQ ID NO: 10;

SEQ ID NO: 11 and SEQ ID NO: 12;

SEQ ID NO: 13 and SEQ ID NO: 14;

SEQ ID NO: 15 and SEQ ID NO: 16; and

SEQ ID NO: 17 and SEQ ID NO: 18;

wherein the presence of the at least one *B. anthracis*-specific amplicon indicates the presence of *B. anthracis* in the sample, and the absence of the at least one *B. anthracis*-specific amplicon indicates the absence of *B. anthracis* from the sample.

2. The method of claim 1, further comprising confirming the absence of *B. anthracis* by detecting at least one *B. anthracis* Near Neighbor-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 19 and SEQ ID NO: 20;

SEQ ID NO: 21 and SEQ ID NO: 22; and

SEQ ID NO: 23 and SEQ ID NO: 24;

wherein detecting at least one *B. anthracis* Near Neighbor-specific amplicon in the sample confirms the absence of *B. anthracis*.

3. The method of claim 1, further comprising confirming the absence of *B. anthracis* by detecting at least one *B. anthracis* Near Neighbor-specific sequence variant (SV) or single nucleotide polymorphism (SNP) using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 25 and SEQ ID NO: 26;

SEQ ID NO: 27 and SEQ ID NO: 28;

SEQ ID NO: 49 and SEQ ID NO: 50;

SEQ ID NO: 51 and SEQ ID NO: 52;

SEQ ID NO: 53 and SEQ ID NO: 54;

SEQ ID NO: 55 and SEQ ID NO: 56;

SEQ ID NO: 57 and SEQ ID NO: 58; and

SEQ ID NO: 59 and SEQ ID NO: 60;

wherein detecting at least one *B. anthracis* Near Neighbor-specific SV in the sample confirms the absence of *B. anthracis*.

4. The method of claim 1, further comprising detecting a virulence locus or virulence plasmid in the sample by detecting a virulence-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 29 and SEQ ID NO: 30;

SEQ ID NO: 31 and SEQ ID NO: 32;

SEQ ID NO: 33 and SEQ ID NO: 34; and

SEQ ID NO: 35 and SEQ ID NO: 36;

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wherein the presence of the virulence-specific amplicon indicates the presence of the virulence locus or virulence plasmid in the sample.

5. The method of claim 1, further comprising detecting at least one drug resistance single nucleotide polymorphism (SNP) from *B. anthracis* in the sample using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 37 and SEQ ID NO: 38;

SEQ ID NO: 39 and SEQ ID NO: 40;

SEQ ID NO: 41 and SEQ ID NO: 42;

SEQ ID NO: 43 and SEQ ID NO: 44;

SEQ ID NO: 45 and SEQ ID NO: 46; and

SEQ ID NO: 47 and SEQ ID NO: 48.

6. The method of claim 1, further comprising detecting *Burkholderia pseudomallei* and/or *Burkholderia mallei* in the sample by detecting at least one *B. pseudomallei* or *B. mallei*-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 61 and SEQ ID NO: 62;

SEQ ID NO: 63 and SEQ ID NO: 64;

SEQ ID NO: 65 and SEQ ID NO: 66;

SEQ ID NO: 67 and SEQ ID NO: 68;

SEQ ID NO: 69 and SEQ ID NO: 70;

SEQ ID NO: 71 and SEQ ID NO: 72;

SEQ ID NO: 73 and SEQ ID NO: 74;

SEQ ID NO: 75 and SEQ ID NO: 76;

SEQ ID NO: 77 and SEQ ID NO: 78;

SEQ ID NO: 79 and SEQ ID NO: 80;

SEQ ID NO: 81 and SEQ ID NO: 82;

SEQ ID NO: 83 and SEQ ID NO: 84;

SEQ ID NO: 85 and SEQ ID NO: 86;

SEQ ID NO: 87 and SEQ ID NO: 88;

SEQ ID NO: 89 and SEQ ID NO: 90;

SEQ ID NO: 91 and SEQ ID NO: 92;

SEQ ID NO: 93 and SEQ ID NO: 94;

SEQ ID NO: 95 and SEQ ID NO: 96;

SEQ ID NO: 97 and SEQ ID NO: 98;

SEQ ID NO: 99 and SEQ ID NO: 100;

SEQ ID NO: 101 and SEQ ID NO: 102;

SEQ ID NO: 103 and SEQ ID NO: 104;

SEQ ID NO: 103 and SEQ ID NO: 104;

SEQ ID NO: 105 and SEQ ID NO: 106;

SEQ ID NO: 107 and SEQ ID NO: 108;

SEQ ID NO: 117 and SEQ ID NO: 118;

SEQ ID NO: 119 and SEQ ID NO: 120;

SEQ ID NO: 121 and SEQ ID NO: 122;

SEQ ID NO: 123 and SEQ ID NO: 124; and

SEQ ID NO: 125 and SEQ ID NO: 126;

wherein the presence of the at least one *B. pseudomallei* or *B. mallei*-specific amplicon indicates the presence of *B. pseudomallei* and/or *B. mallei* in the sample, and an absence of the at least one *B. pseudomallei* or *B. mallei*-specific amplicon indicates an absence of *B. pseudomallei* and *B. mallei* in the sample.

7. The method of claim 6, further comprising confirming the absence of *B. pseudomallei* and *B. mallei* by detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 177 and SEQ ID NO: 178;

SEQ ID NO: 179 and SEQ ID NO: 180;

SEQ ID NO: 181 and SEQ ID NO: 182;

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SEQ ID NO: 183 and SEQ ID NO: 184;
 SEQ ID NO: 185 and SEQ ID NO: 186;
 SEQ ID NO: 187 and SEQ ID NO: 188;
 SEQ ID NO: 189 and SEQ ID NO: 190;
 SEQ ID NO: 191 and SEQ ID NO: 192;
 SEQ ID NO: 193 and SEQ ID NO: 194;
 SEQ ID NO: 195 and SEQ ID NO: 196;
 SEQ ID NO: 197 and SEQ ID NO: 198;
 SEQ ID NO: 199 and SEQ ID NO: 200;
 SEQ ID NO: 201 and SEQ ID NO: 202;
 SEQ ID NO: 203 and SEQ ID NO: 204;
 SEQ ID NO: 205 and SEQ ID NO: 206;
 SEQ ID NO: 207 and SEQ ID NO: 208;
 SEQ ID NO: 207 and SEQ ID NO: 208;
 SEQ ID NO: 209 and SEQ ID NO: 210;
 SEQ ID NO: 211 and SEQ ID NO: 212;
 SEQ ID NO: 213 and SEQ ID NO: 214;
 SEQ ID NO: 215 and SEQ ID NO: 216;
 SEQ ID NO: 217 and SEQ ID NO: 218;
 SEQ ID NO: 219 and SEQ ID NO: 220;
 SEQ ID NO: 221 and SEQ ID NO: 222;
 SEQ ID NO: 223 and SEQ ID NO: 224;
 SEQ ID NO: 225 and SEQ ID NO: 226;
 SEQ ID NO: 227 and SEQ ID NO: 228; and
 SEQ ID NO: 229 and SEQ ID NO: 230;

wherein detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific amplicon in the sample confirms the absence of *B. pseudomallei* and *B. mallei*.

8. The method of claim 6, further comprising confirming the absence of *B. pseudomallei* and *B. mallei* by detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific SNP or SV using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 109 and SEQ ID NO: 110;
 SEQ ID NO: 111 and SEQ ID NO: 112;
 SEQ ID NO: 113 and SEQ ID NO: 114; and
 SEQ ID NO: 115 and SEQ ID NO: 116;

wherein detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific SNP or SV in the sample confirms the absence of *B. pseudomallei* and *B. mallei*.

9. The method of claim 6, further comprising detecting at least one drug resistance SNP or SV from *Burkholderia* spp. in the sample using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 127 and SEQ ID NO: 128;
 SEQ ID NO: 129 and SEQ ID NO: 130;
 SEQ ID NO: 131 and SEQ ID NO: 132;
 SEQ ID NO: 133 and SEQ ID NO: 134;
 SEQ ID NO: 135 and SEQ ID NO: 136;
 SEQ ID NO: 137 and SEQ ID NO: 138;
 SEQ ID NO: 145 and SEQ ID NO: 146;
 SEQ ID NO: 147 and SEQ ID NO: 148;
 SEQ ID NO: 149 and SEQ ID NO: 150;
 SEQ ID NO: 151 and SEQ ID NO: 152; and
 SEQ ID NO: 153 and SEQ ID NO: 154.

10. The method of claim 1, further comprising detecting *Francisella tularensis* in the sample by detecting at least one *F. tularensis*-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 265 and SEQ ID NO: 266;
 SEQ ID NO: 267 and SEQ ID NO: 268; and
 SEQ ID NO: 269 and SEQ ID NO: 270;

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wherein the presence of the at least one *F. tularensis*-specific amplicon indicates that *F. tularensis* is present in the sample, and an absence of the at least one *F. tularensis*-specific amplicon indicates that *F. tularensis* is absent in the sample.

11. The method of claim 10, further comprising confirming the absence of *F. tularensis* by detecting at least one *F. tularensis* Near Neighbor-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 285 and SEQ ID NO: 286;
 SEQ ID NO: 287 and SEQ ID NO: 288;
 SEQ ID NO: 289 and SEQ ID NO: 290;
 SEQ ID NO: 291 and SEQ ID NO: 292; and
 SEQ ID NO: 293 and SEQ ID NO: 294;

wherein detecting at least one *F. tularensis* Near Neighbor-specific amplicon in the sample confirms the absence of *F. tularensis*.

12. The method of claim 10, further comprising confirming the absence of *F. tularensis* by detecting at least one *F. tularensis* Near Neighbor-specific SNP or SV using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 271 and SEQ ID NO: 272;
 SEQ ID NO: 273 and SEQ ID NO: 274;
 SEQ ID NO: 275 and SEQ ID NO: 276;
 SEQ ID NO: 277 and SEQ ID NO: 278;
 SEQ ID NO: 279 and SEQ ID NO: 280;
 SEQ ID NO: 281 and SEQ ID NO: 282; and
 SEQ ID NO: 283 and SEQ ID NO: 284;

wherein detecting at least one *F. tularensis* Near Neighbor-specific SNP or SV in the sample confirms the absence of *F. tularensis*.

13. The method of claim 1, further comprising detecting *Yersinia pestis* in the sample by detecting at least one *Y. pestis*-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 231 and SEQ ID NO: 232;
 SEQ ID NO: 233 and SEQ ID NO: 234;
 SEQ ID NO: 235 and SEQ ID NO: 236; and
 SEQ ID NO: 237 and SEQ ID NO: 238;

wherein the presence of at least one *Y. pestis*-specific amplicon indicates the presence of *Y. pestis* in the sample, and an absence of at least one *Y. pestis*-specific amplicon indicates an absence of *Y. pestis* in the sample.

14. The method of claim 13, further comprising confirming the absence of *Y. pestis* by detecting at least one *Y. pestis* Near Neighbor-specific SNP or SV using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 249 and SEQ ID NO: 250;
 SEQ ID NO: 251 and SEQ ID NO: 252;
 SEQ ID NO: 253 and SEQ ID NO: 254;
 SEQ ID NO: 255 and SEQ ID NO: 256;
 SEQ ID NO: 257 and SEQ ID NO: 258;
 SEQ ID NO: 259 and SEQ ID NO: 260; and
 SEQ ID NO: 261 and SEQ ID NO: 262;

wherein detecting at least one *Y. pestis* Near Neighbor-specific SNP or SV confirms the absence of *Y. pestis*.

15. The method of claim 13, further comprising confirming the absence of *Y. pestis* by detecting at least one *Y. pestis* Near Neighbor-specific amplicon using at least one primer

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pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 263 and SEQ ID NO: 264;

wherein detecting at least one *Y. pestis* Near Neighbor-specific amplicon confirms the absence of *Y. pestis*. 5

16. The method of claim 13, further comprising characterizing and/or subtyping *Y. pestis* in the sample by detecting at least one amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of:

SEQ ID NO: 239 and SEQ ID NO: 240;

SEQ ID NO: 241 and SEQ ID NO: 242;

SEQ ID NO: 243 and SEQ ID NO: 244;

SEQ ID NO: 245 and SEQ ID NO: 246; and 15

SEQ ID NO: 247 and SEQ ID NO: 248.

17. The method of claim 1, wherein the one or more amplicons are generated with at least one multiplex amplification reaction; and each primer in the at least one primer pair comprises a universal tail sequence. 20

18. The method of claim 17, wherein the at least one amplicon is present when a locus read count of the at least one amplicon is at least 10 sequence reads covering at least 75% of a corresponding amplicon reference sequence, and the universal tail sequence for each forward and reverse primer comprises SEQ ID NO: 301 and SEQ ID NO: 303. 25

19. The method of claim 1, wherein the sample is a biological sample obtained from a subject and further comprising administering an effective amount of at least one antibiotic to the subject,

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wherein the at least one antibiotic is selected from the group consisting of a fluoroquinolone, an aminoglycoside, a glycopeptide, a lincosamide, a macrolide/ketolide, a cephalosporin, a monobactam, a nitroimidazole, a penicillin, a streptogramin, a tetracycline, and a physiologically acceptable salt, prodrug, or combination thereof.

20. The method of 19, wherein

the at least one antibiotic is not a fluoroquinolone if a *gyrA* drug resistance SNP is detected; and/or

the at least one antibiotic is not a fluoroquinolone if a *parC* drug resistance SNP is detected; and/or

the at least one antibiotic is not a fluoroquinolone or an aminocoumarin if a *gyrB* drug resistance SNP is detected; and/or

the at least one antibiotic is not a rifamycin if a *rpoB* drug resistance SNP is detected; and/or

the at least one antibiotic is not a β -lactam if a *penA* drug resistance SNP is detected; and/or

the at least one antibiotic is not a trimethoprim and sulfamethoxazole combination, co-trimoxazole, if a *folM* drug resistance SV is detected; and/or

the at least one antibiotic is not a trimethoprim and sulfamethoxazole combination, co-trimoxazole, if a *bpeT* drug resistance SV is detected; and/or

the at least one antibiotic is not a trimethoprim and sulfamethoxazole combination, co-trimoxazole, if a *bpeS* drug resistance SV is detected.

* * * * *